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(12) **United States Patent**
Anderson et al.

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(45) **Date of Patent:** **Sep. 9, 2025**

(54) **ELECTRONIC DEVICE**

(71) Applicant: **Apple Inc.**, Cupertino, CA (US)

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(73) Assignee: **APPLE INC.**, Cupertino, CA (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 338 days.

(21) Appl. No.: **18/151,177**

(22) Filed: **Jan. 6, 2023**

(65) **Prior Publication Data**

US 2023/0229117 A1 Jul. 20, 2023

Related U.S. Application Data

(60) Provisional application No. 63/374,738, filed on Sep. 6, 2022, provisional application No. 63/364,012, filed (Continued)

(51) **Int. Cl.**

G04G 17/04 (2006.01)

G04G 17/08 (2006.01)

G04G 21/08 (2010.01)

(52) **U.S. Cl.**

CPC **G04G 17/045** (2013.01); **G04G 17/08** (2013.01); **G04G 21/08** (2013.01)

(58) **Field of Classification Search**

CPC G04G 17/045; G04G 21/08; G04G 17/02; G04G 17/04; G04G 21/00; G04B 3/048; (Continued)

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Primary Examiner — Edwin A. Leon

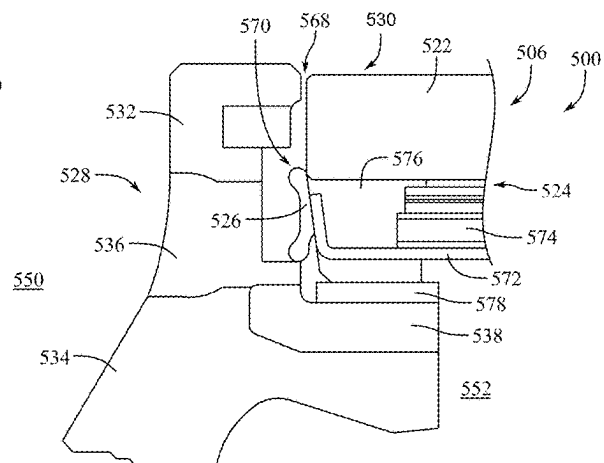
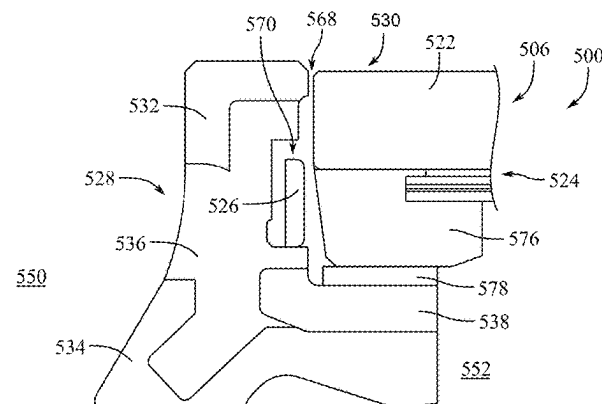
(74) *Attorney, Agent, or Firm* — Dorsey & Whitney LLP

(57)

ABSTRACT

An electronic device includes a housing sidewall defining an opening and a display component, such as a display cover, disposed in the opening to form a gap between the housing sidewall and the display component. In at least one example, the cavity is defined by the sidewall and the display cover with the cavity in fluid communication with an external environment through the gap. In at least one example, an epoxy component at least partially defines the cavity and can be in direct contact with the housing sidewall.

20 Claims, 45 Drawing Sheets



Related U.S. Application Data

on May 2, 2022, provisional application No. 63/266, 829, filed on Jan. 14, 2022.

(58) Field of Classification Search

CPC G06F 1/1637; G06F 1/1656; G06F 1/1671;
G06F 1/1688; G06F 1/163; G06F 1/1692;
H04R 1/403

See application file for complete search history.

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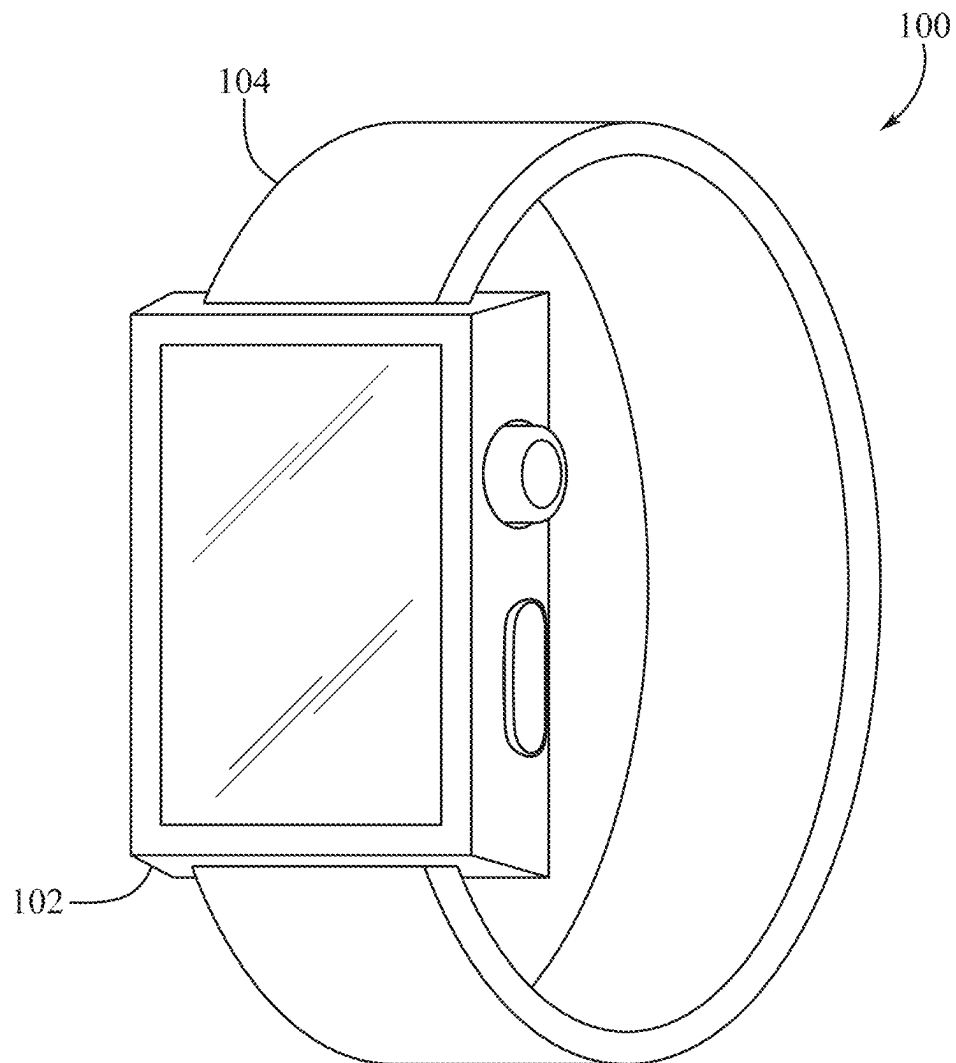
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**FIG. 1A**

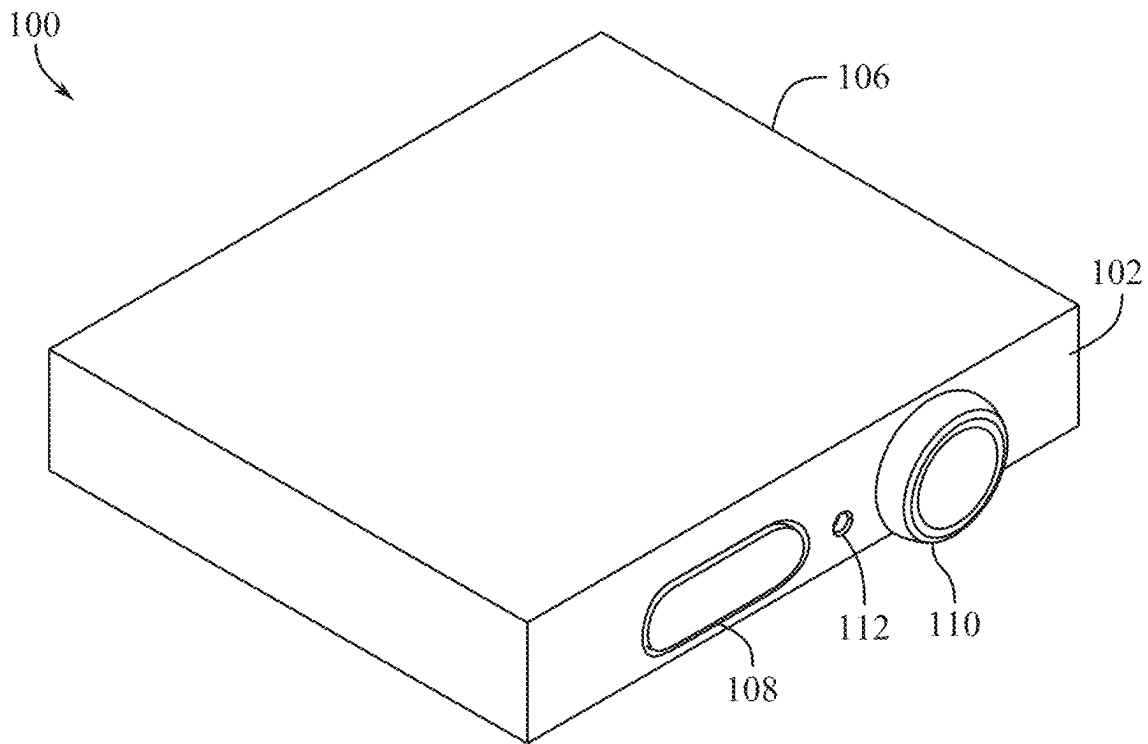


FIG. 1B

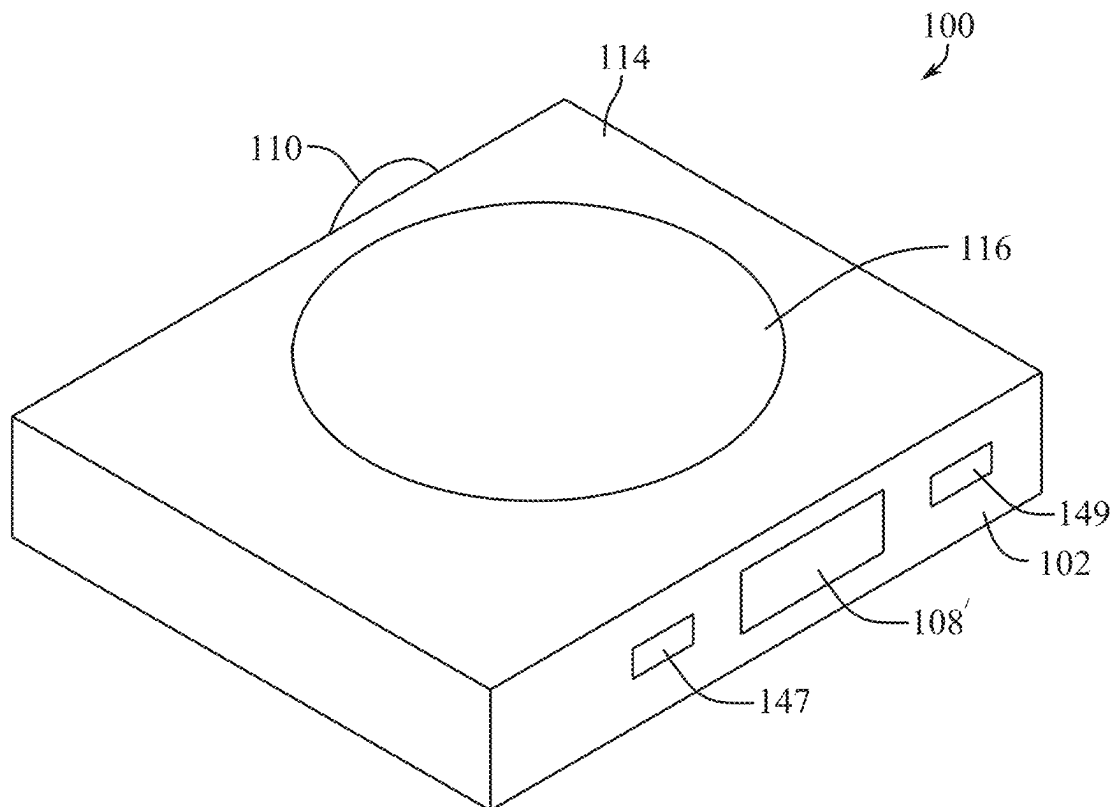
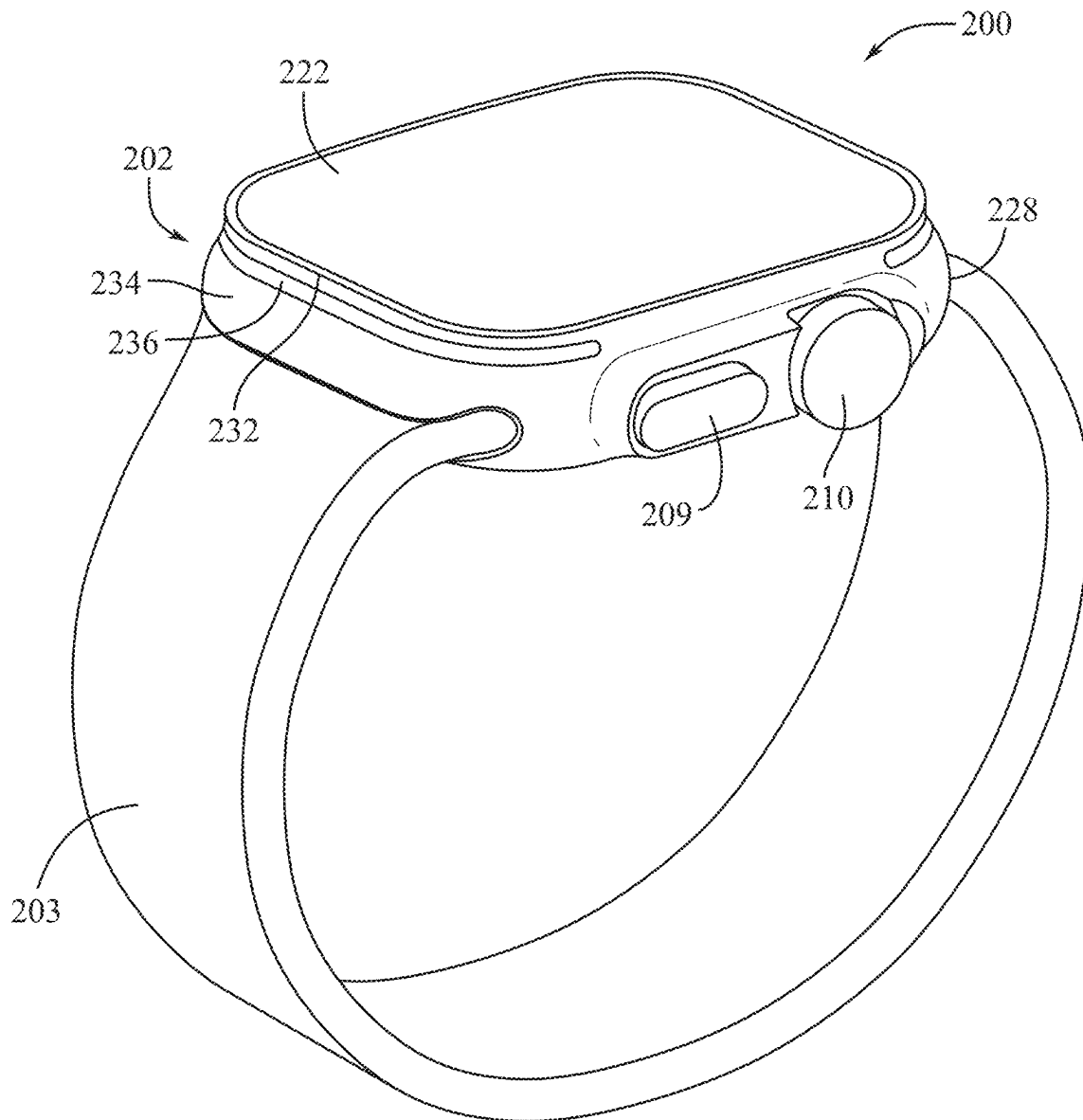


FIG. 1C

**FIG. 2A**

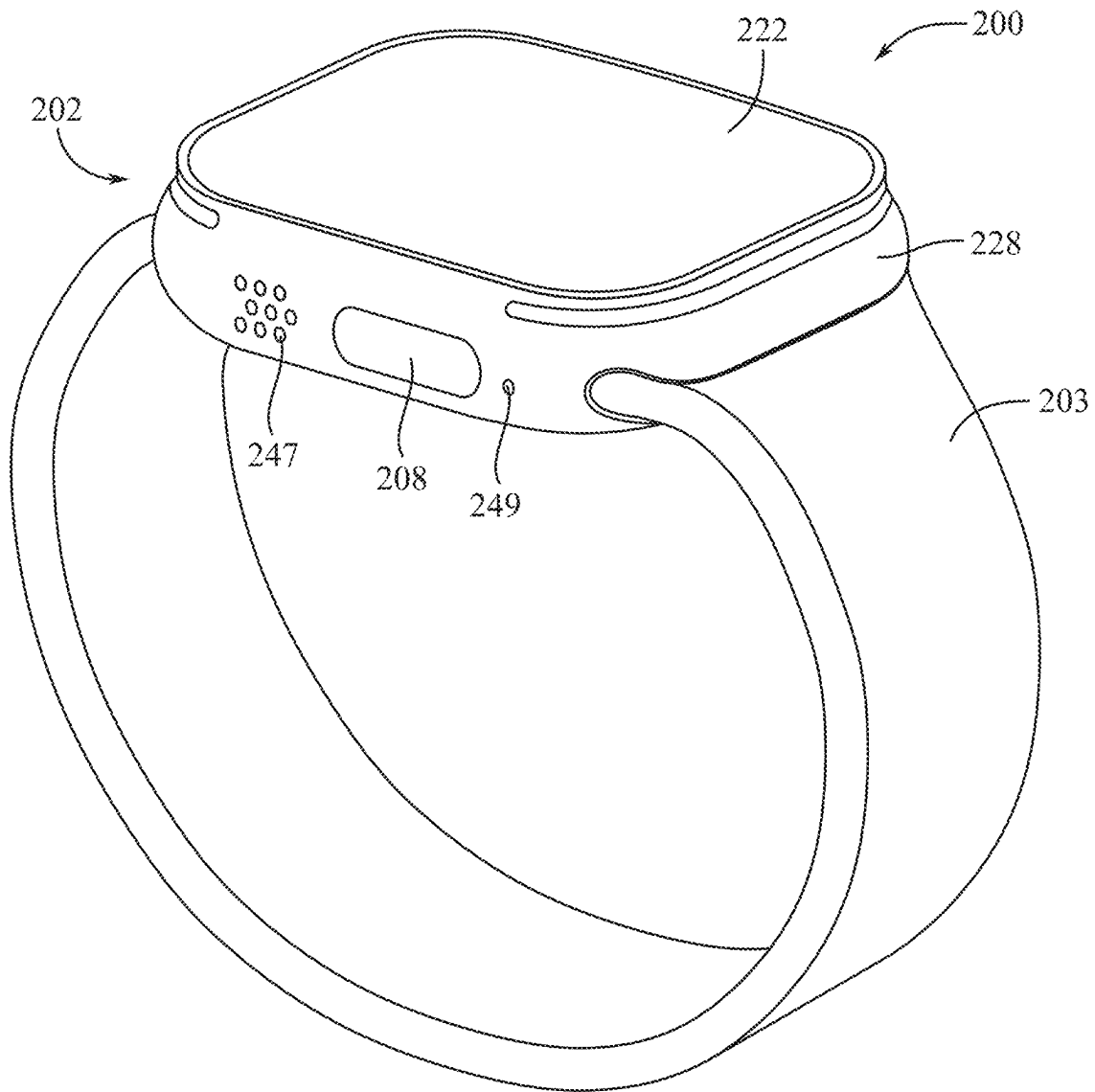
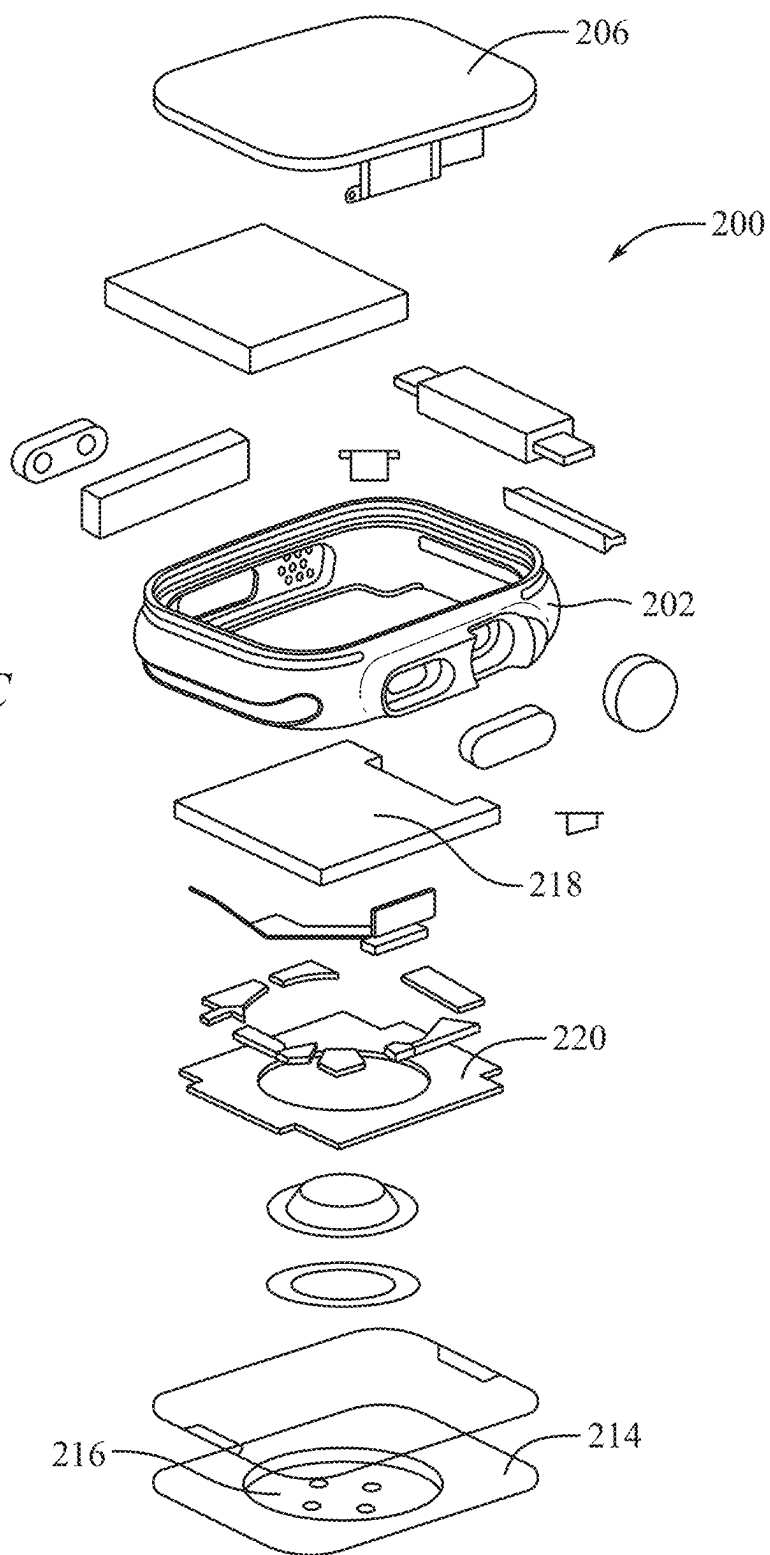
**FIG. 2B**

FIG. 2C



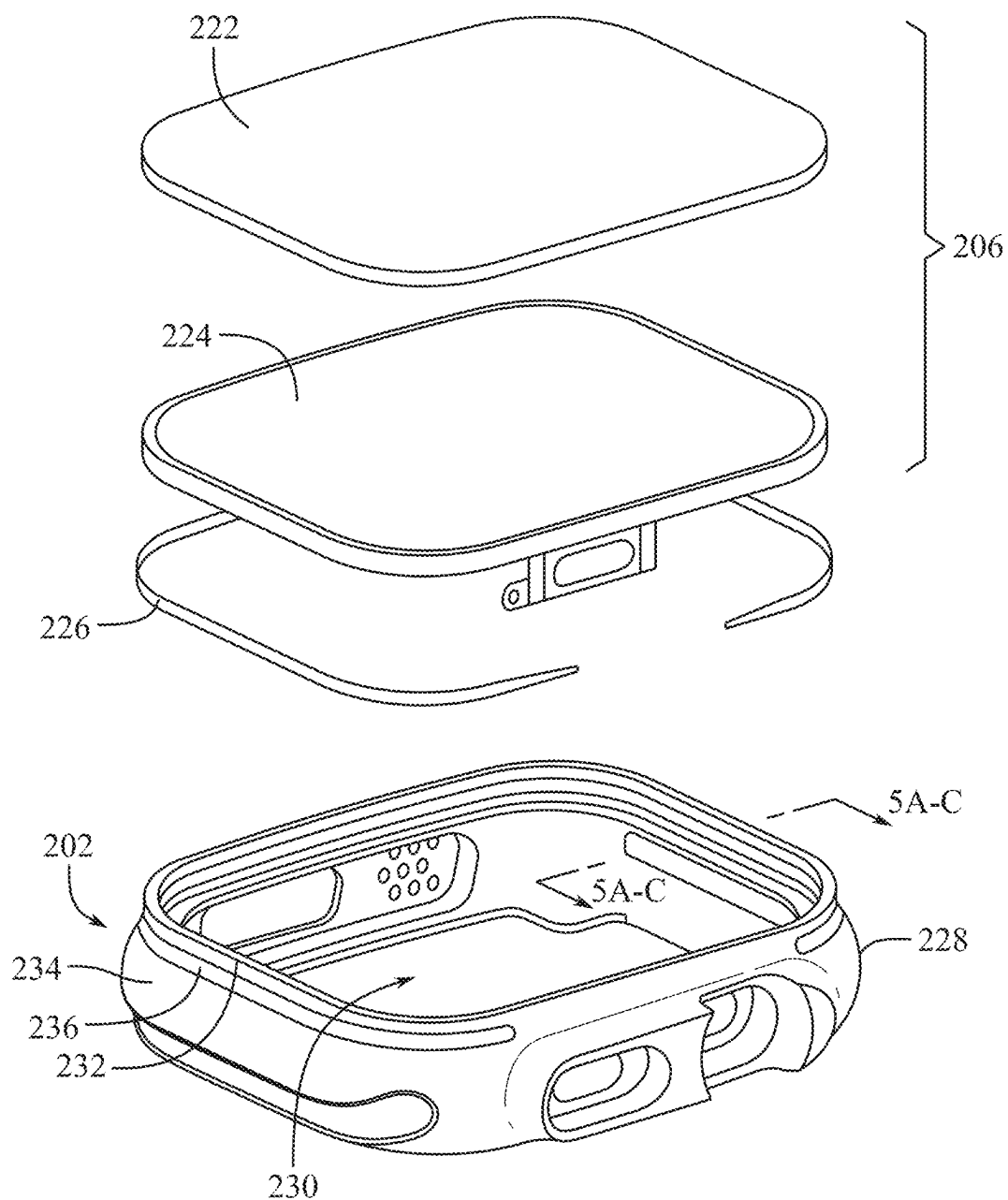
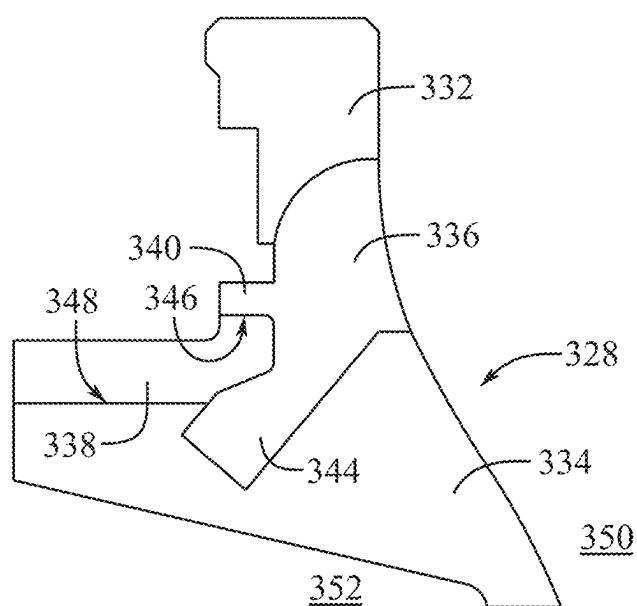
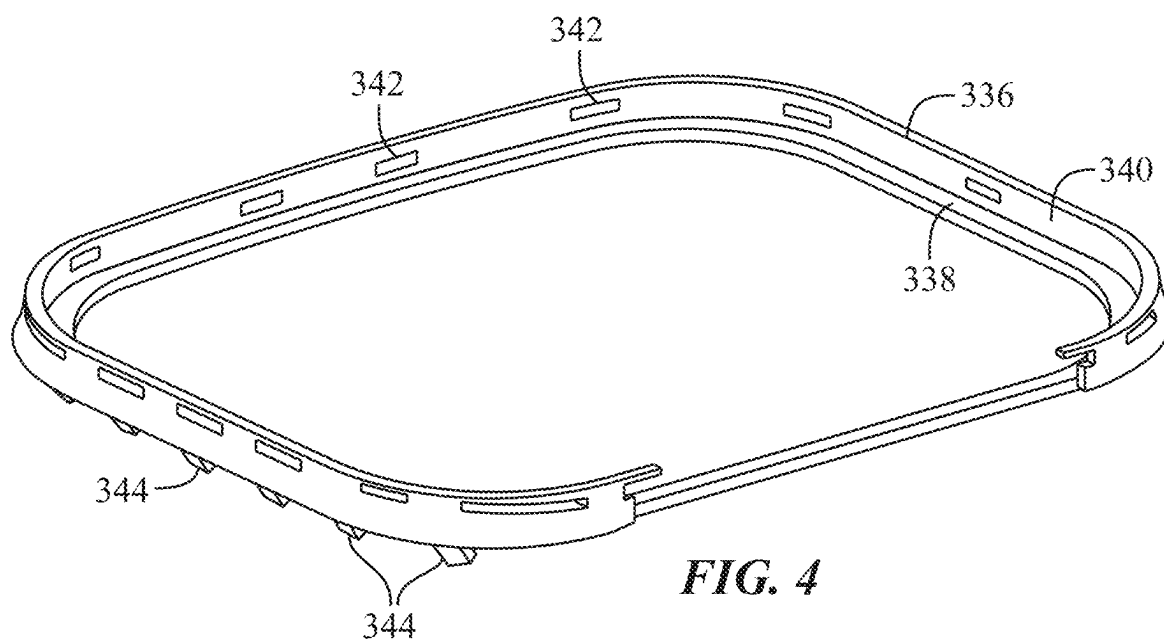


FIG. 3



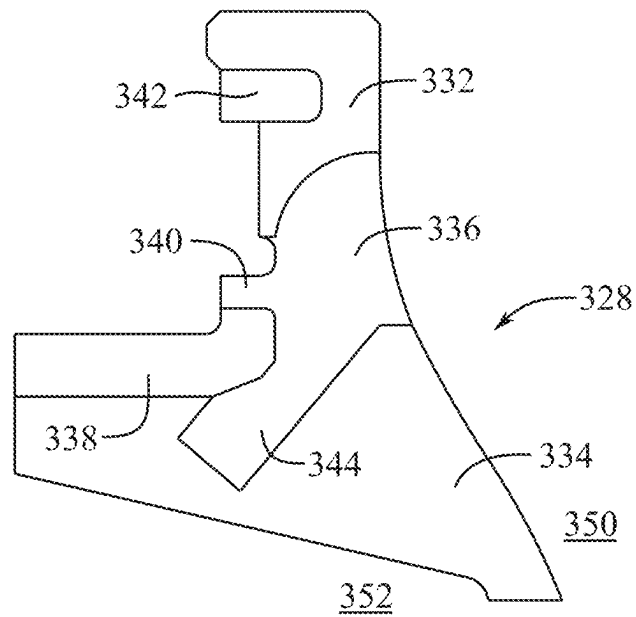


FIG. 5B

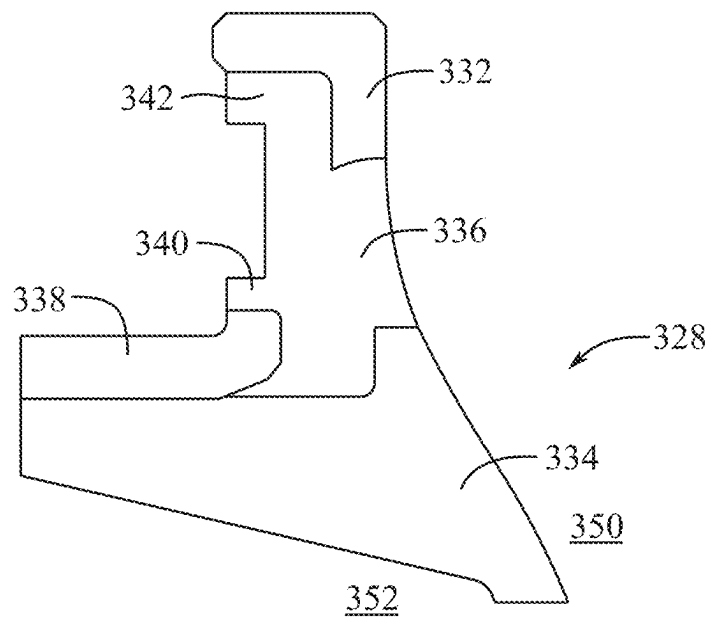
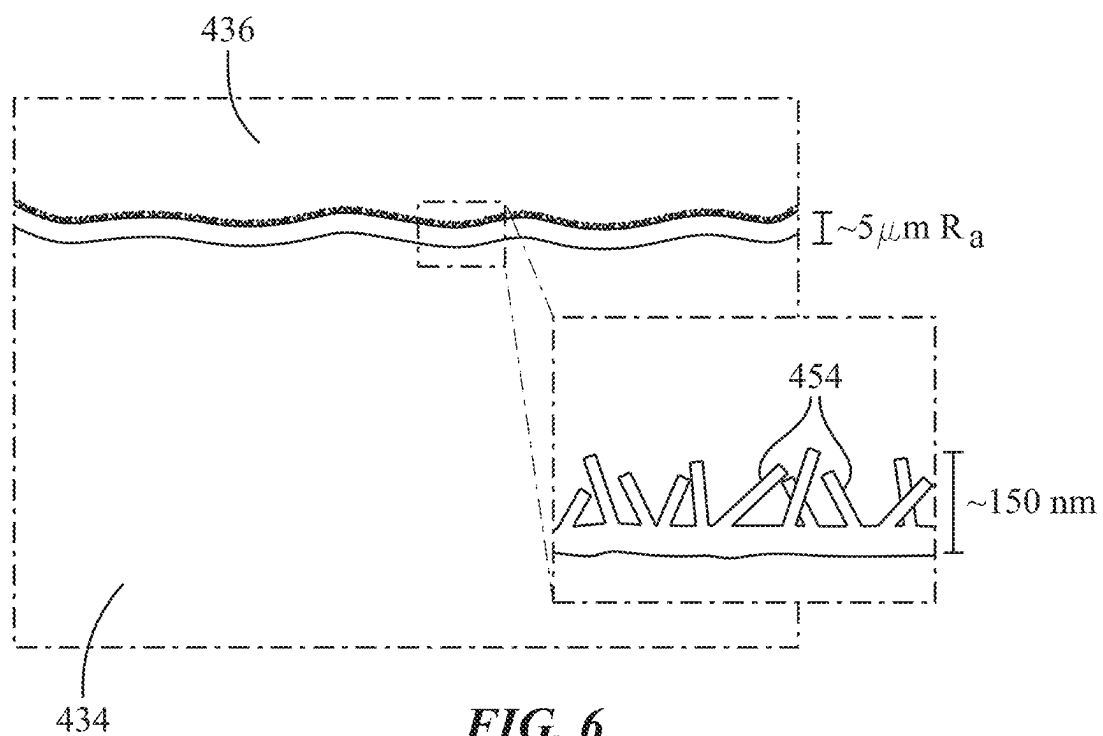
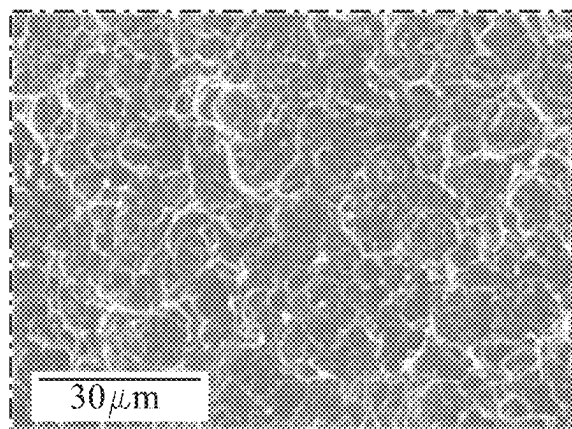
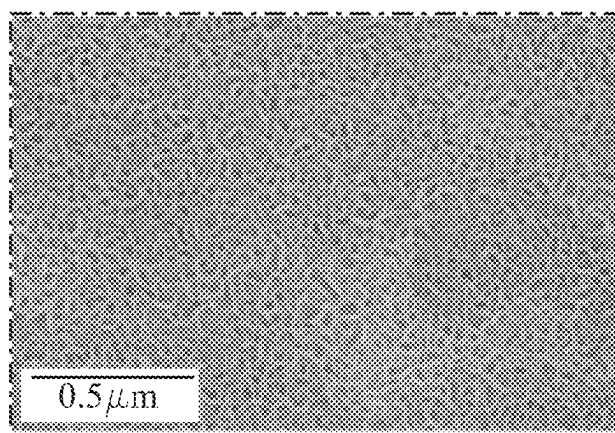
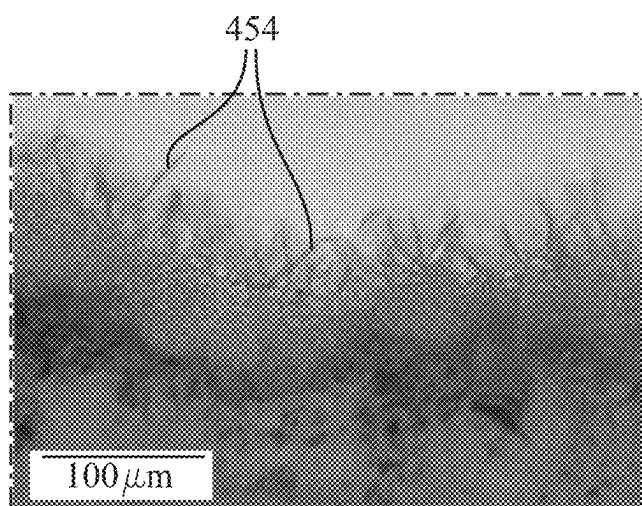


FIG. 5C

**FIG. 6**

*FIG. 7A**FIG. 7B**FIG. 7C*

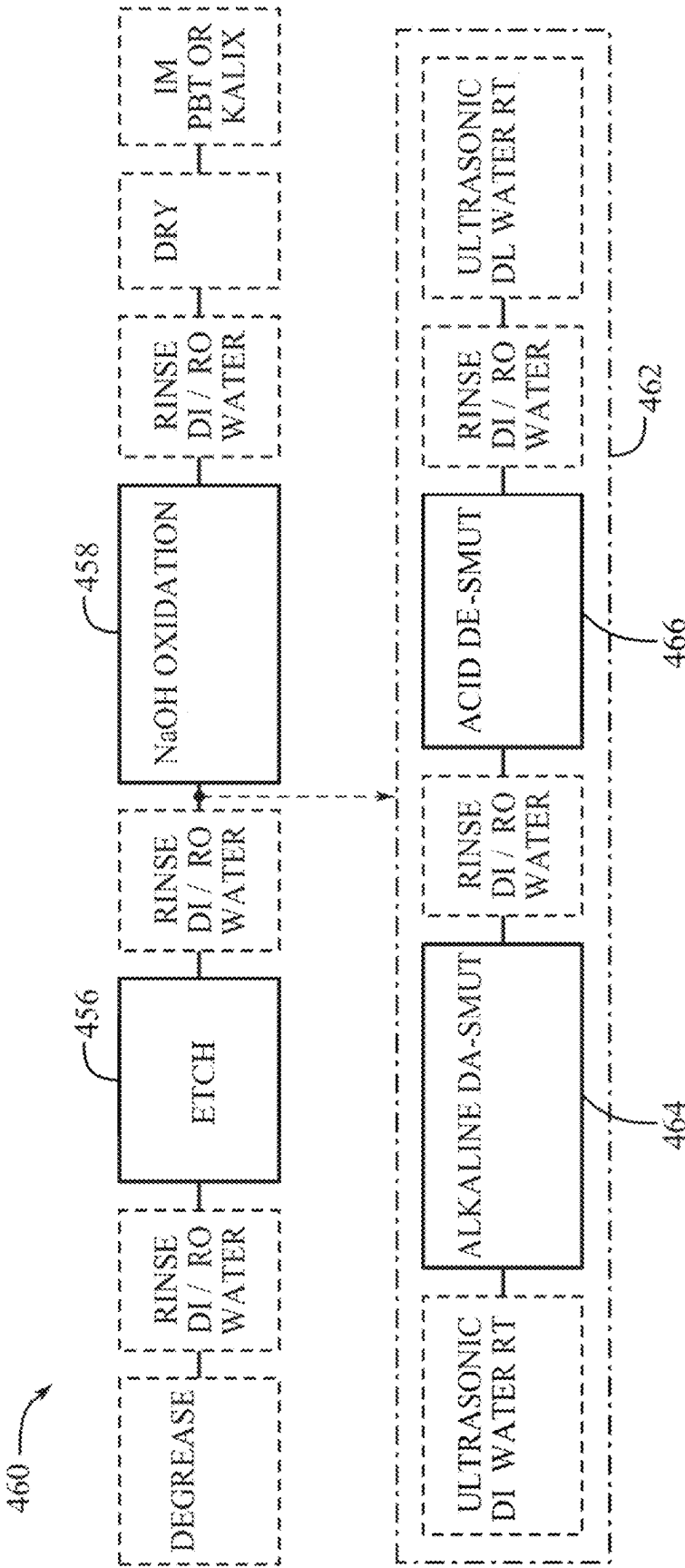


FIG. 8

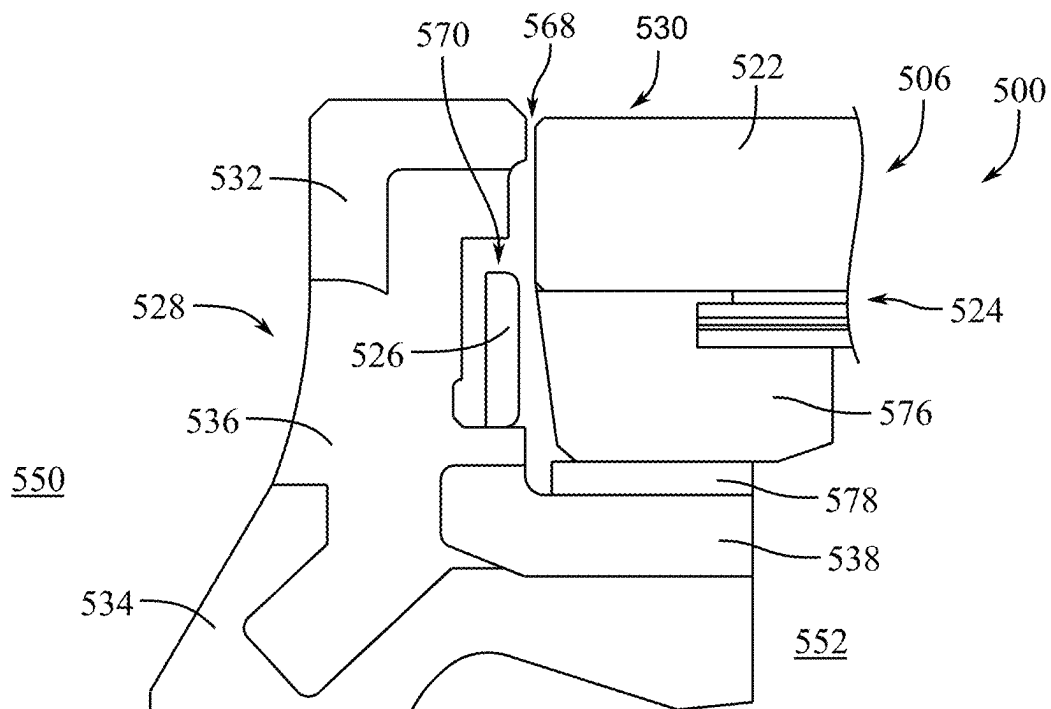


FIG. 9A

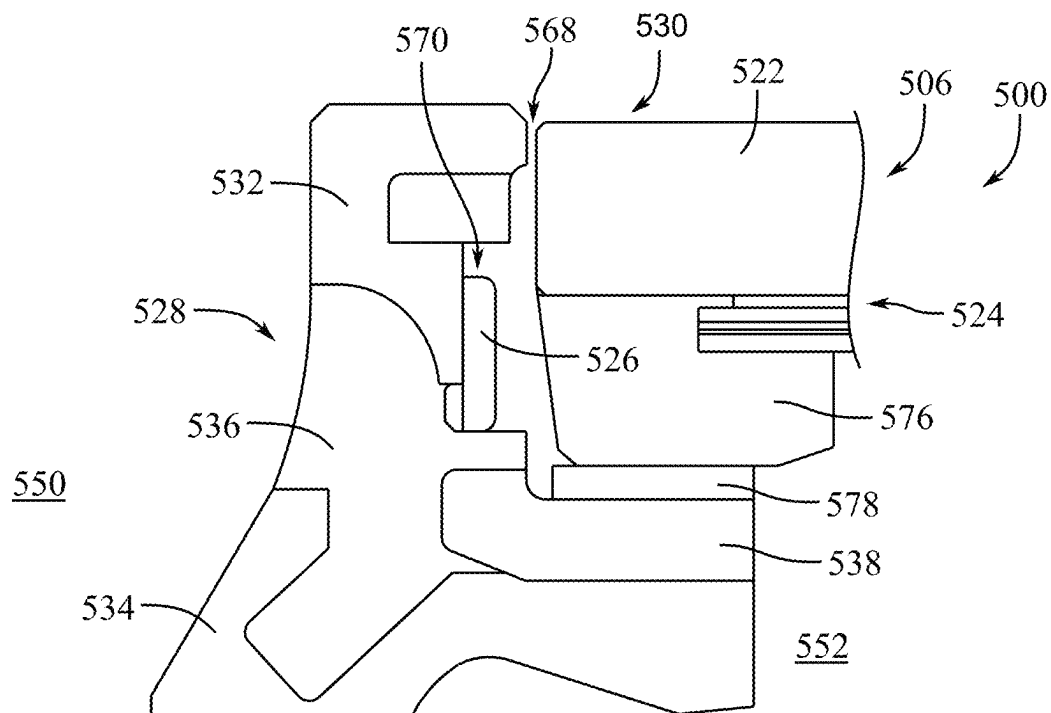
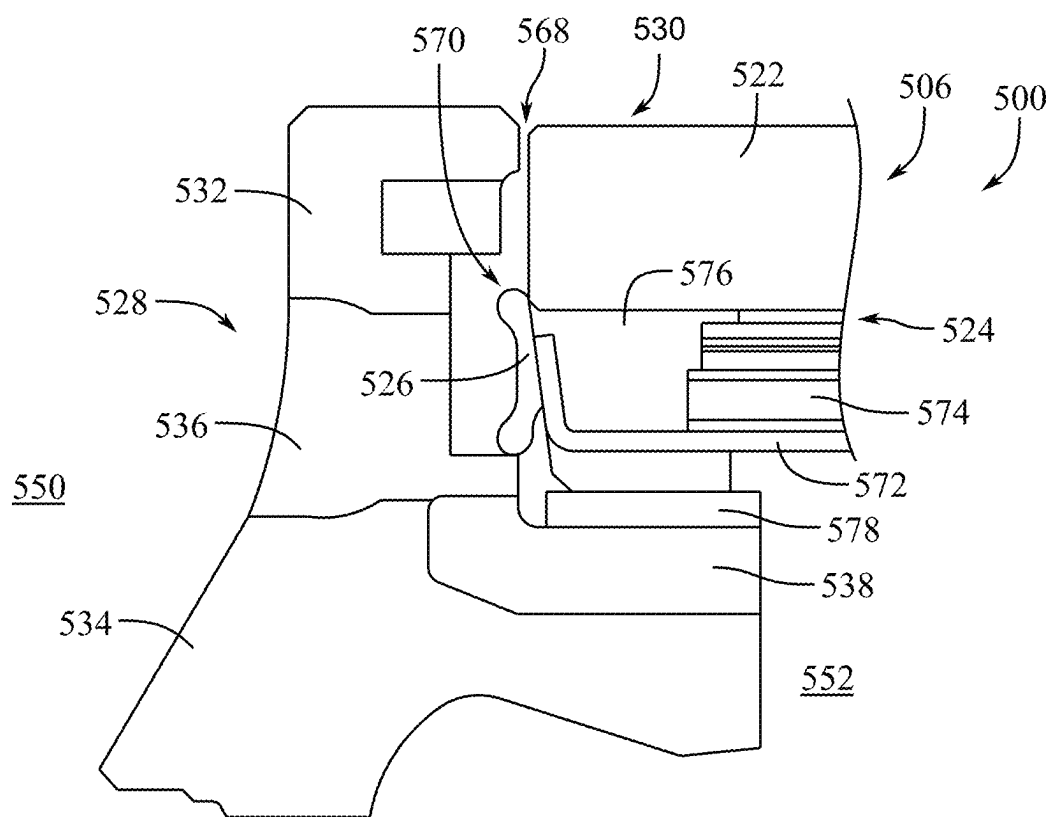
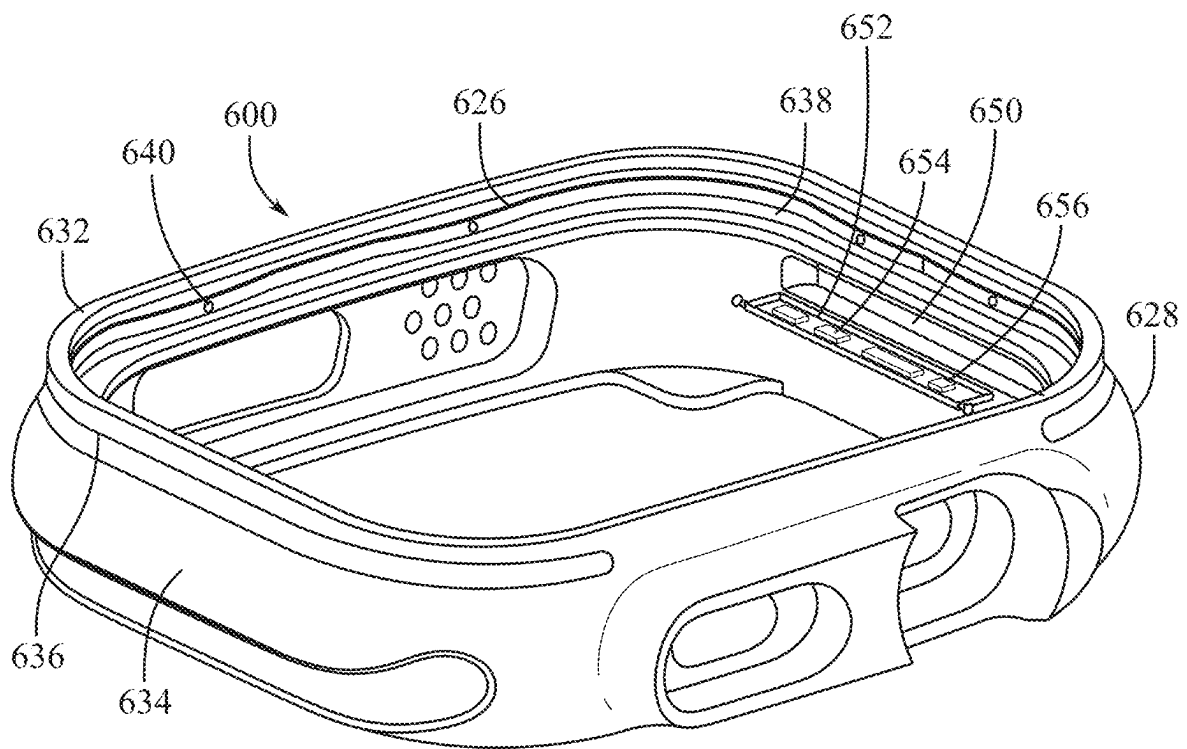
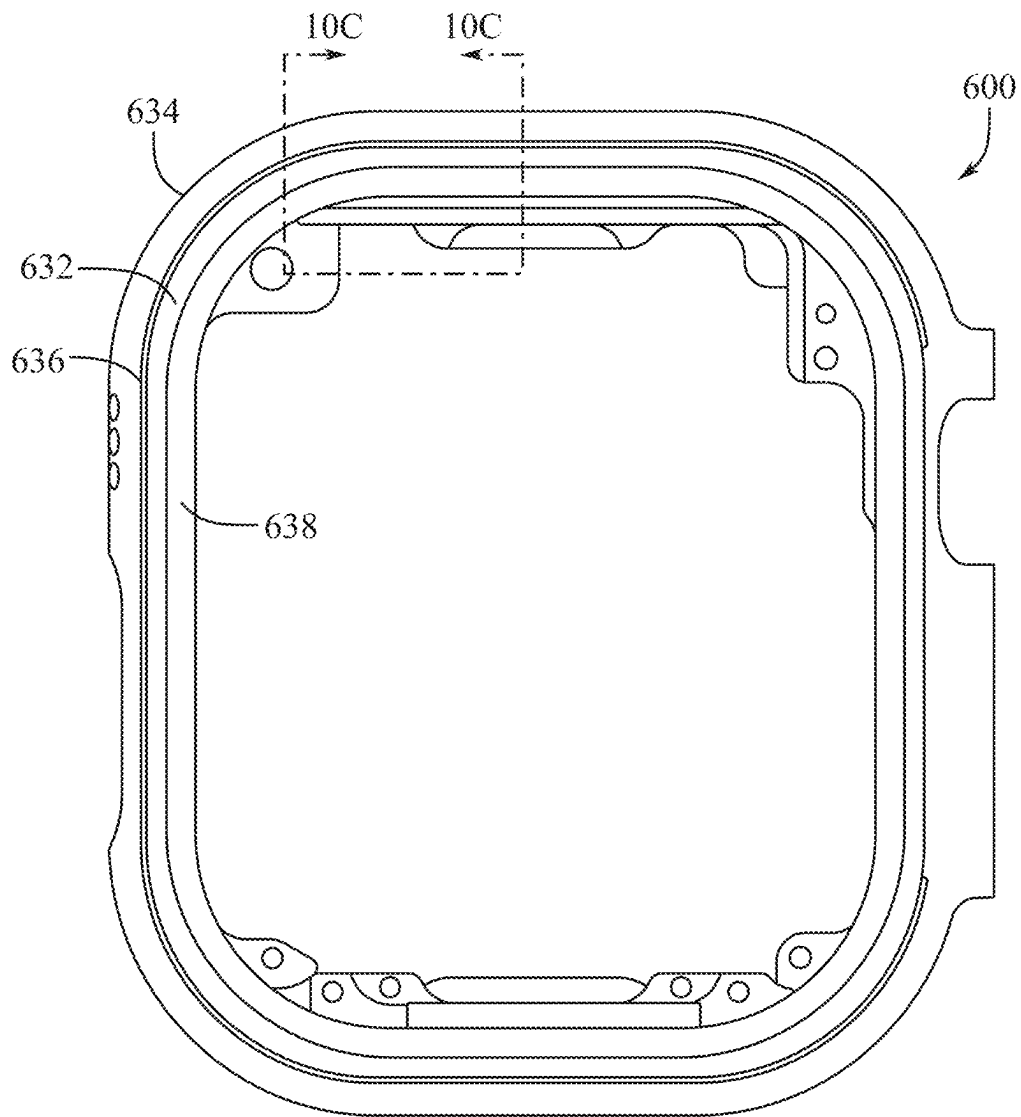


FIG. 9B

**FIG. 9C**

**FIG. 10A**

**FIG. 10B**

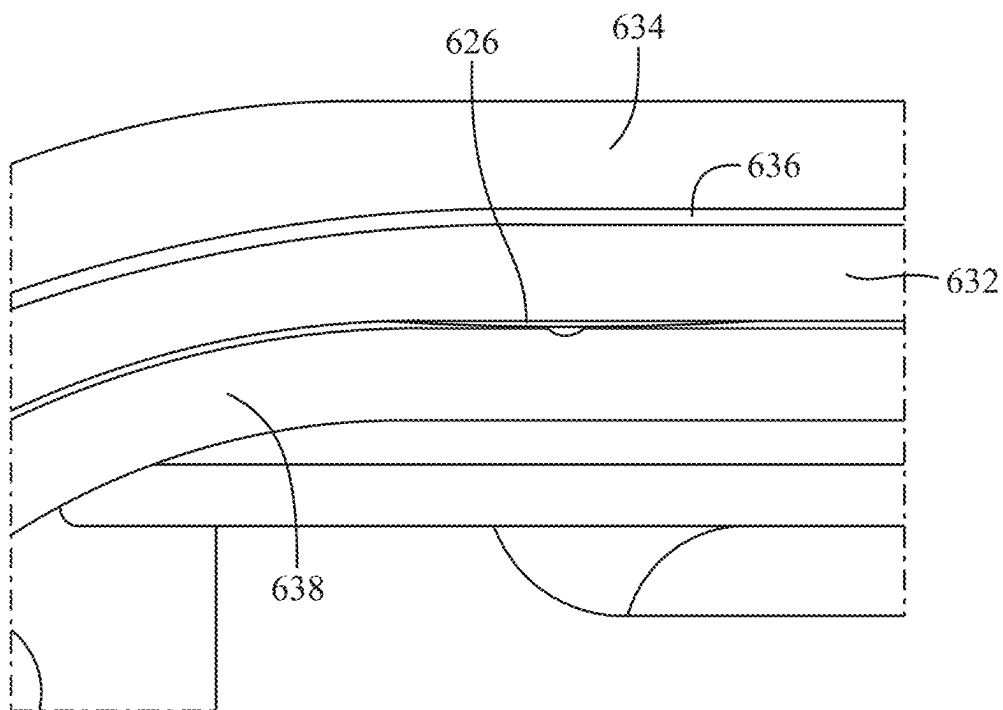


FIG. 10C

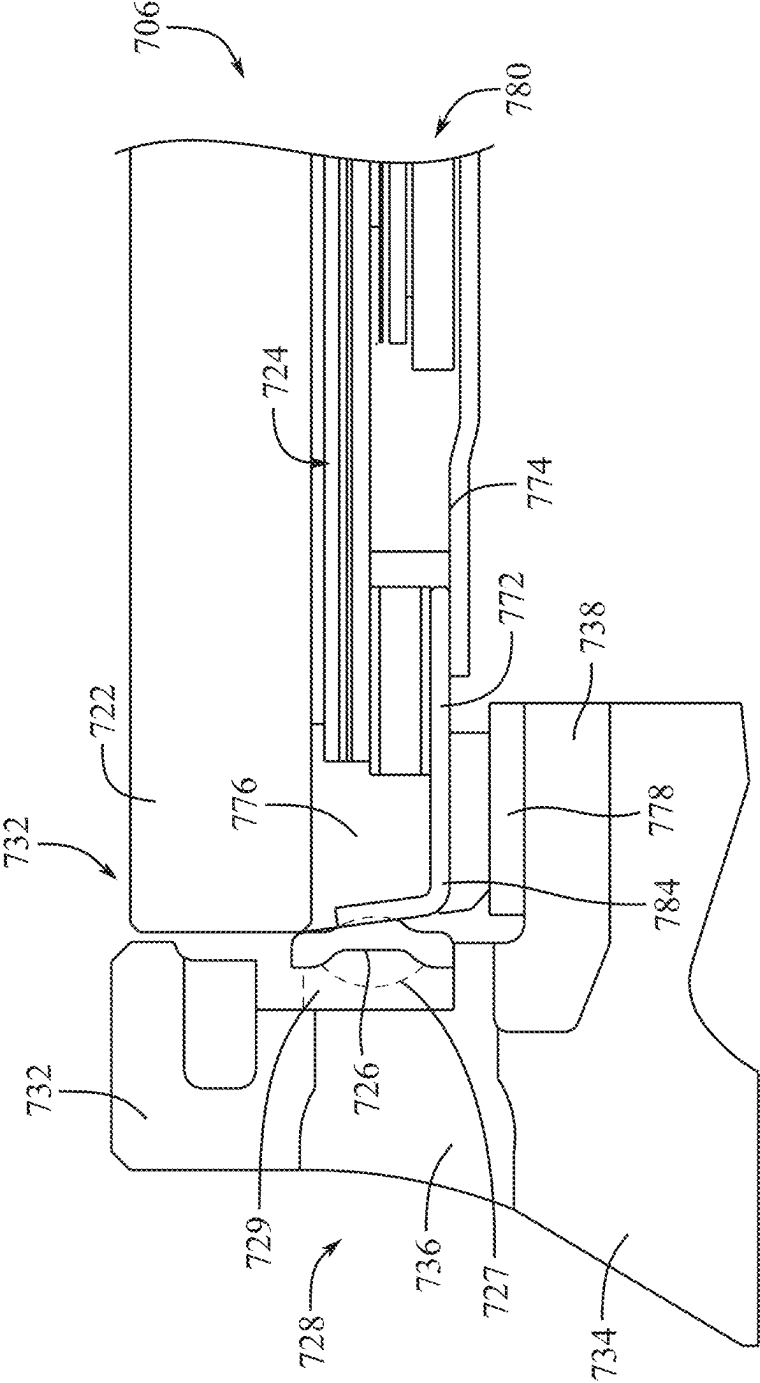
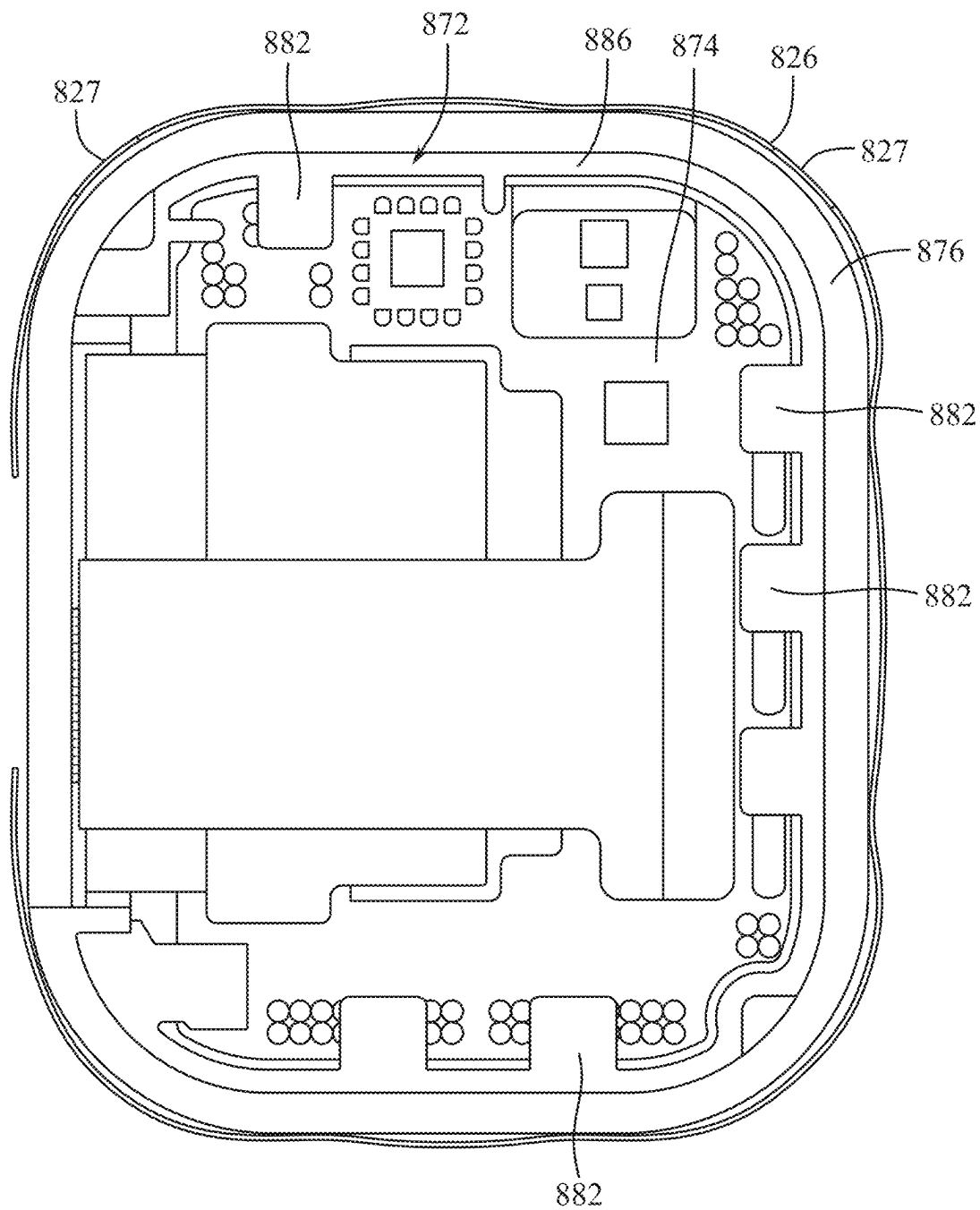


FIG. 11

**FIG. 12**

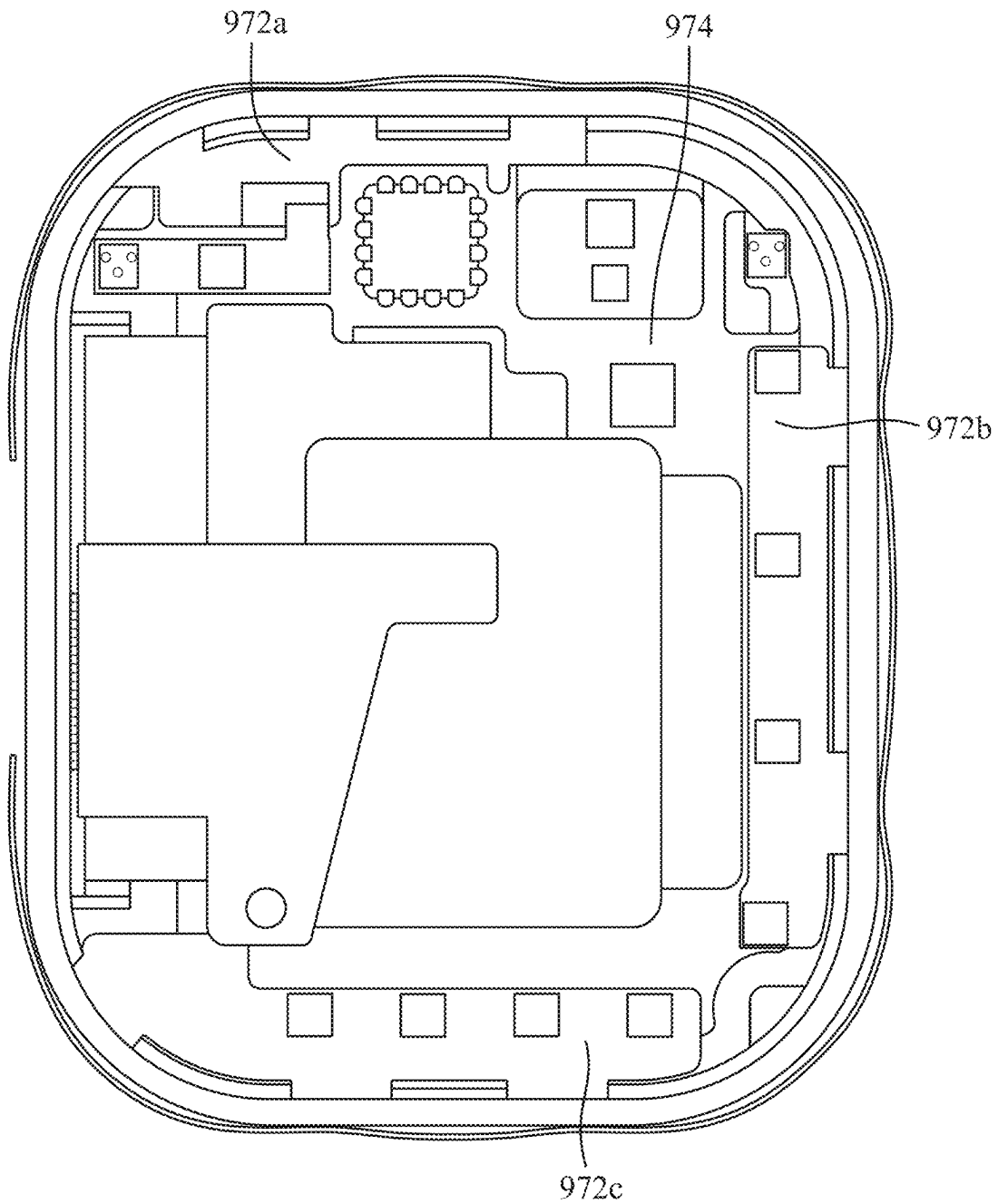


FIG. 13

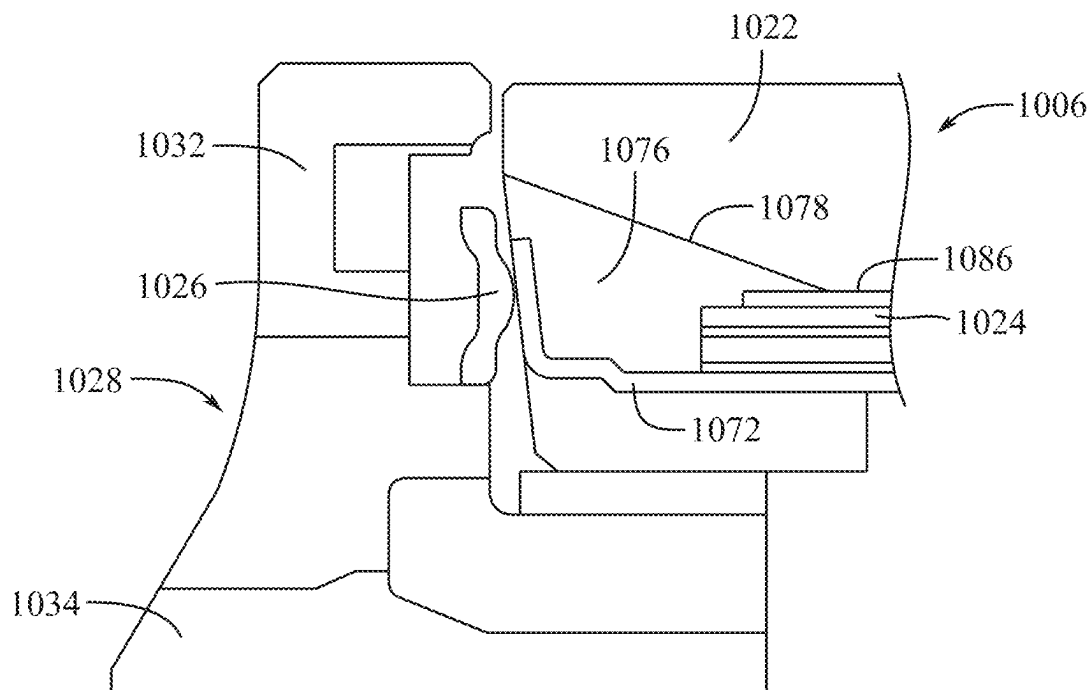


FIG. 14A

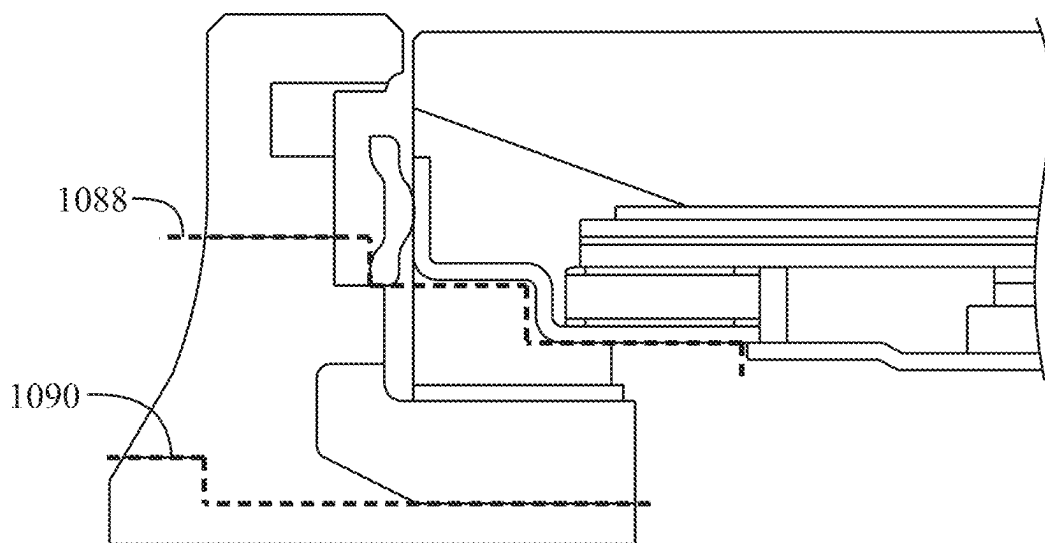
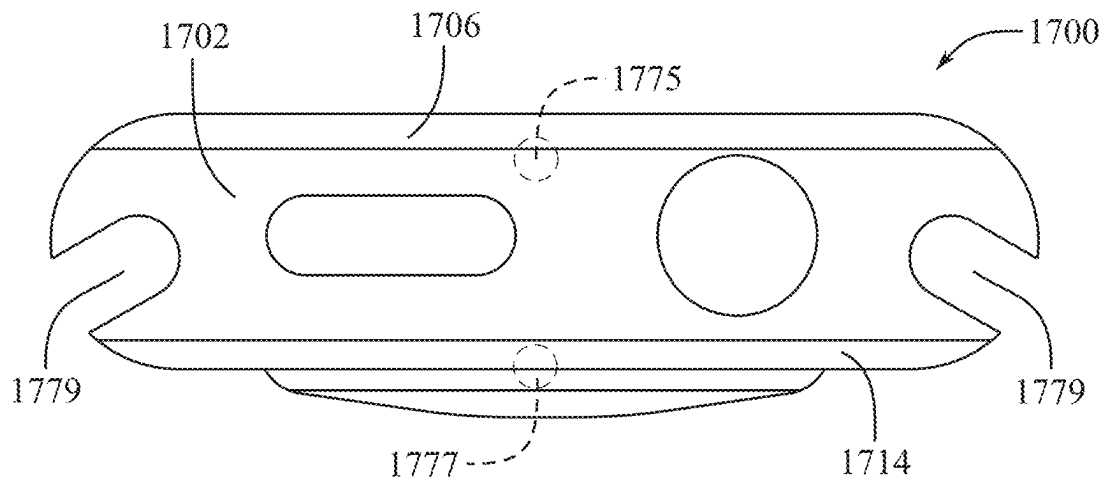


FIG. 14B

**FIG. 15A**

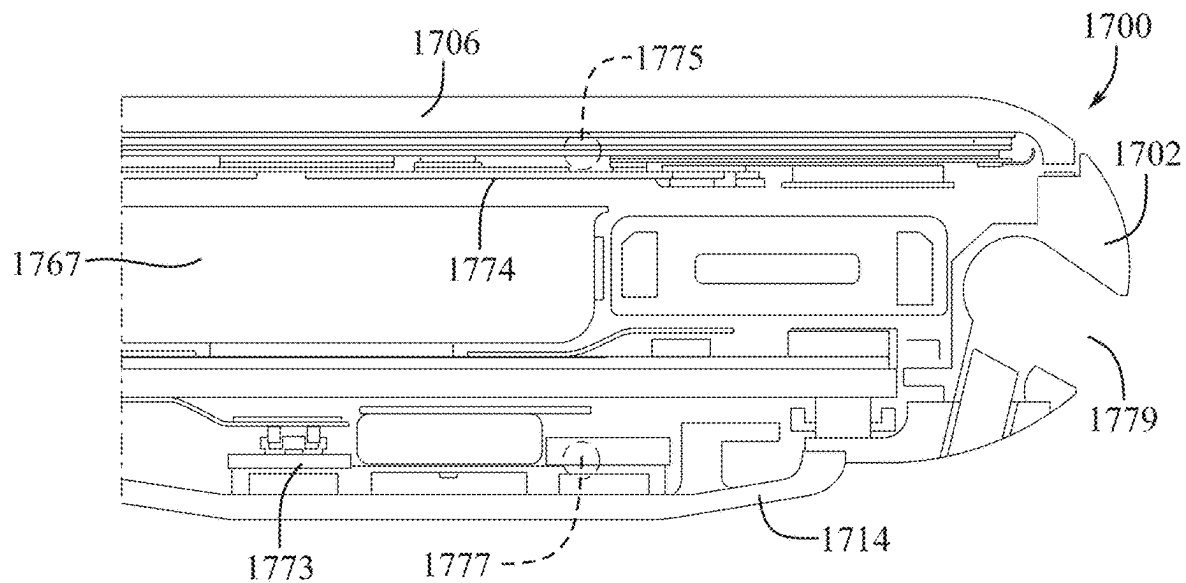


FIG. 15B

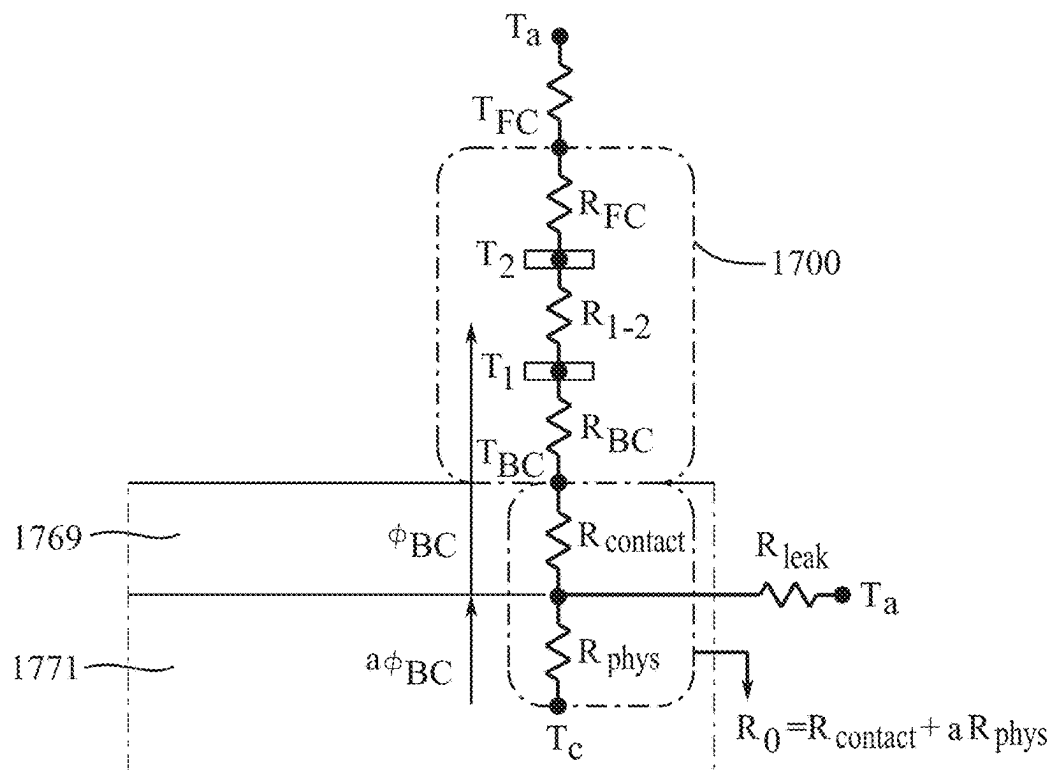


FIG. 15C

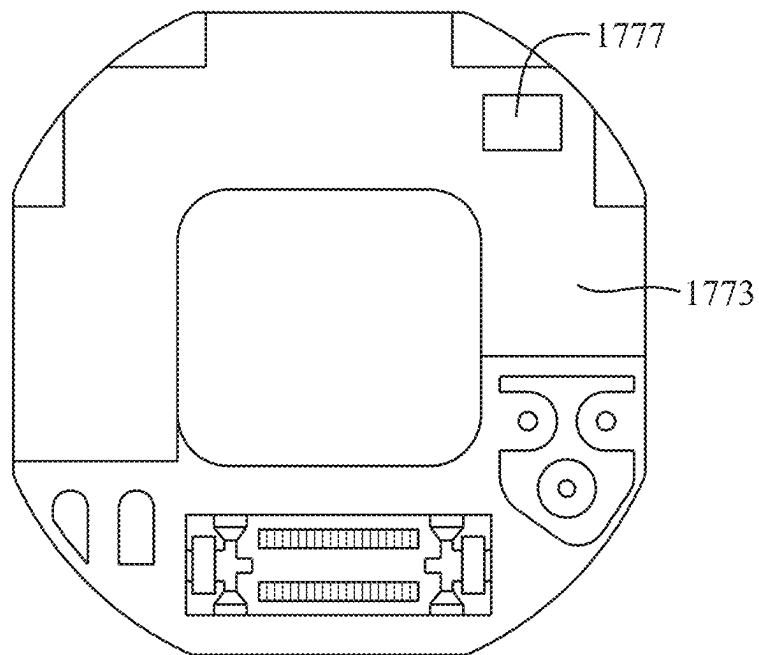


FIG. 16

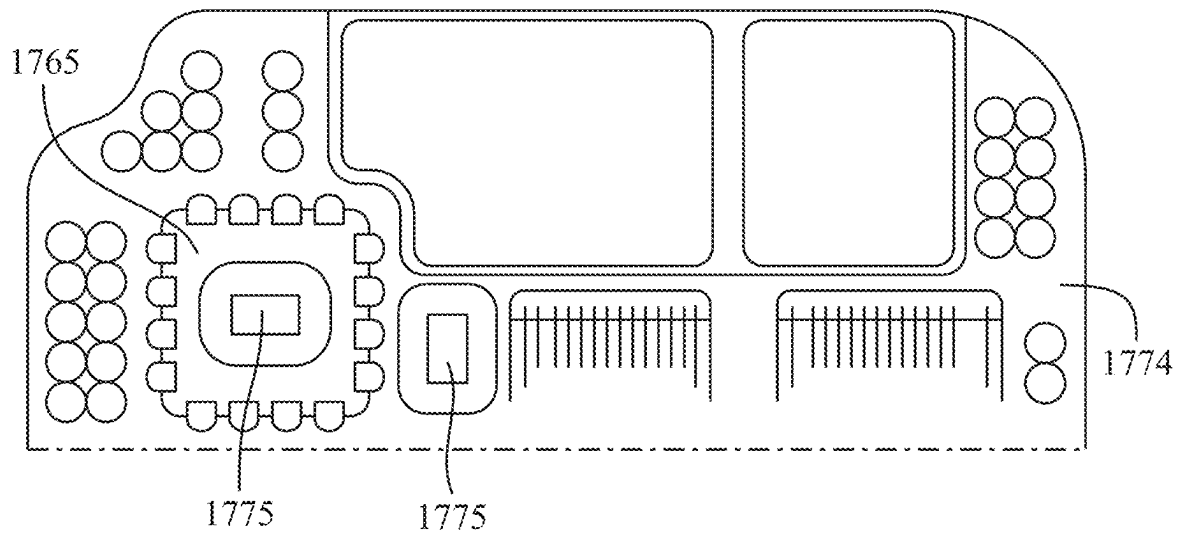


FIG. 17

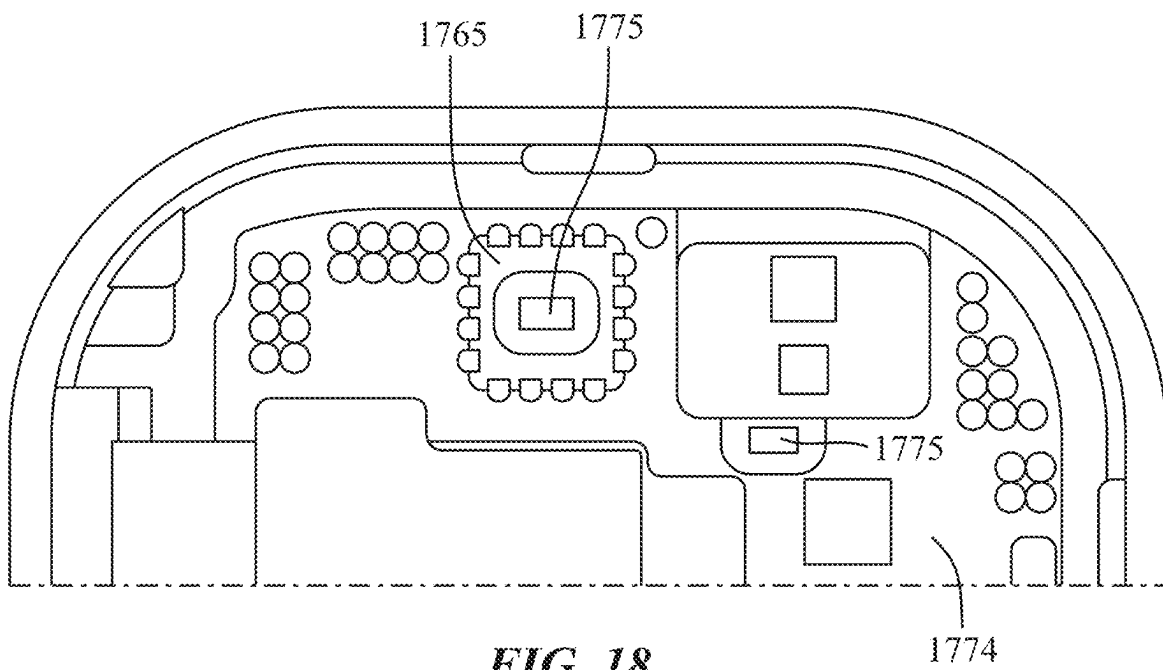


FIG. 18

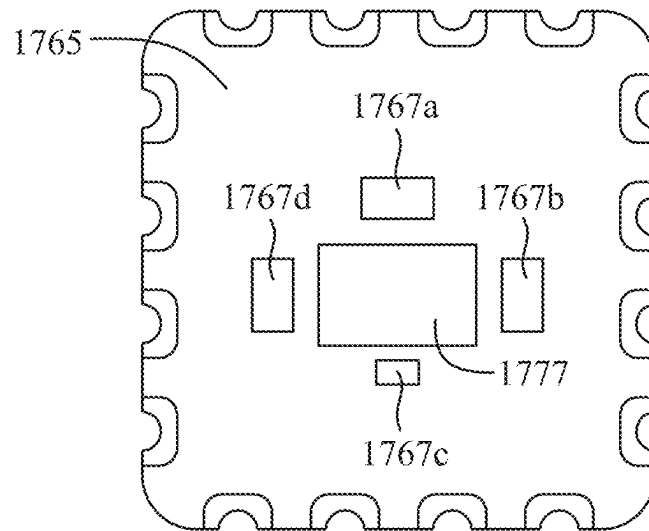


FIG. 19

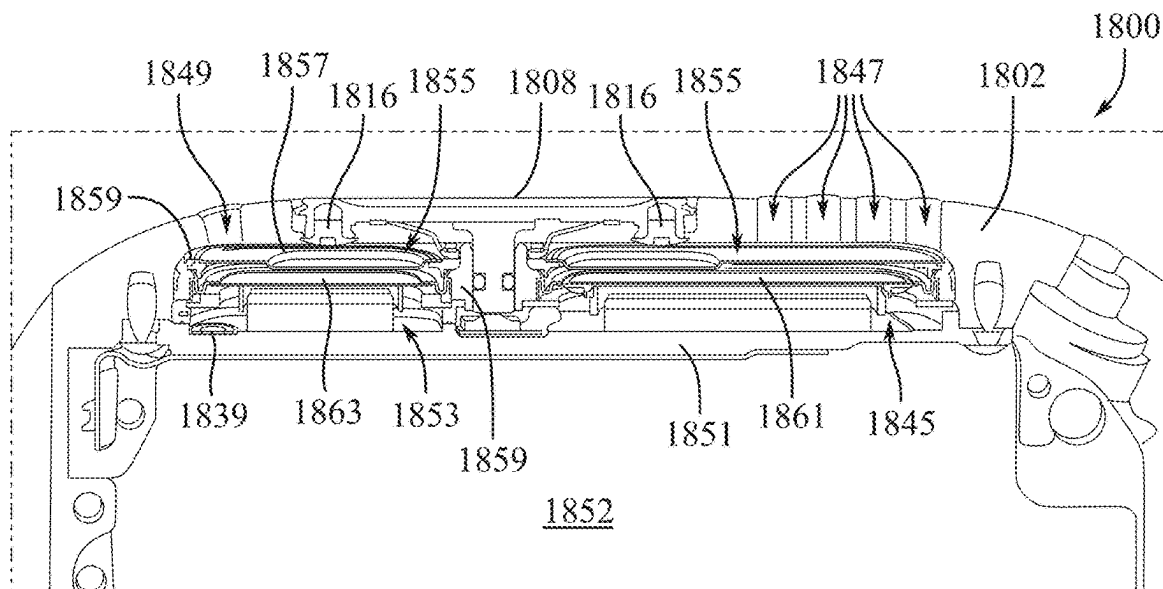


FIG. 20A

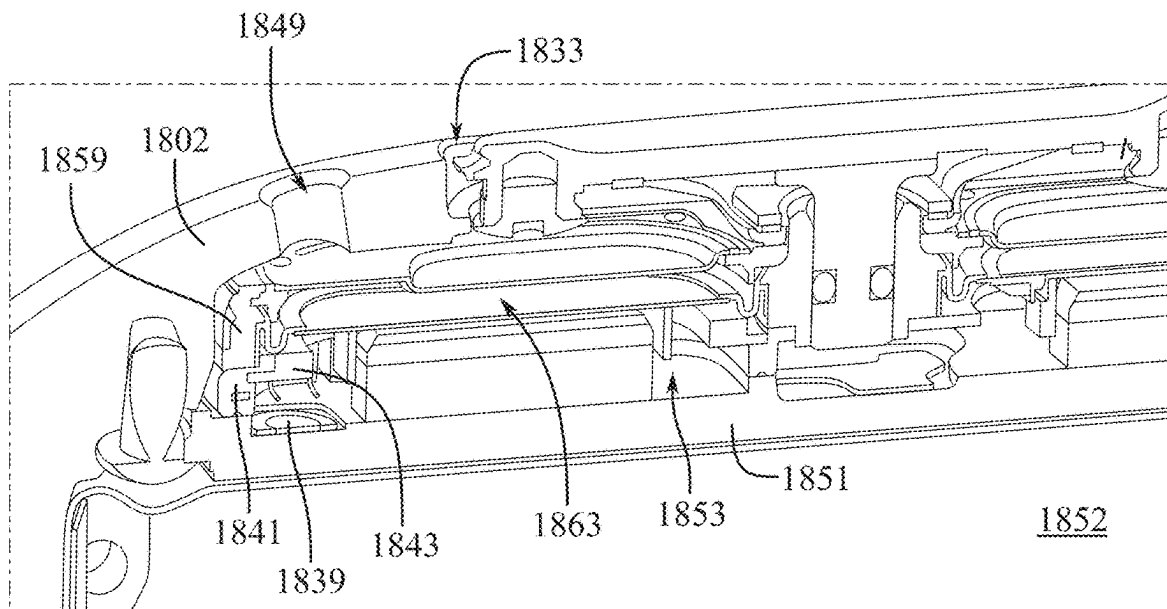
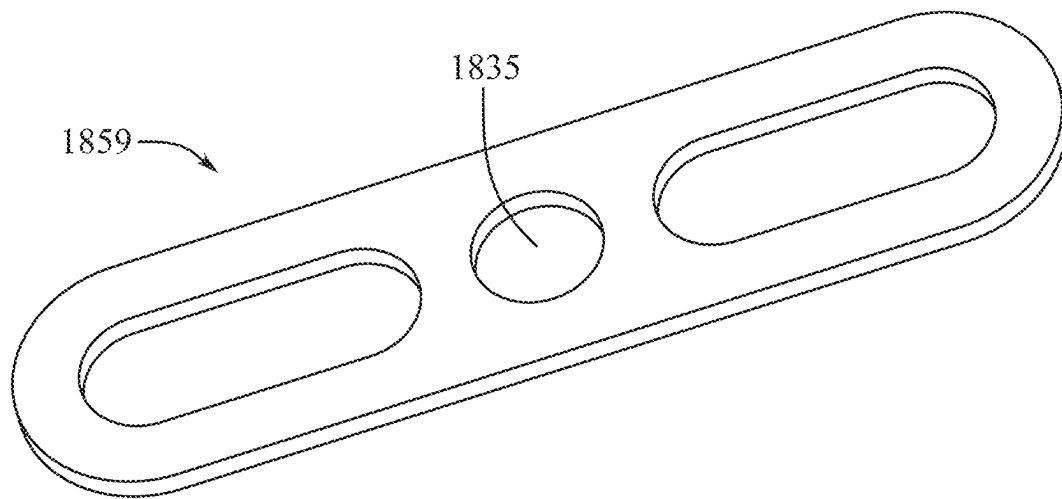
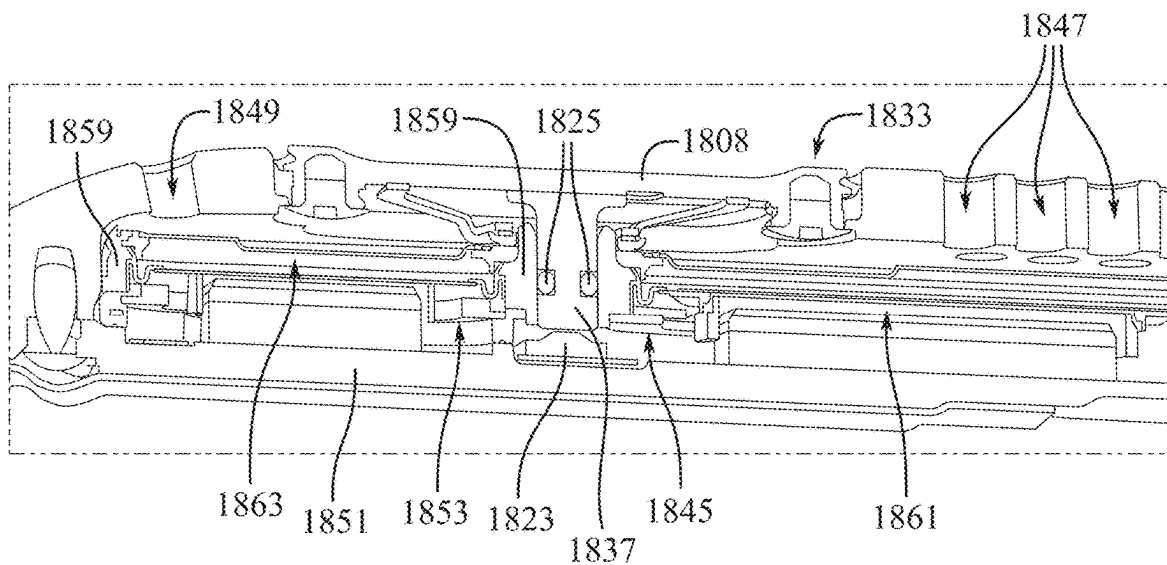
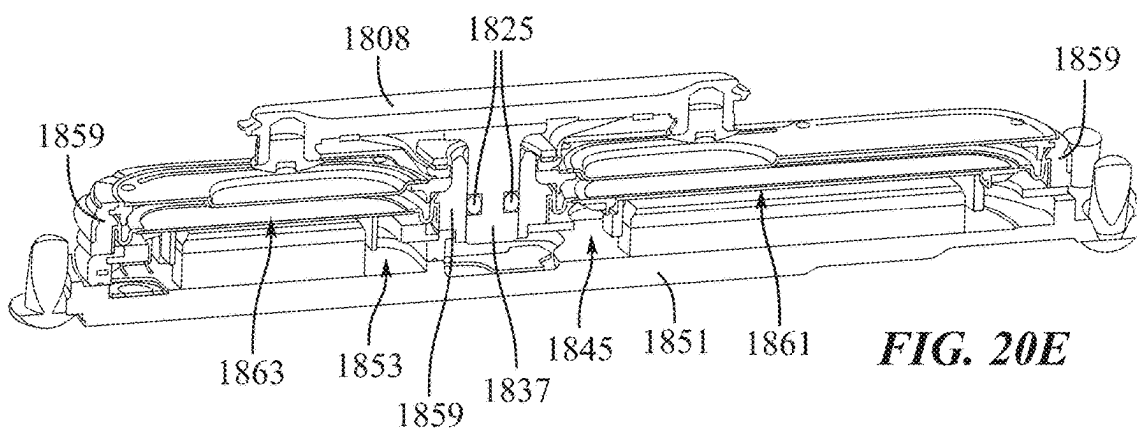
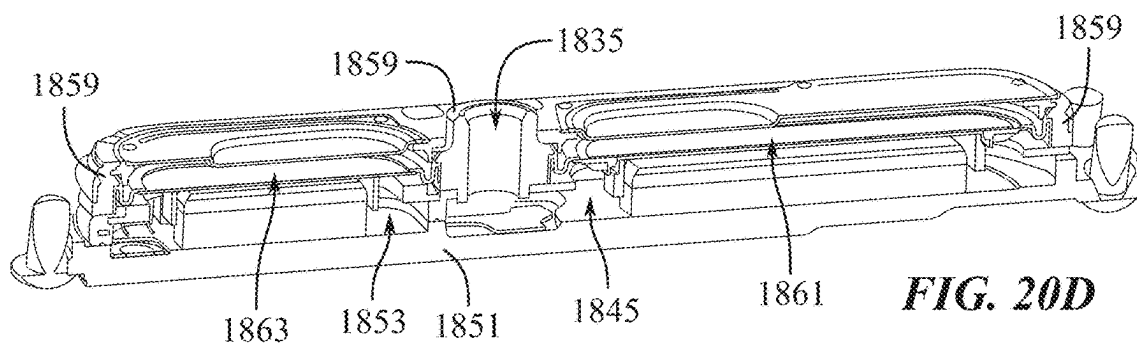
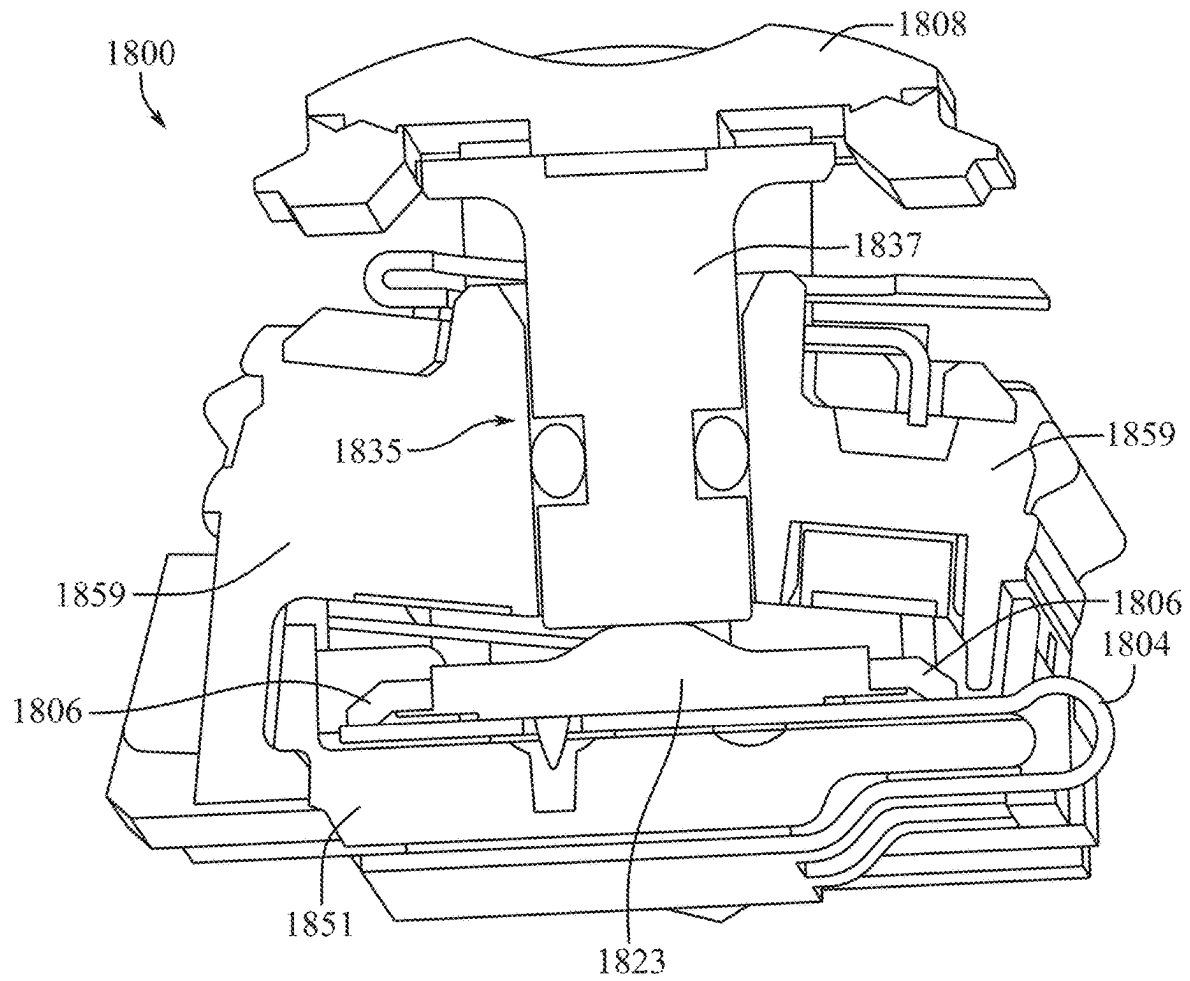


FIG. 20B

*FIG. 20C*



**FIG. 20G**

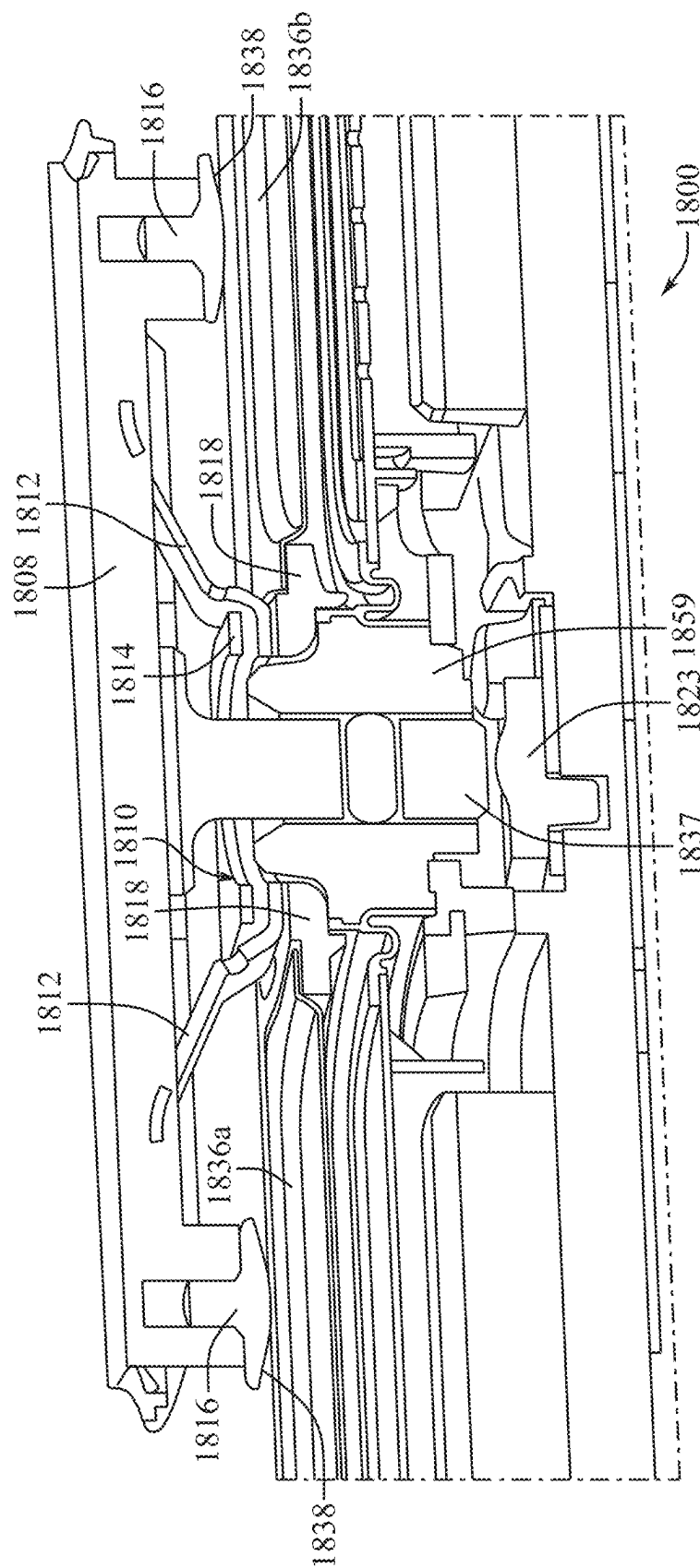
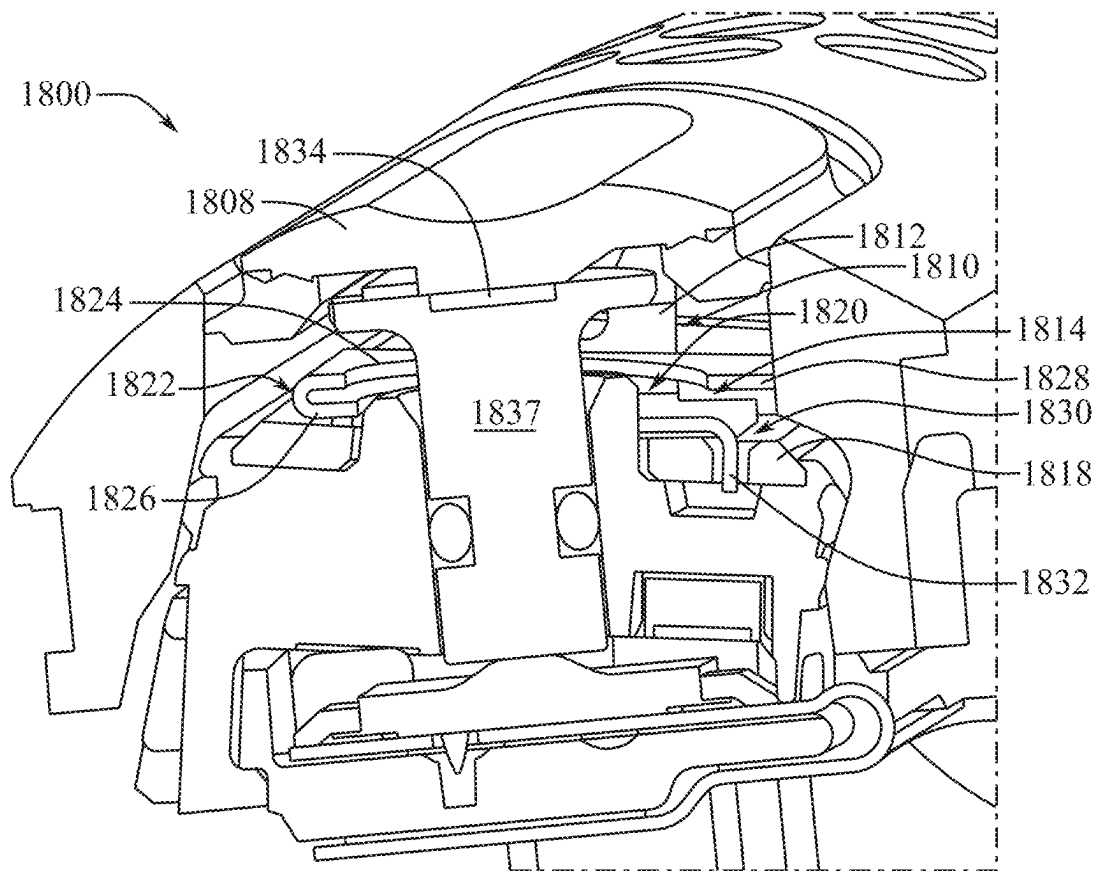


FIG. 20H

**FIG. 20I**

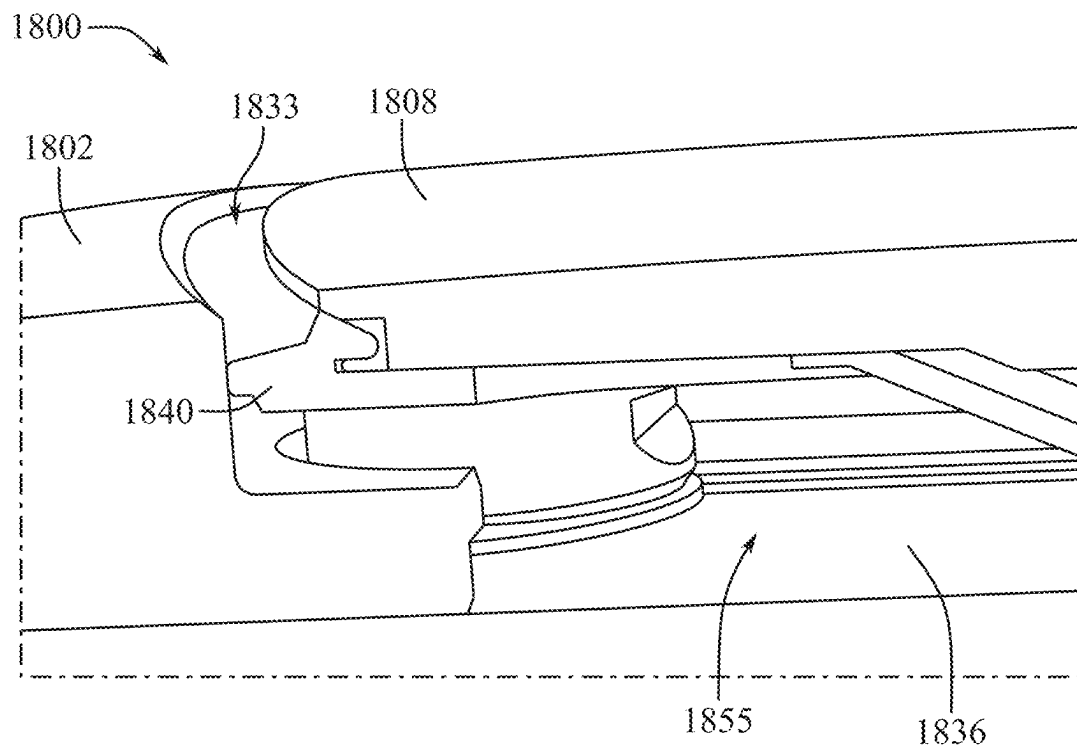
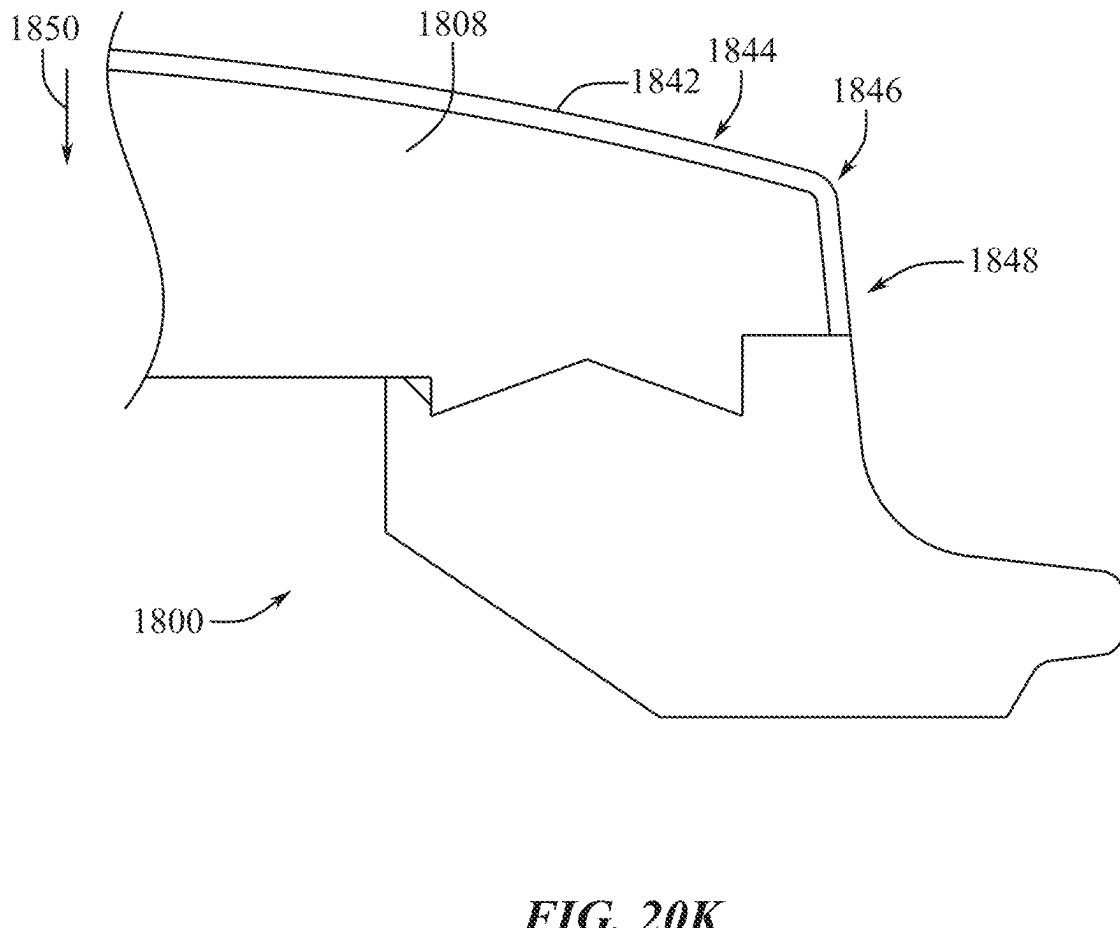


FIG. 20J



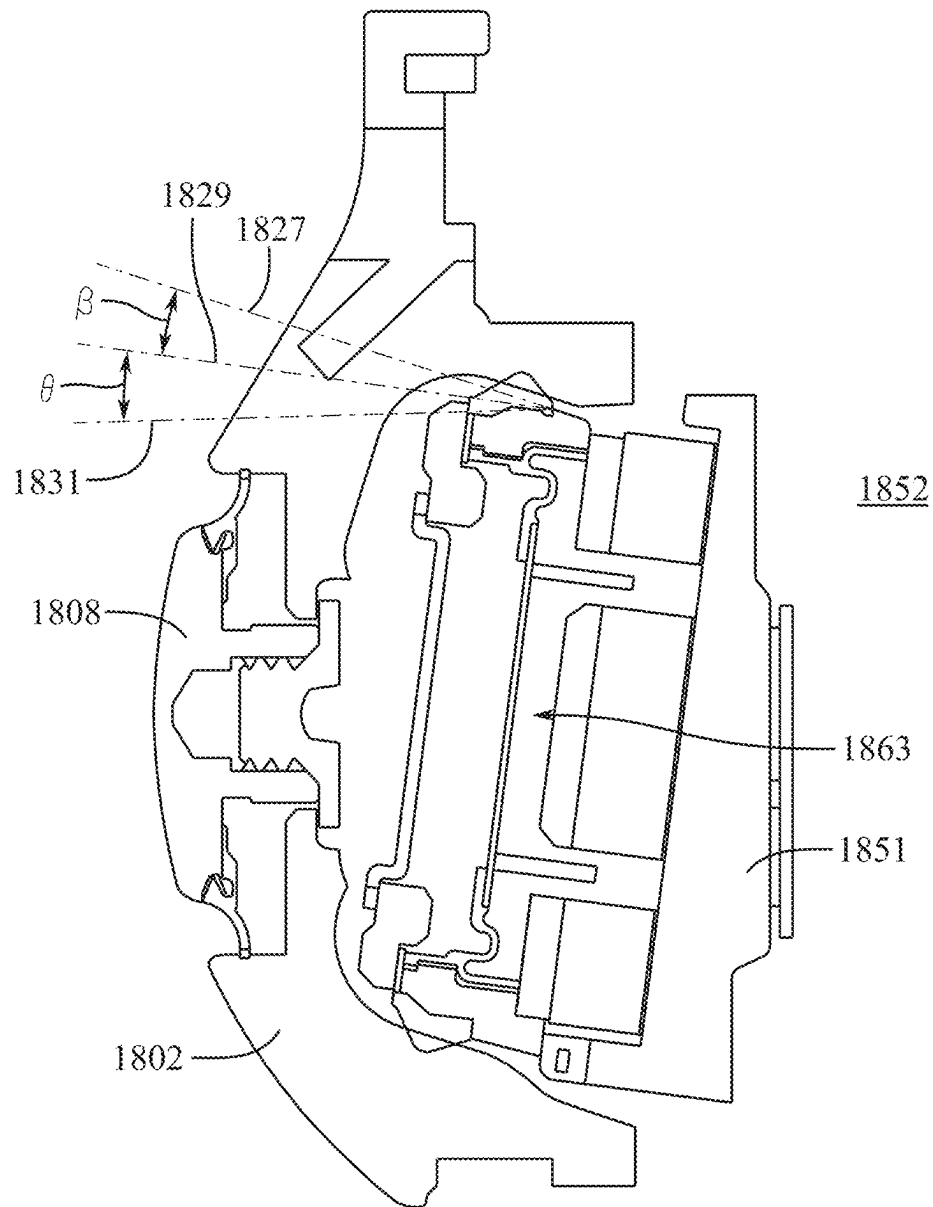
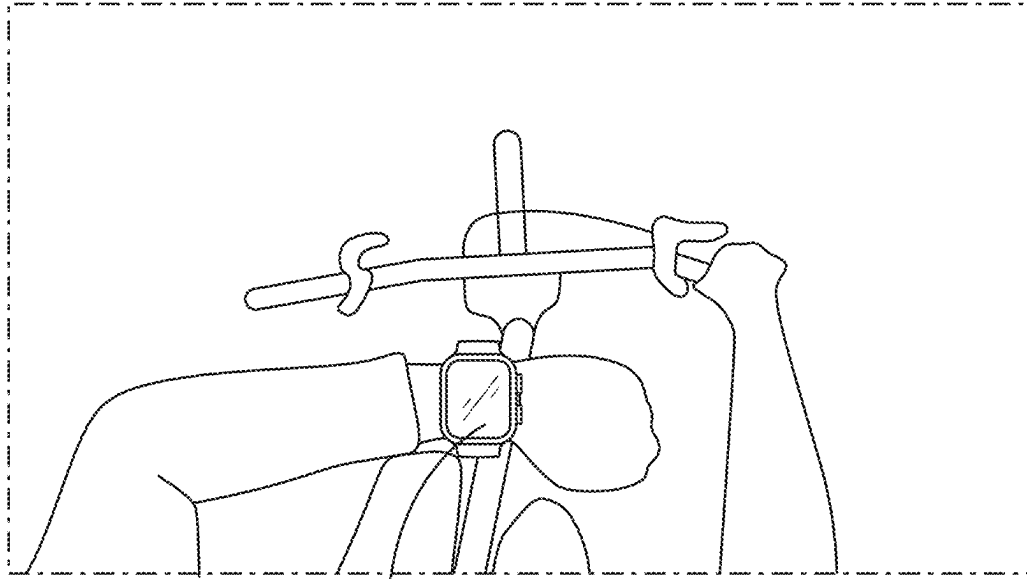


FIG. 21



1900 **FIG. 22**

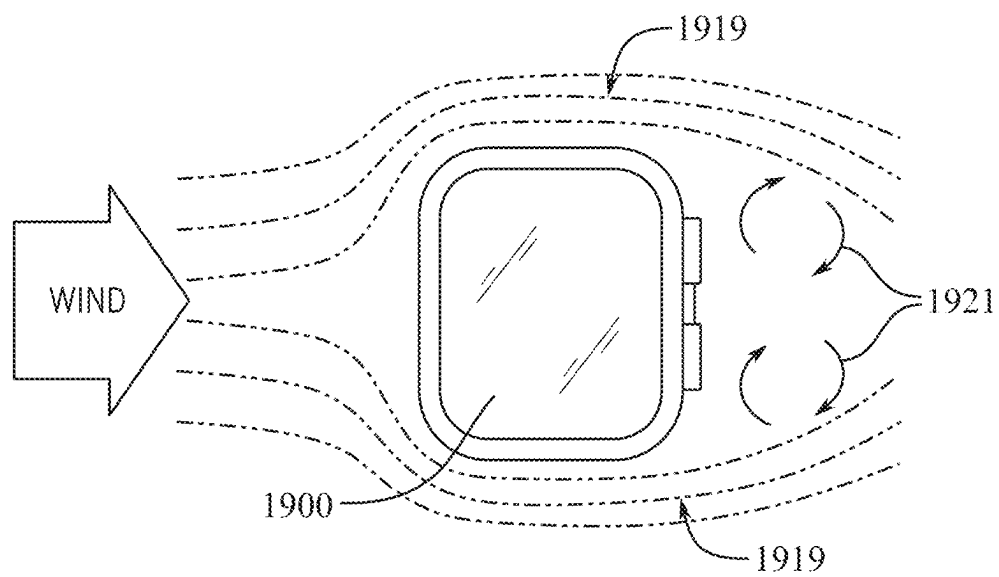


FIG. 23

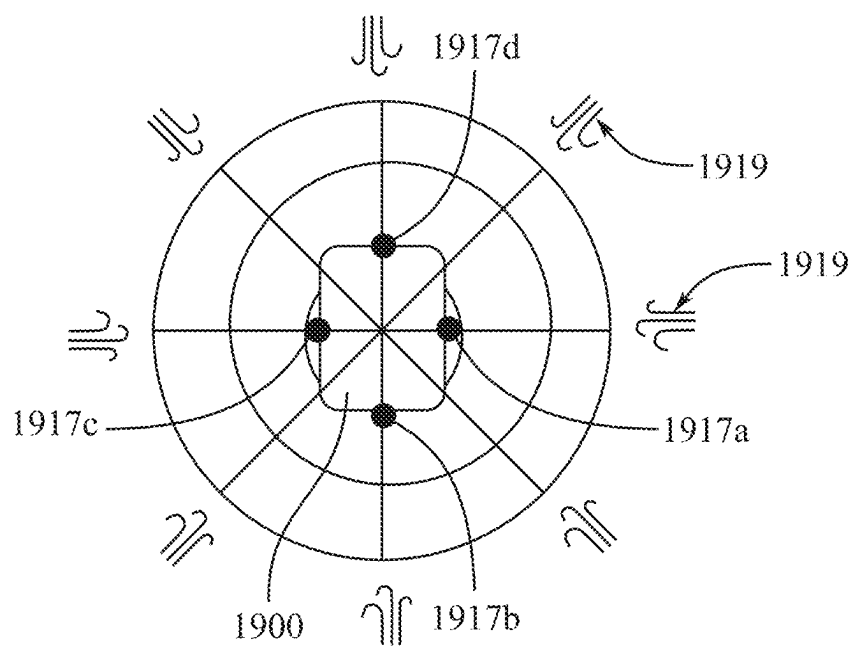


FIG. 24

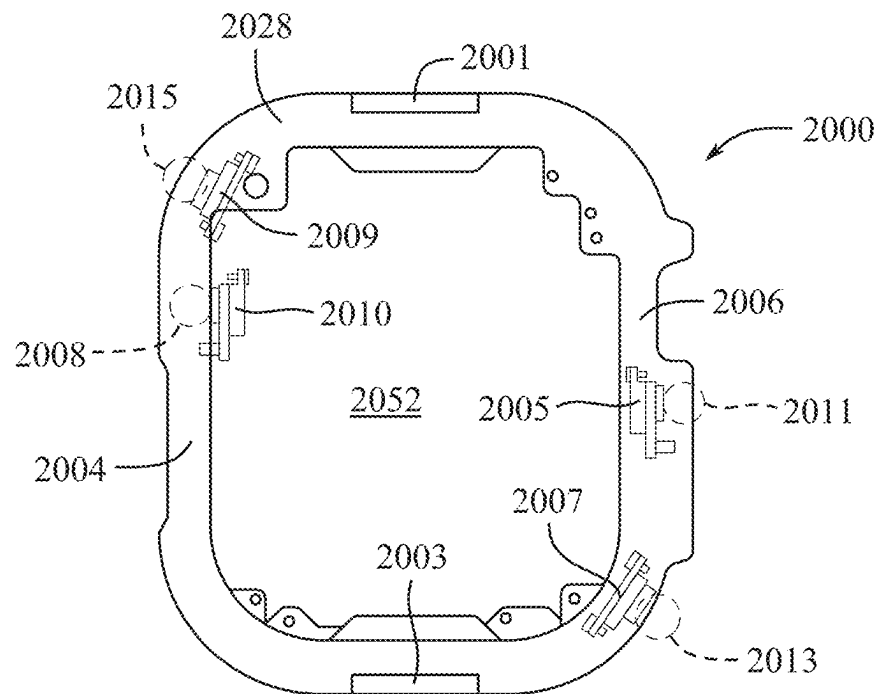


FIG. 25

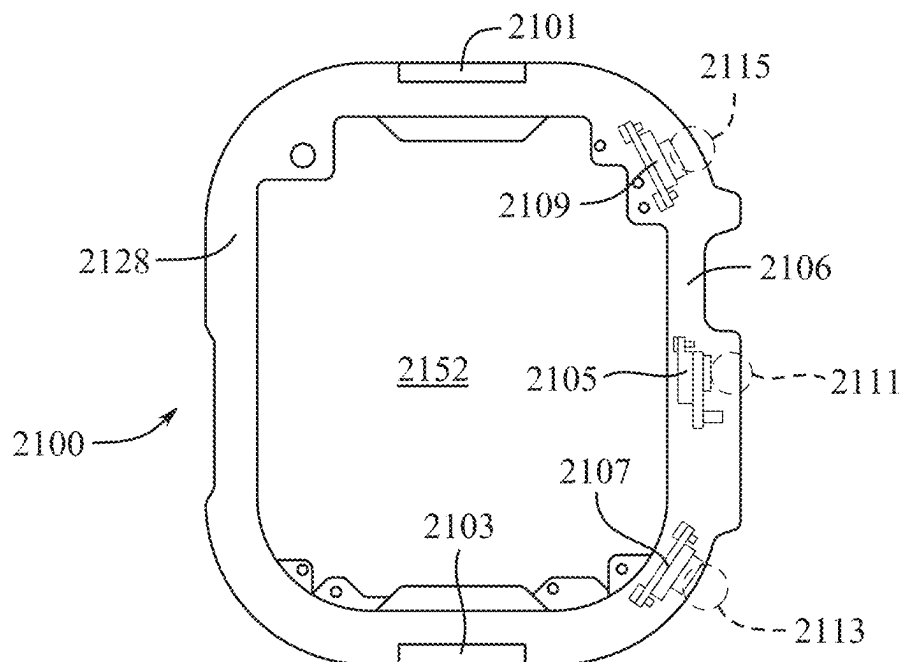


FIG. 26

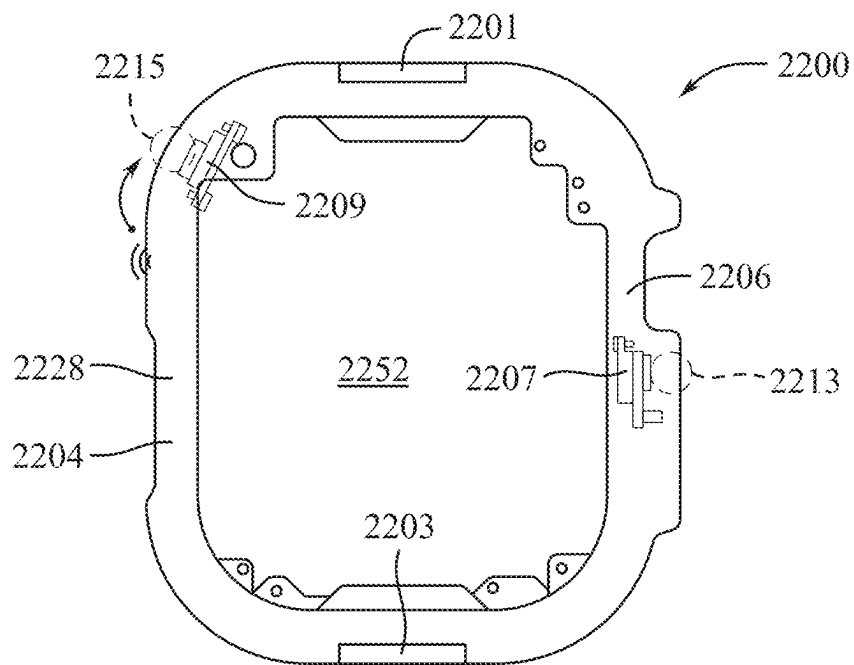


FIG. 27

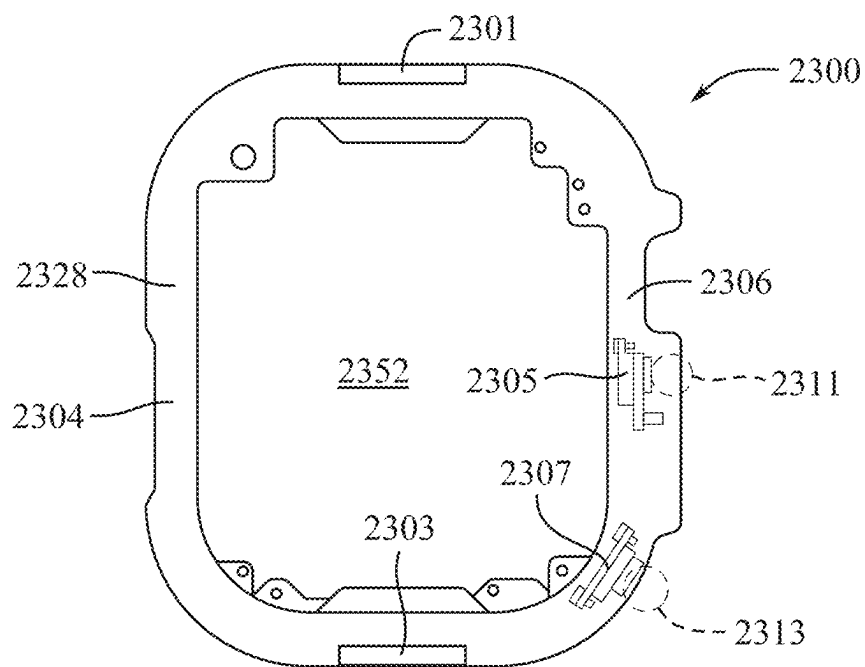


FIG. 28

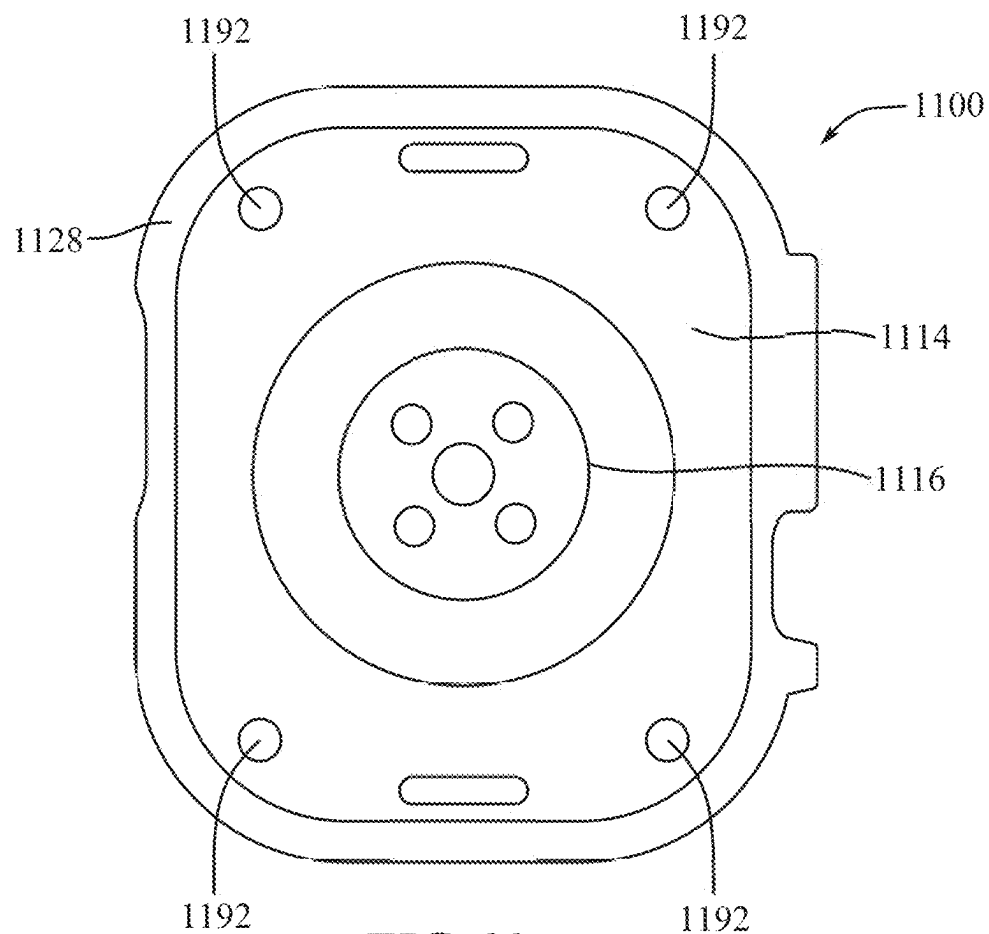


FIG. 29A

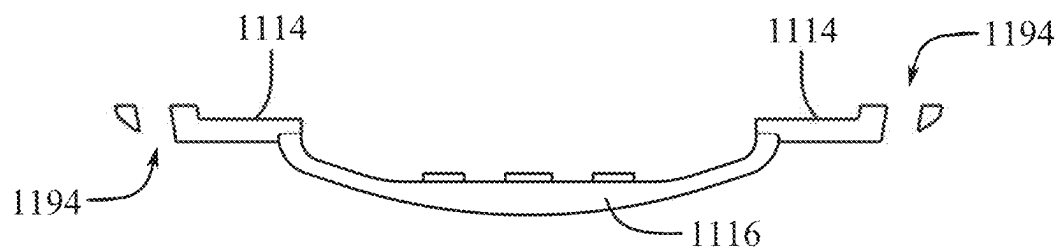


FIG. 29B

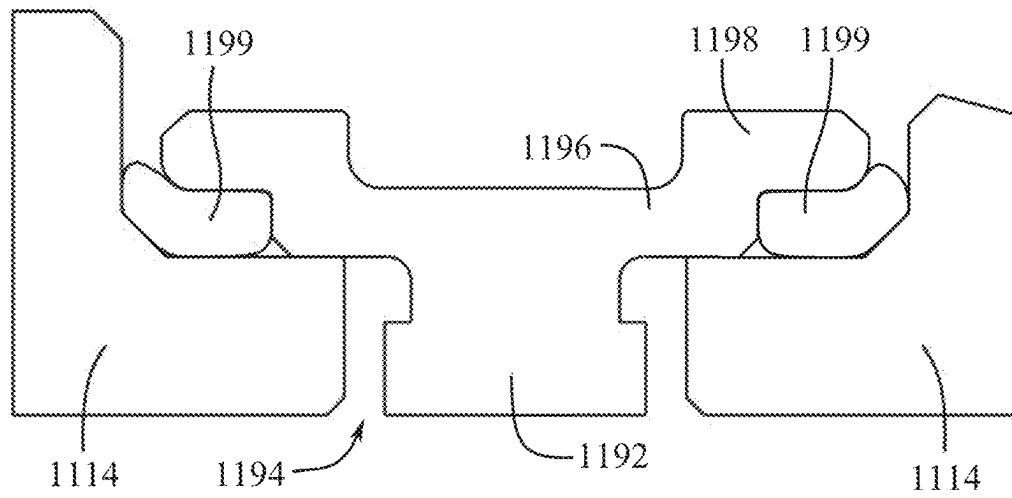


FIG. 30

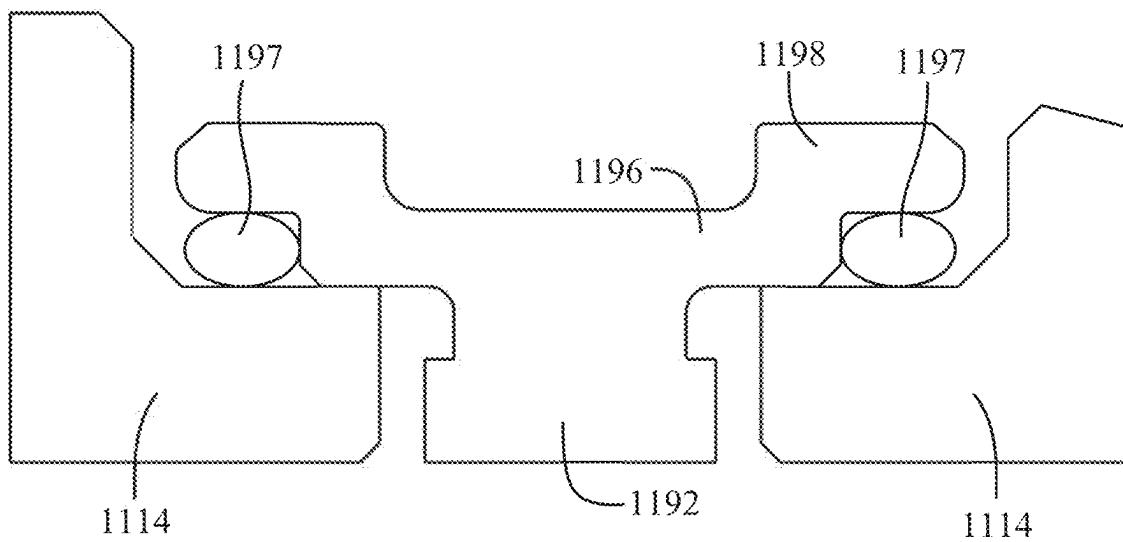


FIG. 31

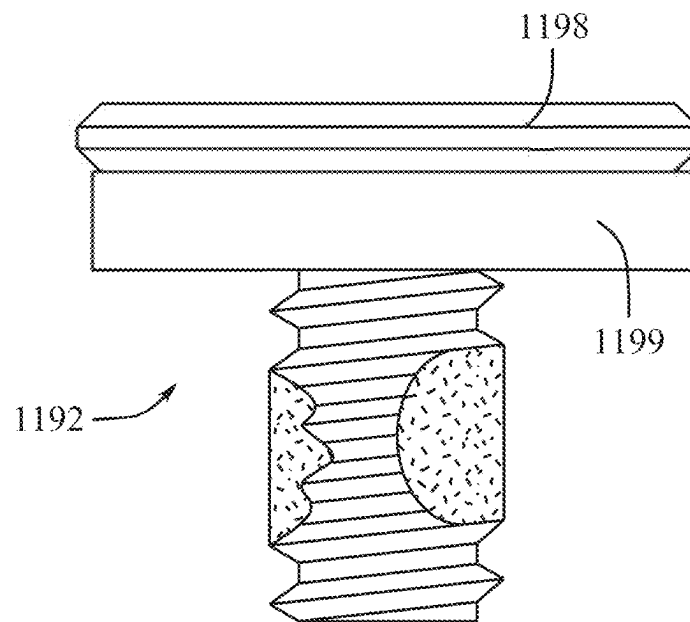
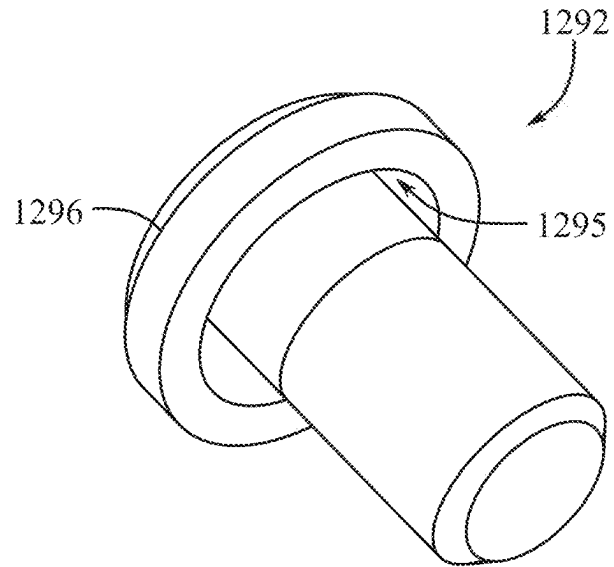
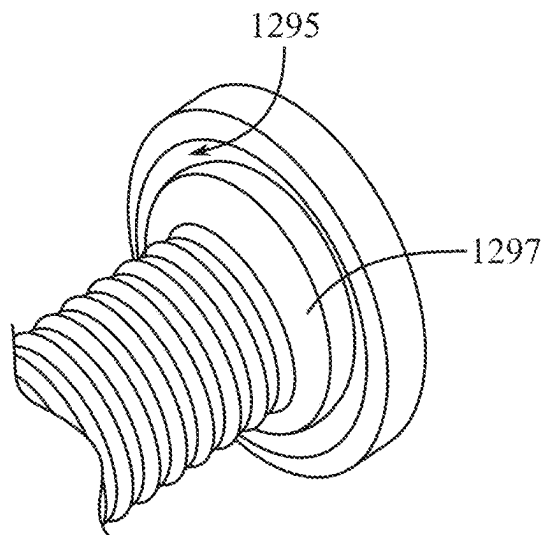
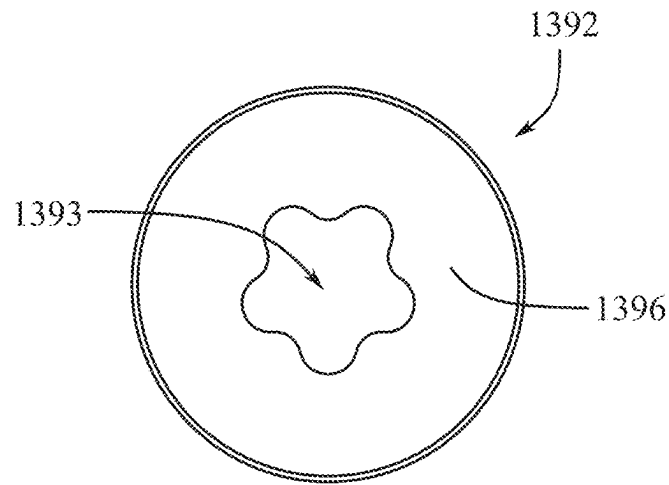
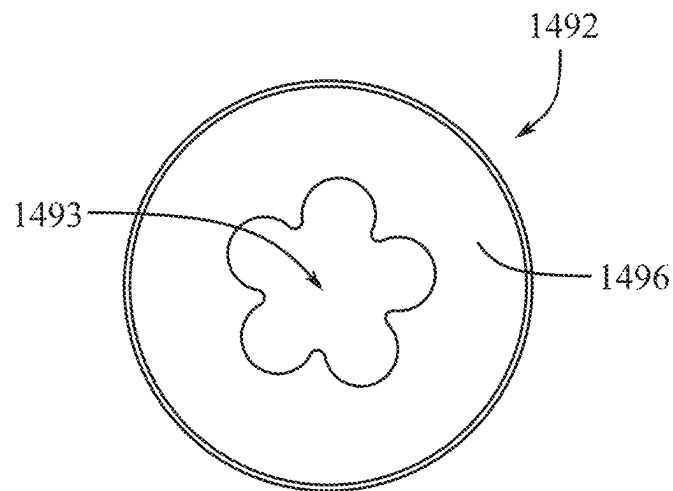


FIG. 32

**FIG. 33A****FIG. 33B**

**FIG. 34****FIG. 35**

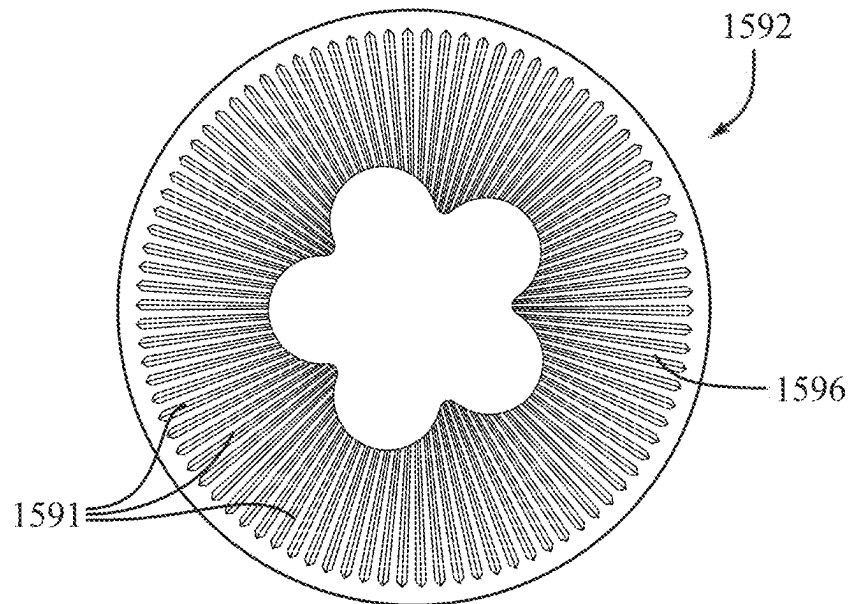


FIG. 36A

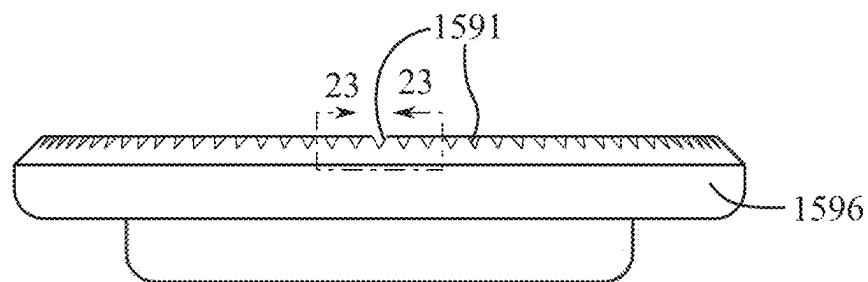


FIG. 36B

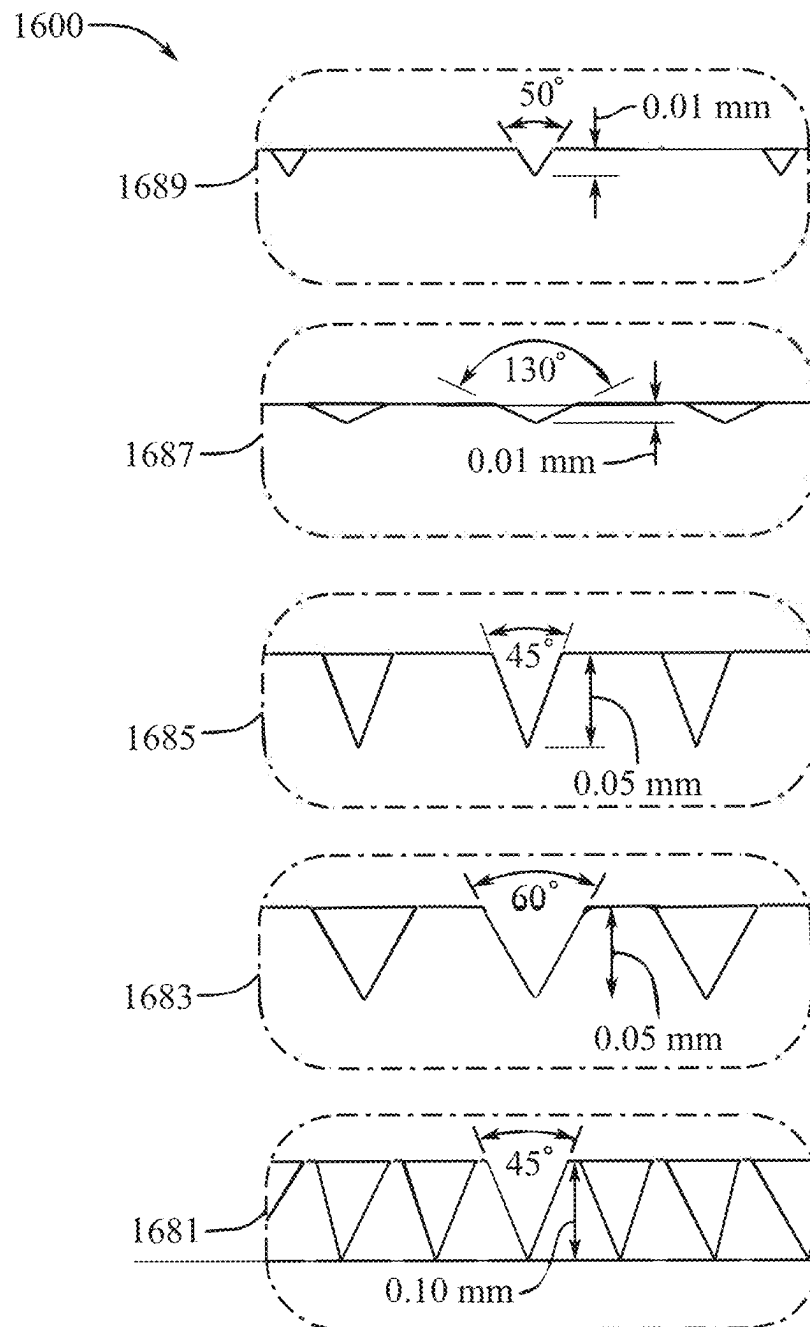


FIG. 37

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ELECTRONIC DEVICE**CROSS-REFERENCE TO RELATED APPLICATION(S)**

This claims priority to U.S. Provisional Patent Application No. 63/374,738, filed 6 Sep. 2022, and entitled "ELECTRONIC DEVICE," to U.S. Provisional Patent Application No. 63/364,012, filed 2 May 2022, and entitled "ELECTRONIC DEVICE," and to U.S. Provisional Patent Application No. 63/266,829, filed 14 Jan. 2022, and entitled "ELECTRONIC DEVICE," the disclosures of which are hereby incorporated by reference in their entireties.

FIELD

The present disclosure relates generally to electronic devices. More particularly, the present disclosure relates to wearable electronic devices.

BACKGROUND

Electronic devices are increasingly being designed with device portability in mind, for example, to allow users to use these devices in a wide variety of situations and environments. In the context of wearable devices, these devices can be designed to include many different functionalities and to be operated in many different locations and environments. The components of an electronic device, for example, the processors, memory, antennas, display, and other components can partially determine a level of performance of the electronic device. Further, the arrangement of these components with respect to one another in the device can also determine the level of overall performance of the electronic device.

Continued advances in electronic devices and their components have enabled considerable increases in performance. Existing components and structures for electronic devices can, however, limit the levels of performance of such devices. For example, while some components can achieve high levels of performance in some situations, the inclusion of multiple components in devices sized to enhance portability can limit the performance of the components, and thus, the performance of the device. Consequently, further tailoring and arrangement of components for electronic devices to provide additional or enhanced functionality, without introducing or increasing undesirable device properties, can be desirable.

SUMMARY

In at least one example of the present disclosure, an electronic device can include a housing including a sidewall defining an upper peripheral edge surrounding an opening and a transparent display cover disposed in the opening, the transparent display cover including an outward facing surface. In such an example, the transparent display cover is positioned such that the upper peripheral edge is level with the outward facing surface.

In one example, the electronic device includes a display assembly including the transparent display cover. In one example, the outward facing surface is planar. In one example, the display assembly further includes a touch interface. In one example, the electronic device further includes a protrusion extending from the sidewall, the protrusion defining a recess. In one example, the electronic device further includes a crown disposed in the recess. In

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one example, the protrusion surrounds a portion of the crown. In one example, the recess includes a first recess, the protrusion further defines a second recess, and the electronic device further includes a button disposed in the second recess.

In at least one example of the present disclosure, an electronic device includes a housing sidewall defining a first speaker vent, a second speaker vent, and an aperture between the first speaker vent and the second speaker vent, and a button disposed in the aperture.

In one example, the housing sidewall defines an internal volume, the electronic device further includes a first speaker disposed within the internal volume, the first speaker positioned to direct sound through the first speaker vent, and the electronic device further includes a second speaker disposed within the internal volume, the second speaker positioned to direct sound through the second speaker vent. In one example, the housing sidewall further defines a front speaker volume disposed between the housing sidewall and the first and second speakers. In one example, the first speaker vent includes at least two openings defined by the housing sidewall and the second speaker vent consists of a single opening defined by the housing sidewall. In one example, the housing sidewall defines a first side of the electronic device and a second side of the electronic device opposite the first side. In such an example, the first side defines the first and second speaker vents and the aperture. In one example, the second side defines a protrusion defining a recessed feature. In one example, the electronic device further includes a crown disposed in the recessed feature.

In at least one example of the present disclosure, an electronic device includes a housing defining an internal volume and an upper peripheral edge, the housing including a first sidewall defining a first speaker vent and a second speaker vent and a second sidewall opposite the first sidewall, the second sidewall including a protrusion defining a recessed feature, a transparent display cover flush with the upper peripheral edge, a button disposed between the first speaker vent and the second speaker vent, and a dial disposed in the recessed feature. In one example, the upper peripheral edge of the housing is configured to protect the transparent display cover from inadvertent contact during use. In one example, the protrusion is configured to protect the dial from inadvertent contact during use. In one example, the first sidewall defines an aperture between the first speaker vent and the second speaker vent and the button is disposed in the aperture. In one example, the protrusion is a first protrusion, the housing includes a second protrusion extending upward toward the upper peripheral edge, the second protrusion including an upper portion separated from a lower portion by a non-metallic material portion, the upper portion defining the upper peripheral edge, and the upper peripheral edge surrounds the display cover.

BRIEF DESCRIPTION OF THE DRAWINGS

The disclosure will be readily understood by the following detailed description in conjunction with the accompanying drawings, wherein like reference numerals designate like structural elements, and in which:

FIG. 1A shows an example of a wearable electronic device;

FIG. 1B shows a top view of a portion of the wearable electronic device;

FIG. 1C shows a bottom view of a portion of the wearable electronic device;

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FIG. 2A shows a perspective view of an example of a wearable electronic device;

FIG. 2B shows a perspective view of an example of a wearable electronic device;

FIG. 2C shows an exploded view of an example of a wearable electronic device;

FIG. 3 shows an exploded view of an example of a wearable electronic device;

FIG. 4 shows a portion of a sidewall of an example of a wearable electronic device;

FIG. 5A shows a cross-sectional view of a sidewall of an example of a wearable electronic device;

FIG. 5B shows another cross-sectional view of a sidewall of an example of a wearable electronic device;

FIG. 5C shows another cross-sectional view of a sidewall of an example of a wearable electronic device;

FIG. 6 shows a cross sectional view of an interface between a plastic and metal portion of a sidewall of an example of an electronic device;

FIG. 7A shows another view of the interface of FIG. 6;

FIG. 7B shows another view of the interface of FIG. 6;

FIG. 7C shows another view of the interface of FIG. 6;

FIG. 8 shows a method of bonding a metal substrate to a non-metal substrate;

FIG. 9A shows a cross-sectional view of a sidewall and internal components of an example of a wearable electronic device;

FIG. 9B shows another cross-sectional view of a sidewall and internal components of an example of a wearable electronic device;

FIG. 9C shows another cross-sectional view of a sidewall and internal components of an example of a wearable electronic device;

FIG. 10A shows a top perspective view of a housing sidewall of an example of an electronic device;

FIG. 10B shows a top view of the housing sidewall of FIG. 10A;

FIG. 10C shows a close up top view of a portion of the housing sidewall of FIG. 10A;

FIG. 11 shows a cross-sectional view of a sidewall and internal components of an example of a wearable electronic device;

FIG. 12 shows a top view of a portion of an example of an electronic device including a printed circuit board (PCB) and surrounding electrical contacts;

FIG. 13 shows a top view of a portion of an example of an electronic device including a printed circuit board (PCB) and surrounding electrical contacts;

FIG. 14A shows a cross-sectional view of a sidewall and internal components of an example of a wearable electronic device;

FIG. 14B shows a cross-sectional view of a sidewall and internal components of an example of a wearable electronic device;

FIG. 15A shows a side view of an example of a wearable electronic device;

FIG. 15B shows a cross sectional view thereof;

FIG. 15C shows a circuit diagram equivalent of the device shown in FIGS. 15A and 15B worn by a user;

FIG. 16 shows a PCB of an example of an electronic device;

FIG. 17 shows a portion of a PCB of an example of an electronic device;

FIG. 18 shows a portion of a PCB of an example of an electronic device;

FIG. 19 shows a temperature sensor disposed on an ALS module of an example of an electronic device;

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FIG. 20A shows a cross sectional view of a portion of an example of an electronic device;

FIG. 20B shows a close up view of the portion of FIG. 20A;

FIG. 20C shows an example of a speaker frame;

FIG. 20D shows an example of a speaker assembly of an electronic device;

FIG. 20E shows the speaker assembly of FIG. 20D with an example of a button;

FIG. 20F shows another cross sectional view of the speaker assembly of FIG. 20E;

FIG. 20G shows another cross sectional view of the speaker and button assembly of FIG. 20E;

FIG. 20H shows another cross sectional view of the speaker and button assembly of FIG. 20E;

FIG. 20I shows another cross sectional view of the speaker and button assembly of FIG. 20F;

FIG. 20J shows another cross sectional view of the speaker and button assembly of FIG. 20F;

FIG. 20K shows another cross sectional view of the speaker and button assembly of FIG. 20F;

FIG. 21 shows another cross sectional view of the speaker assembly of FIG. 20E;

FIG. 22 shows a user riding a bike while wearing an example of a wearable electronic device;

FIG. 23 shows an example of an electronic device subjected to wind;

FIG. 24 shows an example of an electronic device subjected to wind from various directions;

FIG. 25 shows a top view of an example of an electronic device;

FIG. 26 shows a top view of an example of an electronic device;

FIG. 27 shows a top view of an example of an electronic device; and

FIG. 28 shows a top view of an example of an electronic device.

FIG. 29A shows a bottom view of an example of an electronic device;

FIG. 29B shows a cross sectional view of a rear cover of an example of an electronic device;

FIG. 30 shows a partial cross-sectional view of a rear cover and a fastener of an example of an electronic device;

FIG. 31 shows a partial cross-sectional view of a rear cover and a fastener of an example of an electronic device;

FIG. 32 shows an example of a rear cover fastener;

FIG. 33A shows another example of a rear cover fastener;

FIG. 33B shows another example of a rear cover fastener;

FIG. 34 shows a top view of another example of a fastener;

FIG. 35 shows a top view of another example of a fastener;

FIG. 36A shows a top view of another example of a fastener;

FIG. 36B shows a side view thereof;

FIG. 37 shows a method of forming the fastener shown in FIG. 36A and FIG. 36B;

DETAILED DESCRIPTION

Reference will now be made in detail to representative examples illustrated in the accompanying drawings. It should be understood that the following descriptions are not intended to limit the embodiments to one preferred example or embodiment. To the contrary, it is intended to cover alternatives, modifications, and equivalents as can be

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included within the spirit and scope of the described embodiments as defined by the appended claims.

The following disclosure generally relates to electronic devices. More particularly, the present disclosure relates to wearable electronic devices. The wearable electronic devices of the present disclosure include tailored arrangements of components to provide additional or enhanced functionality, without introducing or increasing undesirable device properties or performance. In this way, more functionality and componentry can be included in wearable devices for user's to wear and operate in any condition or activity without limiting the functionality and durability of the devices.

Specific examples and embodiments of electronic devices, including wearable electronic devices, are discussed below with reference to FIGS. 1-28. However, those skilled in the art will readily appreciate that the detailed description given herein with respect to these Figures is for explanatory purposes only and should not be construed as limiting. Furthermore, as used herein, a system, a method, an article, a component, a feature, or a sub-feature comprising at least one of a first option, a second option, or a third option should be understood as referring to a system, a method, an article, a component, a feature, or a sub-feature that can include one of each listed option (e.g., only one of the first option, only one of the second option, or only one of the third option), multiple of a single listed option (e.g., two or more of the first option), two options simultaneously (e.g., one of the first option and one of the second option), or combination thereof (e.g., two of the first option and one of the second option).

FIG. 1A shows an example of an electronic device 100. The electronic device shown in FIG. 1A is a watch, such as a smartwatch. The smartwatch of FIG. 1A is merely one representative example of a device that can be used in conjunction with the systems and methods disclosed herein. Electronic device 100 can correspond to any form of wearable electronic device, a portable media player, a media storage device, a portable digital assistant ("PDA"), a tablet computer, a computer, a mobile communication device, a GPS unit, a remote control device, or other electronic device. The electronic device 100 can be referred to as an electronic device, or a consumer device. In some examples, the electronic device 100 can include a housing 102 that can carry operational components, for example, in an internal volume at least partially defined by the housing. The electronic device 100 can also include a strap 104, or other retaining component that can secured the device 100 to a body of a user as desired. Further details of the electronic device are provided below with reference to FIG. 1B.

FIG. 1B illustrates the electronic device 100, for example a smartwatch, that can be substantially similar to and can include some or all of the features of the devices described herein, including the electronic device 100 shown in FIG. 1A but without the strap 104. The device 100 can include a housing 102, and a display assembly 106 attached to the housing 102. The housing 102 can substantially define at least a portion of an exterior surface of the device 100.

The display assembly 106 can include a glass, a plastic, or any other substantially transparent exterior layer, material, component, or assembly. The display assembly 106 can include multiple layers, with each layer providing a unique function, as described herein. Accordingly, the display assembly 106 can be, or can be a part of, an interface component. The display assembly 106 can define a front exterior surface of the device 100 and, as described herein, this exterior surface can be considered an interface surface.

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In some examples, the interface surface defined by display assembly 106 can receive inputs, such as touch inputs, from a user. In other words, the display assembly 106 can include a touch sensitive layer, or form a touch interface, configured to receive touch inputs from a user.

In some examples, the housing 102 can be a substantially continuous or unitary component and can define one or more openings to receive components of the electronic device 100. In some examples, the device 100 can include input components such as one or more buttons 108 and/or a crown 110 that can be disposed in the openings. In some examples, a material can be disposed between the buttons 108 and/or crown 110 and the housing 102 to provide an airtight and/or watertight seal at the locations of the openings. The housing 102 can also define one or more openings or apertures, such as aperture 112 that can allow for sound to pass into or out of the internal volume defined by the housing 102. For example, the aperture 112 can be in communication with a microphone component disposed in the internal volume. In some examples, the housing 102 can define or include a feature, such as an indentation to removably couple the housing 102 and a strap or retaining component.

FIG. 1C shows a bottom perspective view of the electronic device 100. The device 100 can include a back cover 114 that can be attached to the housing 102, for example, opposite the display assembly 106. The back cover 114 can include ceramic, plastic, metal, or combinations thereof. In some examples, the back cover 114 can include an at least partially electromagnetically transparent component 116. The electromagnetically transparent component 116 can be transparent to any desired wavelengths of electromagnetic radiation, such as visible light, infrared light, radio waves, or combinations thereof. In some examples, the electromagnetically transparent component 116 can allow sensors and/or emitters disposed in the housing 102 to communicate with the external environment. Together, the housing 102, display assembly 106 and back cover 114 can substantially define an internal volume and an external surface of the device 100.

Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. 1A-1C can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown in the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. 1A-1C.

As noted above, portable and wearable electronic devices can be designed to be used in many different environments and during any kind of activity throughout a user's day. For example, wearable electronic watches, headphones, and phones can be carried by a user during exercise, sleep, driving, biking, hiking, swimming, diving, outside in the rain, outside in the sun, and so forth. Wearable electronic devices described herein are configured to withstand the varied and often harsh conditions of various environments, including changing environments and wet environments. Wet environments can include wearing devices in the rain or when submerged during bathing or swimming, for example.

Examples of electronic devices disclosed herein include components, features, arrangements, and configurations that resists damage and corrosion due to exposure to moisture. Some aspects of devices described herein can include gaps between components through which moisture, water, or other fluids could enter. The gaps may be present for aesthetic purposes or for functional purposes. However, one

or more components, including epoxy seals, insulating materials and frames, and other components of devices described herein can be configured to prevent such moisture from entering into the internal volume of the device where sensitive electronic component could be damaged thereby.

Along these lines, FIGS. 2A and 2B show right and left perspective views, respectively, of an example of a wearable electronic device 200 including a housing 202 including sidewalls 228 defining an opening in which the display cover 222 is disposed. The sidewalls can include an upper portion 232 defining an upper peripheral edge surrounding the display cover 222, a lower portion 234, and a middle portion 236 disposed between the upper portion 232 and the lower portion 234. The wearable electronic device 200 can also include a securement strap 203 configured to secure the wearable electronic device 200 to an appendage of the user. In at least one example, the sidewalls 228 of the housing 202 can define an upper peripheral edge of the device 200 surrounding the display cover 222.

In at least one example, the display cover 222 defines a top surface disposed in a plane. The top surface can include an outward facing surface of the display cover 222. In at least one example, the display cover 222 can be a transparent display cover disposed above or over other display layers. In at least one example, the outward facing surface of the display cover 222 can be planar. The plane can be level/flush with or set below and recessed relative to the upper peripheral edge of the sidewalls 228. In this way, when the wearable electronic device 200 comes into contact with a surface or object at or near the upper surface of the display cover 222 and/or the upper peripheral edge of the sidewalls 228, contact and potential damage to the display cover 222 can be reduced. In one example, the display cover 222 is set flush with or below the upper peripheral edges of the sidewalls 228 to protect the display cover 222 from damage.

In at least one example, as shown in FIG. 2A, the sidewalls 228 can define a first side of the wearable electronic device 200 having a turret or protrusion defining a recessed feature in which a crown 210 is positioned. The turret can include a protrusion extending from a surrounding surface of the sidewall 228. The crown 210 can be a part of a turn dial button or other functional knob configured to be manipulated by the user. The crown 210 can be disposed in the recessed portion, as noted above, such that the first side of the sidewalls 228 extend outward in a turret configuration and at least partially around the crown 210. In this way, contacts and bumps against of other objects against the first side of the sidewall 228 during use can contact the sidewall 228 without pressing or turning the crown 210. In this way, the protrusion and the recessed portion of the first side of the sidewalls 228 prevents inadvertent manipulation of the crown 210. In at least one example, the protrusion can include a window exposing a circumferential side surface of the crown 210 where a user can manually manipulate the crown 210 within the recess of the turret protrusion. The button 209 shown in FIG. 2A can also be at least partially surrounded by an outwardly extending portion of the sidewall 228, such that the button 209 is disposed within another recess thereof, to protect the button 209 from inadvertent contacts.

In at least one example, as shown in FIG. 2B, the sidewalls 228 can define a second side opposite the first side shown in FIG. 2A. In such an example, the wearable electronic device 200 can include a first speaker vent 249, a second speaker vent 247, and a button 208 disposed within an aperture defined by the sidewalls 228 between the first speaker vent 249 and the second speaker vent 247. The first

and second speaker vents 249, 247 can provide fluid communication from a common speaker volume behind the sidewall 228 (e.g., within an internal volume defined by the sidewalls 228) and the external environment. The button 208 can be disposed between the first and second speaker vents 249, 247 to save space and provide a compact design without interrupting the functionality of the one or more speakers communicating with the external environment through the first and second speaker vents 247, 249.

FIG. 2C illustrates an exploded view of another example of an electronic device 200, which can also be a portion of a wearable electronic watch or other wearable electronic device. Device 200 includes a display assembly 206, housing 202, back cover 214, and electromagnetically transparent component 216. In addition, the exploded view of FIG. 2A illustrates various internal components that may be disposed within an internal volume defined by the housing 202, back cover 214, electromagnetically transparent component 216, and display assembly 206. For example, the device 200 can include one or more printed circuit boards (PCBs) 218 and one or more antenna components 220, electrical connectors and flexes, buttons, seals, gaskets, memory components, processors, sensors, dials, batteries, and so forth.

Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. 2A-2C can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown in the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. 2A-2C.

FIG. 3 illustrates a close-up view of a portion of the exploded view of the device 200 shown in FIG. 2, including the housing 202 and the display assembly 206, with the display assembly further exploded to illustrate the display cover 222 and display layers 224. In addition, the exploded view of FIG. 3 shows a wave ring 226 (also referred to herein as an “elongate conductive member”), which will be described and discussed in more detail hereafter with reference to other figures. In at least one example, the housing 202 includes sidewall or sidewalls 228 that define an internal volume and an opening 230. When assembled, the display assembly 206 or one or more components of the display assembly 206 can be disposed in the opening to form an outer surface of the device 200 and define the internal volume.

In at least one example, the sidewall 228 can include an upper portion 232 and a lower portion 234. The upper portion 232 and the lower portion 234 can be separated by a middle portion 236 disposed between the upper portion 232 and the lower portion 234. In at least one example, the upper portion 232 and the lower portion 234 of the sidewall 228 can include one or more electrically conductive materials and the middle portion 236 can include one or more electrically non-conductive materials and/or an insulating material. The middle portion 236 can be molded to or otherwise adhered to the upper portion 232 and/or the lower portion 234 such that the upper portion 232, the lower portion 234, and the middle portion 236 form a single, unitary sidewall 228 of the housing 202, as shown.

Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. 3 can be included, either alone or in any combination, in any of the other examples of devices, features, components, and

parts shown in the other figures. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown in the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. 3.

FIG. 4 illustrates an example of a middle portion 336 separated from the rest of a sidewall of a housing, similar to the middle portion 236 shown as part of the sidewall 228 of FIG. 3. In the example shown in FIG. 4, a non-conductive material component or ring 338 is bonded to the middle portion 336. In at least one example, the non-conductive material ring 338 can include an epoxy component 338. The term “epoxy,” as used herein, can include non-conductive adhesives as generally used and understood in the art, including hot-melt adhesives. The middle portion 336 can also include a ridge feature 340 extending at least partially around an internal surface of the middle portion 336. The ridge feature 340 can form a lower surface of the middle portion 336. In addition, at least one example, the middle portion 336 can include one or more upper protrusions 342 spaced about and extending inward relative to an external surface of the sidewall 328. In addition, at least one example of the middle portion 336 can include one or more lower protrusions 344 spaced about and extending from the middle portion 336, as shown in FIG. 4. FIGS. 5A, 5B, and 5C illustrate partial cross-sectional views, as indicated in FIG. 3, but around various points along the sidewall 228 in order to illustrate the positions and configurations of the middle portion 336, epoxy component 338, upper portion 332, and lower portion 334.

Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. 4 can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown in the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. 4.

In the first cross-sectional view of FIG. 5A, the ridge feature 340 and lower protrusion 344 of the middle portion 336 can be seen. In at least one example, the epoxy component 338 is bonded to an upper surface 348 defined by the lower portion 334 of the sidewall 328. The epoxy component 338 can also be bonded directly to the middle portion 336, for example including the lower protrusion 344 and a lower surface 346 of the middle portion 336 defined by the ridge feature 340. FIG. 5B illustrates another cross-sectional view at a point along the sidewall 328 where an upper protrusion 342 and a lower protrusion 344 of the middle portion 336 can be seen and FIG. 5C illustrates another cross-sectional view at a point along the sidewall 328 where an upper protrusion 342 of the middle portion 336 can be seen.

As shown in FIGS. 5A, 5B, and 5C, any possible path for moisture to travel from an external environment 350 to the internal volume 352 through the sidewall 328 of the device is blocked by the epoxy component 338 bonded directly to either the lower portion 334 of the sidewall 328 or the middle portion 336. Thus, the tight bond of the epoxy component 338 against one or more portions 334, 336 prevents water and moisture from entering into the internal volume 352 from the external environment 350 through the sidewall 328. In addition to the moisture-tight bond between the epoxy component 338 and the sidewall 328, in at least

one example, the middle portion 336 of the sidewall 328 can be bonded to the upper and lower portions 332, 334, respectively, such that the bond therebetween substantially or completely prevents moisture from passing through or into the sidewall 328 at the interface between the middle portion 336 and the upper and lower portions 332, 334, respectively. FIGS. 6-8 illustrate the interface and methods of bonding the middle portion 336 to the lower portion 334 and/or the upper portion 332 of the sidewall 328.

Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. 5A-5C can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown in the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. 5A-5C.

One example of a plastic-metal interface, such as an interface between a conductive bottom portion 434 and a non-conductive middle portion 436, is shown in FIG. 6. In at least one example, the bottom portion 434 can include titanium and the middle portion 436 can include a polymer material. In one example, polymer material can include polybutylene terephthalate (PBT) including glass filled PBT. As shown, the titanium material of bottom portion 434 can be treated to form enhanced polymer-Ti bonding at the interface.

In at least one example, surface features including nano-pores and protrusions 454 can be present and the polymer can flow into and around the pores and protrusions during formation to increase the bonding therebetween. In at least one example, an etching treatment can be carried out to form the features 454 shown in FIG. 6. In at least one example, the etching treatment can include etching with sulfuric acid to form roughened macro-pockets or features in the titanium substrate of the bottom portion 434. Also, an oxide layer can be formed using a sodium hydroxide oxidation step resulting in the nano-pores and protrusions 454 shown in FIG. 6.

Along these lines, FIGS. 7A-7C show images of the bottom portion 434 at various stages of the process described above, including a 1,000×SEM image of an acid etched surface in FIG. 7A. FIG. 7B shows a 50,000×SEM image of an oxide surface with uniform oxide formation exhibiting plate-like morphology. FIG. 7C shows a layered double hydroxide interface including protrusions 454, similar to those illustrated in FIG. 6, into and around which the plastic polymer material of the middle portion 436 can flow and interlock with the metal material of the bottom portion 434 for enhanced bonding, as described above.

FIG. 8 shows a flowchart of an example of a method 460 of forming the interface shown in FIGS. 6-7C. In at least one example, a step 456 includes the sulfuric acid etching step noted above and another step 458 includes the sodium hydroxide oxidation step noted above. In at least one example of the method 460, between steps 456 and 458, a de-smutting method 462 can be carried out to de-smut the surface. In at least one example of the de-smutting method 462 shown in FIG. 8, a step 464 can include an alkaline de-smut, for example using sodium hydroxide (NaOH) and another step 466 can include an acid de-smutting step, for example using nitric acid (HNO₃).

Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. 6-8 can be included, either alone or in any combination, in any of the other examples of devices, features, components,

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and parts shown in the other figures. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown in the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. 6-8.

FIGS. 9A-9C illustrate a cross-sectional view of a device 500 similar to FIGS. 5A-5C but with a display assembly 506 disposed in the opening 530 formed by the sidewall 528. In at least one example, the display assembly 506, which includes the display cover 522 and one or more other display layers 524 disposed below the display cover 522, can be disposed in the opening 530 such that a gap 568 is formed between the display assembly 506 and the sidewall 528. The gap 568 can be understood as a space between the display assembly 506, or the display cover 522 thereof, and the sidewall 528, or upper portion 532 thereof, wherein the display cover 522 does not contact the sidewall 528. In at least one example, an upper surface of the display cover 522 can be flush with, or disposed lower than, an upper surface of the upper portion 532.

In at least one example, a cavity 570 is formed in which the wave ring 526 is disposed. The cavity can be defined by the sidewall 528, including the upper portion 532 and the middle portion 536, the epoxy component 538, and the display assembly 506 or at least the display cover 522 thereof. In at least one example, the cavity can also be defined by an insulating material 576 disposed between the display assembly 506 and/or display cover 522 thereof and the epoxy component 538. One or more other components, including an lth (or last) antenna layer 578 or other layers. As noted above, the epoxy component 538 can bond to other layers and components, including the lth antenna layer 578, middle portion 536, lower portion 534, and/or the insulating material 576 to prevent moisture from entering the internal volume 552 from an external environment 550 of the device 500, such that any moisture or fluids entering the cavity 570 through the gap 568 do not continue on into the internal volume 552. In this way, the cavity can be fluid-tight.

FIGS. 9A, 9B, and 9C show cross-sectional views at various locations around the sidewall 528 to illustrate how the wave ring 526 disposed in the cavity 570 can contact the upper portion 532 of the sidewall 528 at one or more locations along a length of the wave ring 526, as shown in FIG. 9B, and contact an electrical contact 572 on the other side of the cavity 570 at one or more other locations along the length of the wave ring 526, as shown in FIG. 9C.

Accordingly, in at least one example of the present disclosure, the housing sidewall 528 can define an opening 532 and a display component, such as the display cover 522, can be disposed in the opening 532 to form the gap 568 between the housing sidewall 528 and the display component. In at least one example, the cavity 570 is defined by the sidewall 528 and the display cover 522 with the cavity 570 in fluid communication with the external environment 550 through the gap 568. In at least one example, the epoxy component 538 at least partially defines the cavity 570 and can be in direct contact with the housing sidewall 528.

In at least one example of the electronic device 500, the housing sidewall 528 has an upper sidewall portion 532 and a lower sidewall portion 534 bonded to a middle sidewall portion 536 disposed between the upper and lower sidewall portions 532, 534, respectively. The housing can define the opening 532 and the display assembly 506 can be disposed in the opening 532 to form the gap 568 between the housing and the display assembly 506. Also, in at least one example, the epoxy component 538 can serve as a seal disposed

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underneath the display assembly 506 and extend laterally across the gap 568 with the epoxy component seal 538 bonded directly to the middle portion 536 of the sidewall 528.

In at least one example of the present disclosure, the electronic device 500 can include the sidewall 528 defining the internal volume 552 and the opening 532. In at least one example, the sidewall 528 can include an upper portion 532, a lower portion 534, and a middle portion 536 disposed between and bonded to the upper portion 532 and the lower portion 534. The device 500 can also include the display cover 522 disposed in the opening 532 and defining the internal volume 552, the side cavity 570 defined by the display assembly 506 and the sidewall 528, with the cavity 570 in fluid communication with an external environment 550 through the gap 568 formed between the display assembly 506 and the sidewall 528, and an epoxy layer 538 contacting the lower portion 534 and the middle portion 536, and at least partially defining the cavity 570.

As noted above and as shown in FIGS. 9A-9C, the device 500 can include the epoxy component 538 at least partially disposed between the display cover 522 of the display assembly 506 and the lower portion 534, or between one or more other components of the display assembly 506, including the display layers 524, and the lower portion 534. One or more other components can also be disposed or stacked between the epoxy component 538 and the display assembly 506 or cover 522, for example the lth antenna layer 578. In addition, as shown in FIG. 9C, one or more examples of the device 500 can include an insulating material 576. The insulating polymer 576 can include and support a printed circuit board (PCB) 574 disposed in the internal volume 552.

Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. 9A-9C can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown in the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. 9A-9C.

As noted above, FIGS. 9A, 9B, and 9C show cross-sectional views at various locations around the sidewall 528 to illustrate how the wave ring 526 disposed in the cavity 570 can contact the upper portion 532 of the sidewall 528 at one or more locations along a length of the wave ring 526, as shown in FIG. 9B. Accordingly, in at least one example, the middle portion 536 can include gaps or windows through which columns or other portions of the upper portion 532 of the sidewall 528 are exposed through the middle portion 536 such that the wave ring 526 can contact the upper portion 532 directly, as shown in FIG. 9B. Also, the wave ring 526 can contact an electrical contact 572 on the other side of the cavity 570 at one or more other locations along the length of the wave ring 526, as shown in FIG. 9C. The electrical contact 572 can extend through the insulating material 576 and electrically connect to the PCB 574. In this way, the upper portion 532 of the sidewall 528 can be electrically connected to the PCB 574 through the wave ring 526. While the contact between the wave ring 526 and the electrical contact 572 is shown in FIGS. 9A-9C (as well as in FIGS. 12 and 13 below) as resulting from an undulation or circumferential variation (or wave) in the wave ring 526, the electrical contacts can be made by any number of discrete or continuous geometries such as indents or protrusions on the wave ring 526, or by discrete or continuous protrusions on

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the housing. According to this example, any number of mating geometries can be used between the wave ring 526 and the housing.

In at least one example, the upper portion 532 of the sidewall 528 can be electrically isolated from the lower portion 534 via the intermediary and non-conductive middle portion 536. In this way, the upper portion 532 can be a resonating element of an antenna of the device 500 with the lower portion 534 of the sidewall 528 acting as an electrical grounding plane relative to the resonating plane of the upper portion 532. As noted above, the upper portion 532 can be electrically connected to the PCB 574 of the device 500 such that signals received and sent by the resonating upper portion 532 can be directed to the PCB 574 and can be processed with one or more processors or other electronic components of the device 500, including any processors or other electronic components mounted on the PCB.

The wearable electronic devices described herein can include antennas configured to send and receive electromagnetic signals during use. Incorporating effective antennas into small, compact devices such as wearable electronic watches can be challenging because the greater the distance between a resonating plane and a grounding plane of an antenna, among other factors, the better the performance of the antenna will be. However, space is often limited to create the required Z-distances necessary in compact wearable electronic devices. In devices described herein, the housing and sidewalls of the device can be electrically separated into multiple portions to create resonating elements and grounding elements of an antenna with sufficient separation (Z-distance) therebetween for the housing itself to act as an antenna. However, this design has its own challenges, including electrically connecting the resonating element to a PCB, processor, or other electronic device without reducing the Z-distance of the antenna. Wearable electronic devices described herein are configured to overcome these challenges.

Along these lines, FIGS. 10A and 10B show a top perspective view and a top view, respectively, of a subassembly of a device 600, according to the present disclosure. The subassembly includes housing sidewall 628 that includes the upper portion 632, lower portion 634, and middle portion 636 separating the upper portion 632 from the lower portion 634. As noted above, the upper portion 632 and the lower portion 632 can include electrically conductive material and the middle portion 636 can include electrically insulating or non-conductive material such that the upper portion 632 of the sidewall 628 forms a resonating element of an antenna separated by a distance in the vertical or "Z" direction (or a "Z-distance") relative to the electrical grounding plane of the lower portion 634.

The epoxy component 638 can also be bonded to an inside of the sidewall 628 and to the middle portion 636 and the lower portion 634. In addition, the wave ring 626 is also shown in FIG. 10A. A close-up view of the subassembly from a top view is shown in FIG. 10C, with the close-up region indicated in FIG. 10B, to illustrate a portion of the wave ring 626 extending away from the sidewall 628 to make contact with an electrical contact in the internal volume of the device. As shown in FIG. 10B, the interior portion of the wave ring 626 includes a number of dimples or protrusions 640 configured to engage electrical contacts 572 that can extend through the insulating material 576 and electrically connect to the PCB 574 (as shown in FIG. 9C). As noted above, in addition or alternative to the protrusions 640 on the wave ring 626, protrusions can be formed on the

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housing to facilitate or ensure a secure connection between the wave ring 626 and the electrical contacts 572.

FIG. 10A also illustrates a cutout 650 formed in the inner surface of the housing sidewall 628. As shown, the cutout 650 extends into the housing sidewall 628 below the middle portion 636 and the epoxy portion 638. According to one example, the cutout 650 is formed by machining or otherwise removing a slot of the housing sidewall 628 to create a cavity on the upper portion of the interior volume, prior to assembly of the device 600. As shown, the cutout 650 provides for additional volume within the housing sidewall 628 that can be used for connections or other housings. According to one example, the cutout provides an isolated volume where component connections can be made, such as for microphones or pressure sensors that are typically located near the exterior of the device 600, without consuming valuable interior volume. In one example, board to board connections, such as hot-bar soldering, can be performed on the main PCB within the main cavity, and a flexible cable can then extend from the main PCT to the cutout 650 formed in the inner surface of the housing sidewall 628. Within the cutout 650, according to one example, a tray 652 that can be fastened to the housing can include a circuit board 654 with any number of components 656, such as connectors, components, gyroscopes, accelerometers, etc. that can then be connected to the PCB via the flexible cable. According to this example, the transition of a number of connections to previously unused portions of the housing sidewall 628 allows for added room within the housing for additional battery volume or the inclusion of additional connectors or features on the main PCB.

Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. 10A-10C can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown in the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. 10A-10C.

FIG. 11 shows another cross-sectional view similar to the cross-sectional views shown in FIGS. 9A-9C, where a sidewall 728 includes an upper portion 732, a middle portion 736, and a lower portion 734. The sidewall 728 defines an opening 732 in which a display assembly 706 is disposed, the display assembly 706 including a display cover 722 and one or more other display layers or components 724. An epoxy component 738 can contact the lower portion 734 and the middle portion 736 as shown and one or more other layers or components can be stacked or disposed between the display cover 722 and the epoxy component 738, for example the lth antenna layer 772 and the insulating material 776. In at least one example, the insulating material 776, which can also be referred to herein as a mounting component 776, can structurally support one or more other components, including the electrical contact 772 extending through and/or the display cover 722 or other components 724 of the display assembly 706.

In at least one example, the display mounting component 776 may be formed from a molded material, such as a molded insulating material, including a polymer (e.g., a low-injection-pressure-overmolded polymer). The material that forms component 776 may be epoxy, polyurethane, and/or other polymer materials. Thermoplastic and/or thermoset polymer may be used in forming component 776. Heat and/or light (e.g., ultraviolet light) may be used in

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curing the polymer forming component 776. As one illustrative example, component 776 may be formed from a thermoset structural adhesive such as a one-part heat-cured epoxy. Other polymer(s) may be used, if desired. Vacuum may be applied to the interior of a mold to help draw liquid polymer into a desired shape within a mold during formation of component 776.

One or more surfaces of component 776 can serve as a reference surface (datum) that helps establish a desired physical relationship between component 776 and other portions of a device including the display assembly 706. As an example, component 776 can be attached to an opposing surface of a housing using a layer of adhesive. The shape and location of component 776 relative to display cover layer 722, display layers 724, and other structures in display 706 can help establish a desired position for display 706 relative to a device housing. The upper surface of component 776 can be molded directly to the underside of display cover 722 to help form an environmental seal. In some examples, however, the display assembly 706 can include a separate seal that can aid in forming an environmental seal between the display assembly 706 and a housing.

The location of the display mounting component is indicated in FIGS. 11 and 12. As can be seen, the display mounting component 776 extends around a periphery of the PCB 774. As the PCB 774 is smaller than the display cover 722, the molded insulating material of the display mounting component 776 may be adjacent to an edge of the PCB 774.

In some examples, however, a display assembly 706 for an electronic device can include a PCB 774 that has one or more major dimensions, such as a width and/or height, which are substantially similar to the corresponding major dimension of the display cover 722 or other components 724 of the display assembly 706. By using a PCB 774 with these dimensional relationships, a tail of the display layer 724 can be made flush with a major surface of the PCB 774 so that only a single shut-off is needed during the molding operation which can be used to form the display mounting component. Accordingly, the molded insulating material of the display mounting component 776 can be disposed on a major surface of the PCB 774 and adjacent to a periphery thereon while also at least partially surrounding the flexible tail of an associated display layers 724.

In the present example, the molded material of the display mounting component 776 can also serve to affix the display assembly 706 to the device housing sidewall 728, or at least to a component of the sidewall 728 such as the lower portion 734, and/or to provide an environmental seal between the display cover 722 and the device housing sidewall 728. In some examples, the display mounting component 776 can at least partially define an exterior surface of the device, such as at an upper surface of the upper portion 732. Thus, in some examples, a portion of the insulating molded material of the display mounting component 776 that defines the exterior surface of the device can be positioned between the display cover 722 and a sidewall 728 of the housing. Further, in some examples, the portion of the exterior surface defined by the display mounting component 776 can be substantially level, in line with, and/or flush with portion of the exterior surface defined by the housing sidewall 728 and/or display cover 722.

As noted above, in order to electrically connect the upper portion 732 of the sidewall 728, as a resonating element of an antenna, the electrical connector 772 can extend through the insulating material 776 from the wave ring 726 to the PCB 774 and the wave ring 726 can contact the upper portion 732 of the sidewall 728 at another point or location

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along the length of the wave ring 726, similar to that shown in FIG. 9B and described above. In this way, while the PCB 774 is disposed lower than the upper portion 732 of the sidewall 728, the upper portion 732 of the sidewall 728 can predominantly define an upper resonating plane separated from the grounding plane defined by the lower portion 734 of the sidewall 728. This increased Z-distance between the upper and lower portions 732, 734 of the sidewall 728 correspondingly increases the performance of the antenna of which the upper portion 732 forms a part, or at least forms a part of a resonating element thereof.

As illustrated in FIG. 11, the wavering 726 can vary in profile. According to one example, the wavering 726 is asymmetric in shape, placing the protrusion or dimple lower on the body so that it provides a better contact with the extension feature 784 of the electrical connector 772. In other words, due to the asymmetric profile of the wave ring 726, the dimple or protrusion sits lower in the channel formed by the sidewall 728, and can make a more secure connection to an extension feature 784 having a lower profile. Additionally, as shown in FIG. 11, the wave ring 726 can include any number of optional back dimples 727 to extend toward the wave ring 726 and towards the extension feature 784, ensuring a consistent contact therewith, as well as promoting contact with the housing. Alternatively, or additionally, the sidewall 728 can have selective protrusions or bump-outs 729 that reduce the distance between the wave ring 726 and the extension feature 784, ensuring a secure contact. Such added connection securement features can be especially beneficial in corners of the sidewall 728 where antenna feeds can be located.

Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. 11 can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown in the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. 11.

FIG. 12 shows a top view of a subassembly of a device according to the present disclosure, including a wave ring 826, insulating material 876, electrical contact 872, and PCB 874, which can be similar or part of any of the devices, systems, or subassemblies described herein with reference to other figures. As shown in FIG. 12, portions of the wave ring 826 can be selectively overmolded 827 or otherwise insulated in areas where metal on metal contact with the housing is not desired, such as where plastic to metal housing interlocks may occur. In the illustrated example of FIG. 12, the electrical contact 872 forms a singular unitary piece extending around the PCB 874 with discrete connection points 882 extending onto and contacting the PCB 882 or electrical circuits/pathways or other components of or on the PCB 882.

Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. 12 can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown in the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. 12.

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FIG. 13 shows another example of top view of a subassembly similar to that shown in FIG. 12 but with multiple discrete and separate electrical connectors 972a, 972b, and 972c making contact with the PCB 774. Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. 13 can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown in the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. 13.

In at least one example, an electronic device can include a sidewall 728 including an antenna, the sidewall defining an internal volume. The device can also include a PCB 774 disposed in the internal volume, an insulating material 776 disposed in the internal volume, and an electrical connector 772 contacting the PCB 774, the electrical connector 772 extending through the insulating material 776 and forming an electrical contact between the antenna and the PCB 774.

In one example, the device can include a conductive housing sidewall 728 defining an internal volume, a PCB 774 disposed in the internal volume, an electrical connector 772 contacting the PCB 774 and extending through the insulating material 776, and an elongate conductive member 726 (also referred to herein as the wave ring 726) disposed between the housing sidewall 728 and the electrical connector 772, the elongate conductive member 726 contacting the electrical connector 772 and the housing sidewall 728.

In one example, an electronic device can include a housing sidewall 728 including a lower portion 734 and an electrically conductive upper portion 732 separated from the lower portion 734 by a non-conductive material 736, the housing sidewall 728 defining an internal volume and an opening 732, a display component 722 disposed in the opening 732, a PCB 774 disposed in internal volume below the display component 722, an insulating material 776 disposed in the internal volume between the housing sidewall 728 and the PCB 774, and a connector 772 forming an electrical pathway between the upper conductive portion 732 of the sidewall 728 and the PCB 774. In such an example, the upper portion 732 can form a ring surrounding a periphery of the display component 722.

In at least one example, the insulating material 776 is molded to the electrical connector 772. In one example, as shown in FIG. 12, the insulating material 776 can include a continuous member extending around a periphery of the PCB 774. Similarly, in at least one example, the insulating material 776 can form or include a closed ring disposed between the PCB 774 and the antenna 732 formed by the upper portion 732 of the sidewall 728. In one example, the elongate conductive member 726 contacts the electrical connector 772 at a first location along a length of the elongate conductive member 726 and contacts the housing sidewall 728 at a second location along the length of the elongate conductive member 726. In at least one example, the elongate conductive member 726 contacts the electrical connector 772 at a third location along the length of the elongate conductive member 726 and contacts the housing sidewall 728 at a fourth location along the length of the elongate conductive member 772.

In at least one example, the electrical connector 772 includes a continuous member 886, shown in FIG. 12, that has one or more discrete extensions features 784, as shown in FIG. 11, extending from the continuous member 886 and through the insulating material 776. The extension features

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784 extend through the insulating material 776 to contact the wave ring 726 (or “elongate conductive member”), as shown in FIG. 11.

As noted above, and with reference now to FIG. 14A, an increased Z-distance between the upper portion 1032 as a resonating element of an antenna relative to the electrical ground plane of the lower portion 1034 of the sidewall 1028 can increase the performance of the antenna. In order to increase the Z-distance, the display cover 1022 of the device can include a lower beveled surface to make room for the electrical connector 1072 to extend further upward when making contact with the wave ring 1026, thus increasing the overall plane of the resonating element of the antenna above the grounding plane of the lower portion 1034 of the housing.

For example, as shown in FIG. 14A, the display cover 1022 includes a lower beveled surface 1084, which provides a space for the electrical connector 1072 to make an electrical connection with the wave ring 1026, and thus the upper portion 1032 of the sidewall 1028 that contacts the wave ring 1026, further upward to increase the Z-distance. The Z-distance between the resonating plane 1088 and the grounding plane 1090 is illustrated in FIG. 14B. As shown in FIG. 14A, the electrical connector 1072 extends upward toward the display cover 1022 and into the space that would otherwise be occupied by the display cover 1022 if not for the beveled surface 1084.

The wave ring 1026 can likewise be raised to the level of contact with the electrical connector 1072 shown in FIG. 14A to raise the average height/level of the resonating plane relative to the grounding plane formed at least in part by the lower portion 1034 of the sidewall 1028. In this way, the Z-distance shown in FIG. 14B can be maximized to increase the average height/level of the resonating plane relative to the grounding plane for improved antenna performance.

In at least one example, the insulating material 1076, elsewhere referred to herein as the display mounting component, can extend into the spaced adjacent to the beveled surface of the display cover 1022 to support the display cover and other components of the display assembly 1006, including the various display layers 1024 shown. In addition, in at least one example, a mask layer that can include a PVD layer, ink layer, or other masking layer can be disposed on a lower flat surface 1086 of the display cover 1022 adjacent the beveled surface 1085. The mask can provide an aesthetic feature that reduces unwanted light scattering and reflections at the transition between the lower surface of the display cover 1022 and the beveled surface 1084. In at least one example, the mask can be between about 50 microns and 150 microns thick, for example about 100 microns thick.

Thus, in at least one example, an electronic device described herein can include a sidewall 1028 defining an opening, a display component such as the display cover 1022 disposed in the opening. The display cover 1022 can include a lower beveled edge forming a beveled surface facing the internal volume of the device. In such an example, the insulating material 1076 can contact the beveled surface of the display cover 1022 forming a beveled boundary 1078 as shown in FIG. 14A.

Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. 14A and 14B can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof

shown in the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. 14A and 14B.

Referring briefly back to the exploded view of a device shown in FIG. 2, at least one example of the device 200 can include a back cover 214 and an electromagnetically transparent component 216.

Because wearable electronic devices are in contact with the user's body during use, it can be advantageous to use the device for detecting a user's body temperature, including surface and core temperature. However, the temperature of the device and the environment in which it is used can change from moment to moment during use such that detecting the user's core temperature with a wearable device can be challenging. However, devices described herein can overcome this challenge by incorporating more than one temperature sensor in the device at different locations and applying one or more algorithms that include the temperature sensed by each sensor as an input to determine a core temperature of the user.

FIG. 15A illustrates an example of a device, for example a wearable electronic watch device 1700. In at least one example, the electronic device 1700 can include a housing 1702 defining front and rear openings, with a display component 1706 disposed at the front opening and a rear cover 1714 disposed at the rear opening. The device 1700 of FIG. 15A can also include strap retention features 1779 defined by the housing 1702 for securing a strap to the device 1700. When a strap is connected to the device 1700 via the strap retention features 1779, the device 1700 can be configured to be worn by a user, for example on the wrist of a user, with the strap securing the rear cover 1714 against the skin of the user.

In such an example, the device 1700 can be configured to detect a wrist or skin temperature of the user and extrapolate or detect/measure the user's core temperature. In order to do this, in at least one example, the device 1700 can include two or more temperature sensors on or within the device 1700. For example, a first temperature sensor 1777 can be located at, near, or adjacent the rear cover 1714, as indicated by the lower dot shown in FIG. 15A, also referred to as the bottom or lower side of the device 1700. The dot is not a representation of a sensor itself but indicates an approximate location of the first temperature sensor 1777. In addition, the device 1700 can include a second temperature sensor 1775 located at, near, or adjacent the display component 1706 on an opposite side from the first temperature sensor, also referred to as a top side of the device 1700.

In at least one example, a processor (not shown in FIG. 15A but disposed inside the device 1700) can be electrically connected to the first temperature sensor 1777 and the second temperature sensor 1775 and configured to determine a core temperature of a user based on a first temperature detected by the first temperature sensor 1777 and a second temperature detected by the second temperature sensor 1775.

FIG. 15A shows a partial cross-sectional view of the device 1700 shown in FIG. 15A to illustrate various internal components thereof. As shown, the device 1700 can include a housing 1702 defining front and rear openings and an internal volume, with a display component 1706 disposed at the front opening and a rear cover 1714 disposed at the rear opening. The internal components can include various processors, batteries, microphones, speakers, wires and electrical flexes, antennas, display components, and so forth. In addition, the internal components of the device 1700 can include a first PCB 1773 disposed near, adjacent, and above

the rear cover 1714. In at least one example, the first PCB 1773 can be adhered to the rear cover 1714. The device 1700 can also include a second PCB 1774 disposed near, adjacent, and beneath the display component 1706.

As shown in the cross-sectional view of FIG. 15B, the first temperature sensor 1777 can be disposed on the first PCB 1773 and the second temperature sensor 1775 can be disposed on the second PCB 1774. In at least one example, one or more other electronic components, including heat generating electronic components can be disposed between the first temperature sensor 1777 and the second temperature sensor 1775 or, if not between the temperature sensors, 1777, 1775, a part of a thermal path defined from one temperature sensor to the other through one or more internal component of the device 1700. For example, a battery 1767 can be disposed within the internal volume of the device 1700 at least partially between the first temperature sensor 1777 and the second temperature sensor 1775.

While the first temperature sensor 1777 can be near the user's wrist to determine the temperature at or near the user's wrist, the device 1700 can include other internal components that may generate or absorb heat such that the system temperature of the device 1700 can affect the accuracy of the measurement of the user's wrist with the first temperature sensor 1777. Accordingly, in at least some examples, the device 1700 can include the second temperature sensor 1775 that takes into account the system temperature of the device 1700 and one or more algorithms can be used to determine the user's core temperature using measurement taken from both the first temperature sensor 1777 and the second temperature sensor 1775. In at least one example, the first temperature sensor 1777 and the second temperature sensor can be in electrical communication with one another.

In at least one example, the device 1700 can include one or more processors in electrical communication with the first temperature sensor 1777 and the second temperature sensor 1775. The one or more processors can determine the user's core temperature from measurement taken by both the first and second temperature sensors 1777, 1775 with one or more algorithms applied to the measurement to take into account the system temperature and any thermal path existing through the device 1700 and its internal components disposed therein, some of which may be disposed between the first and second temperature sensors 1777, 1775 or, if not between the temperature sensors, 1777, 1775, a part of a thermal path defined from one temperature sensor to the other through one or more internal component of the device 1700. In this way, determining the user's core temperature can be based, at least in part, on the heat generated by the heat generating components or heat absorbing component or any other components disposed in the internal volume of the device 1700.

Along these lines, FIG. 15C illustrates a circuit diagram equivalent to the device 1700 contacting a user 1771 with a contact interface shown at 1769. The illustrated diagram identifies temperature sensors T_1 and T_2 , which can equate to the first and second temperature sensors 1777, 1775 shown in FIG. 15B, respectively. The heat transfer path from the user 1771, through the device 1700, and out to an external environment can be modeled as a series of resistances illustrated by resistors R_{phys} , $R_{contact}$, R_{BC} , R_{1-2} , and R_{FC} , as shown in FIG. 15C with R_{phys} equating to the resistance of the user, $R_{contact}$ equating to the resistance at the contact interface, R_{BC} equating to the resistance of the back cover or any other components of the device 1700 between the first temperature sensor T_1 (1777 in FIG. 15B)

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measuring temperature T_1 and the contact interface **1769**, R_{1-2} equating to any resistance in the system of the device **1700** including the thermal path between the first temperature sensor **1777** and the second temperature sensor **1775**, and R_{FC} equating to the resistance of the display component **1706** or any other components of the device **1700** between the second temperature sensor (**1775** in FIG. 15B) measuring temperature T_2 and the external surface or external environment of the device **1700**.

Using the modeled circuit diagram of heat flow from the wrist through the device **1700** as shown in FIG. 15C, one or more algorithms can be used to determine the core temperature of the user. For example, a first algorithm modeling the temperature T_{BC} at the rear cover **1714** of the device **1700** can include:

$$T_{BC}=T_1+a_o(T_1-T_2)$$

where:

$$a_o=R_{BC}/R_{1-2}$$

An algorithm modeling a corrected temperature can include:

$$T_c=T_1+c_o(T_1-T_2)$$

where:

$$c_o=a_o+h_o$$

and where:

$$h_o=R_d/R_{1-2}$$

Furthermore, the model can also include self-heating constants (i.e., c_1 , a_1 , h_1).

In at least one example, more than two temperature sensors can be disposed in the device **1700** with measurement taken and input into one or more algorithms to determine the core temperature of the user when the device **1700** contacts the user. As noted above, in at least one example, one or more electrical and/or heat generating components can be disposed in internal volume of the device **1700** and at least partially between the various sensors or at least as part of a thermal path between the various sensors.

In at least one example, the first temperature sensor **1777** can be disposed directly against or adhered directly to the rear cover **1714** or another portion of the housing **1702** near or adjacent or in contact with the user during use. Similarly, in at least one example, the second temperature sensor **1775** can be disposed directly against, or adhered directly to, the display component **1706** or other portion of the housing **1702**. In at least one example, regardless of where each temperature sensor **1777**, **1775** is disposed, a thermally conductive adhesive, such as a thermally conductive pressure-sensitive adhesive, can be used to secure the temperature sensors **1777**, **1775** to another component within the internal volume of the device **1700**.

Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. 15A-15C can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown in the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. 15A-15C.

FIG. 16 illustrates a PCB **1773** with an example of a location of a temperature sensor **1777**. The PCB **1773** can be

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disposed near the rear cover **1714** similar to the first PCB **1773** shown in FIG. 15B. FIG. 26 illustrates a PCB **1774** similar to the second PCB shown in FIG. 15B that can be disposed in the internal volume of the device **1700** near the display component **1706**. In the illustrated example of FIG. 17, two locations of a temperature sensor **1775** are shown where the temperature sensor **1775** can be disposed on the PCB **1774** to be at or near the display component **1706** of the device **1700**. In one example shown, the temperature sensor **1775** can be disposed on the ALS module **1765**. FIG. 18 shows another example of locations of the temperature sensor **1775** on an example of a PCB **1774** with one example location of the temperature sensor **1775** being on an ALS module **1765** of the PCB **1774**.

In at least one example, the temperature sensors **1777**, **1775** described herein can be adhered or otherwise secured to a PCB or other component, including the housing **1702** of the device **1700** without any under-fill material between the temperature sensor **1777**, **1775** and the housing **1702** or PCB **1774**. In at least one example, the temperature sensors **1777**, **1775** can be mounted to a PCB **1774**, housing **1702**, or other portion of the device **1700** without any encapsulation over the temperature sensor **1777**, **1775**. The absence of under-fill material and/or encapsulation over the temperature sensor **1777**, **1775** reduces the complexity and uncertainty of the thermal path between the sensors **1777**, **1775** and/or between the sensors **1777**, **1775** and the user's body or the external environment, thus simplifying the modeling and processing of the user's core temperature.

In at least one example, as shown in FIG. 19, a temperature sensor **1777** is disposed on the ALS module **1765**. In at least one example, as noted above, the temperature sensor **1777** can be adhered to the ALS module **1765** using an SMT/solder or other adhesive or bonding medium without any encapsulation material disposed over and/or encapsulating the temperature sensor **1777**. In such an example, in order to protect the temperature sensor **1777** from physical damage, one or more shields **1767a**, **1767b**, **1767c**, and **1767d** can be disposed around the temperature sensor **1777** such that other components are likely to come into contact with the shields **1767a-d** before coming into contact with the temperature sensor **1777** during assembly or use. The shields **1767a-d** can vary in number, size, position, and configuration but generally are taller than the temperature sensor **1777** such that the shields **1767a-d** physically protect the temperature sensor **1777**. In at least one example, the shields **1767a-d** include inexpensive, non-electrically functioning or connected components. In this way, the shields **1767a-d** can absorb contact, be dented, chipped, or otherwise physically damaged during assembly or use of the device **1700** without negatively affecting the functionality of the device **1700** and the temperature sensor **1777**.

In many scenarios or environments where a user may want to wear the devices disclosed herein, it may be advantageous to output high-frequency, high decibel sounds to alert others of an emergency situation such as a fall or injury to the user. These sounds or alerts can be referred to herein as siren alerts and/or sounds. In order to produce high-frequency siren sounds alongside typical lower frequency outputs for normal use, including music, voice outputs, and so forth, at least one example of a device can include a dual speaker system as shown in FIGS. 20A-21. The speaker assemblies shown herein generally include two speakers sharing a front volume but having separate back volumes associated with each speaker. Separate vents to the external environment, in addition to the separate back volumes, can enable tuning of the speaker assembly to produce clear

frequencies in various high and low ranges from the various speakers in the device **1800** shown.

Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **16-19** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown in the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **16-19**.

FIG. **20A** shows an example of an electronic device **1800** that includes an outer housing **1802** defining an internal volume **1852**, a first speaker **1863** and a second speaker **1861** disposed in the internal volume **1852**, the first speaker **1863** including a frame **1859** disposed around a periphery of a diaphragm **1857** of the first speaker **1863**, a front volume **1855** defined by the outer housing, the first speaker **1863**, and the second speaker **1861**, a first back volume **1853** defined by the first speaker **1863** and the frame **1859**, and a second back volume **1845** defined by the second speaker **1861** and the frame **1859**.

In at least one example, the electronic device **1800** can include an outer housing **1802**, an inner housing **1851** spaced apart from the outer housing **1802**, a speaker assembly disposed between the inner and outer housings **1802**, **1851**. The speaker assembly can include the first speaker **1863**, the second speaker **1861**, and the speaker frame **1859** supporting the first speaker **1863**. The device can further include the first back volume **1853** defined by the inner housing **1851** and the first speaker **1863** and the second back volume **1845** defined by the inner housing **1851** and the second speaker **1861**, with the second back volume **1845** separated from the first back volume **1853** by the speaker frame **1859**.

Another example of the electronic device **1800** can include the outer housing **1802**, the inner housing **1851**, the speaker assembly disposed between the inner and outer housings **1851**, **1802** including the first speaker **1863** and the second speaker **1861**. The device **1800** can also include a front volume **1855** defined by the outer housing **1802** and the speaker assembly, the back volume defined by the inner housing and the speaker assembly and separated into first and second isolated portions **1853** and **1845**, respectively. In at least one example, the device **1800** can also include first vent **1849** defined by the housing **1802** through which a first end of the front volume **1855** is in fluid communication with an external environment and a second vent **1847** defined by the housing **1802** through which a second end of the front volume **1855** is in fluid communication with the external environment.

In at least one example, the front volume **1855** can be isolated from the first and second back volumes **1853** and **1845**, respectively. The speakers **1863**, **1861** can be disposed between the front volume **1855** and the first and second back volumes **1853**, **1845** and the frame **1859** can structurally support the first speaker **1863**. In at least one example, the frame **1859** forms an air-tight seal between the first back volume **1853** and the second back volume **1845**. In addition, as noted above, the inner housing **1851** can at least partially define the first back volume **1853**. For example, the frame **1859** can include a collar **1843** and a molded seal **1841** extending from the collar **1843** toward the internal volume **1852** and contacting the inner housing **1851** to seal the first back volume **1853** as shown behind/below the first speaker **1863**. The collar **1843** can include a metal ring disposed

around the speaker **1863** and configured to redirect magnetic flux around the speaker **1863**. In at least one example, the speaker frame **1859** structurally supports the second speaker **1861**.

In one example, the first speaker **1863** is smaller than the second speaker **1861**. The first speaker **1863** can be referred to as a tweeter and be configured to output higher frequency sound waves than the larger second speaker **1861**. Accordingly, to accommodate the smaller volumetric air displacement caused by the first speaker **1863**, in at least one example, the first back volume **1853** can be smaller than the second back volume **1845**.

In at least one example, as shown in FIGS. **20A** and **20B**, the electronic device **1800** can also include a valve **1839** disposed in an aperture defined by the inner housing **1851** to vent air from the first isolated portion **1853** of the back volume to the internal volume **1852**. In at least one example, the pressure valve **1839** can be configured to equalize pressure between the internal volume **1852** and the back volume **1853**. In at least one example, the valve **1839** can include a mesh and a channel passing through and defined by the inner housing **1851**.

The shared front volume **1855** can be in fluid communication with an external environment through various vents passing through the outer housing **1802**. The location and configuration of each vent can be designed to accommodate high siren-type frequencies output by the smaller tweeter speaker (first speaker **1863**) and lower frequencies output by the second speaker **1861**. In this way, a broader range of frequencies can be output by the speaker assembly clearly and effectively.

In at least one example, the first vent **1849** is formed of a single aperture defined by the outer housing **1802**. The second vent **1847** can include two or more apertures defined by the outer housing **1802**. In at least one example, a distance between any two adjacent apertures of the second vent **1847** can be less than a distance between any aperture of the second vent **1847** and the single aperture of the first vent **1849**.

As noted above, the arrangement and configuration of the speaker assembly of the electronic device **1800** shown in FIGS. **20A** and **20B** enables the speaker assembly to output frequencies in the normal range of daily use, including music, voice, and other typical audio outputs, as well as loud, high frequencies in the range of above 3 kHz, 3.5 kHz, or even above 4.5 kHz from the first smaller speaker **1863** to be used as a siren. The siren can be used in conjunction with a fall-detection system of the device **1800** to alert others if the user has fallen or been injured. During other activities, for example during mountain biking, the siren can output warning signals when coming around a blind corner on a trail or the like. A guardian mode of the device **1800** could activate the siren as an assault whistle or a mugging deterrent.

In order to fit the dual speaker assembly within a tight space between the inner housing **1851** and the outer housing **1802** of the device **1800**, some of the components discussed above and shown in FIGS. **20A-20B** are configured to interface with and be disposed with other component of the device **1800** to form a tight, compact, space-saving device **1800**. For example, the speaker assembly shown in FIGS. **20A** and **20B** can be disposed within the device **1800** in generally the same location as a button of the device, such that the button and the speaker assembly share the same location or portion of the internal volume of the device **1800**. In such an example, the button may include one or more

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components disposed between through, or with one or more components of the speaker assembly.

In order to accommodate the speaker assembly and the button together in the same area, the speaker frame **1859**, as shown in FIG. 20C, can include an opening **1835** defined by the frame **1859**. The opening can be positioned to receive one or more components of a button that pass through the frame **1859**. FIG. 20D illustrates the frame **1859** supporting the first speaker **1863** and the second speaker **1861** and defining the opening **1835**. The opening **1835** can be defined/dispersed between the first speaker **1863** and the second speaker **1861**. As shown in FIG. 20E, a button **1808** can include a plunger **1837** that is aligned with and/or extends through the opening **1835** between the first speaker **1863** and the second speaker **1861**.

In at least one example of the electronic device **1800**, the external housing **1802** can define an internal volume **1852** and an aperture **1833**, as labeled in FIG. 20B. The button **1808** can be disposed in the aperture **1833**. The button **1808** can include the plunger **1837** extending into or toward the internal volume **1852** and the speaker frame **1859** can be disposed in the internal volume **1852** and define the opening **1835**. In such an example, the plunger can extend through the opening.

In one example, the frame **1859** can structurally support the first speaker **1863** and the second speaker **1861**. The frame **1859** can be disposed in the internal volume **1852** with the frame **1859** defining the opening **1835** (otherwise referred to herein as a “hole”) between the first and second speakers **1863**, **1861**, respectively. In at least one example, the plunger **1837** can be aligned with the hole/opening **1835**. A portion of the internal volume **1852** between the inner housing **1851** spaced part from the outer housing **1802** can define a speaker volume including the front volume **1855** and the first and second back volumes **1853**, **1845**, respectively. The plunger **1837** can be aligned with the hole **1835** and extend into the speaker volume toward the inner housing **1851**.

In at least one example, the speaker frame **1859** supports the first speaker **1863** and the second speaker **1861** and the opening **1835** is defined between the first speaker **1863** and the second speaker **1861**. Accordingly, in at least one example, the plunger **1837** extends between the first speaker **1863** and the second speaker **1861**. In at least one example, the plunger **1837** can extend through the front volume **1855** and into the back volume **1845**.

In order to seal off the front volume **1855** from the second back volume **1845**, the device **1800** can include a gasket **1825** surrounding the plunger **1837** and forming a fluid-tight seal between the frame **1859** and the plunger **1837**. Thus a fluid-tight seal is formed by the gasket **1825** between the front volume **1855** and the second back volume **1845**. In at least one example, the gasket can include an O-ring disposed around the plunger **1837**. The plunger **1837** can define a recess in which the O-ring can be disposed and positioned between the plunger **1837** and the speaker frame **1859**. The materials, size, and shape of the O-ring **1825** can be selected to keep fluid out of the volumes surrounding the speakers **1863**, **1861** and to tune the tactile sensation experienced by the user when depressing the button **1808**.

In addition, as the button is pressed downward, the plunger can make contact with an electrical contact **1823** disposed on the inner housing, as shown in at least FIG. 20F. The plunger is aligned with the electrical contact **1823** so that an electrical pathway or circuit can be completed between the plunger **1837** and the electrical contact **1823** when the button **1808** is pressed down during operation.

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Accordingly, the plunger can include or be formed from electrically conductive material.

Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. 20A-20F can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown in the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. 20A-20F.

FIG. 20G illustrates a side, cross-sectional view of a portion of the device **1800** showing a viewing plane orthogonal to the cross-sectional viewing plane of FIG. 20F and extending through the plunger **1837**. FIG. 20G shows the button **1808** and the plunger **1837** extending through the hole **1835** defined by the speaker frame **1859** and the electrical contact **1823** against which a lower surface of the plunger **1837** presses or contacts when the button **1808** is depressed, as shown, to complete an electrical circuit between the plunger **1837** and the electrical contact **1823**. When the button **1808** is not depressed, the plunger **1837** and the electrical contact **1823** are separated so that no electrical connection is made therebetween. The electrical contact **1823** can also be referred to herein as a “tactile switch” or a “tac switch.”

The tac switch **1823** can electrically couple and/or physically contact an electrical flex **1804** partially disposed on a top surface of the inner housing **1851** and at least partially extending under the tac switch **1823** between the tac switch **1823** and the inner housing **1851**. The flex **1804** can extend around an edge of the inner housing **1851** and continue underneath or on a lower surface, opposite the top surface of the inner housing **1851**, as shown. In one example, the bend formed in the flex **1804** as the flex rounds the edge of the inner housing **1851** from one surface to the other can, on its own, bias the portion of the flex **1804** disposed between the tac switch **1823** and the inner housing **1851** away from the inner housing **1851**.

In order to counteract this biasing force away from the inner housing **1851**, the device **1800** can include a foot **1806** pressing downward onto the flex **1804** to keep the flex **1804** in position between the tac switch **1823** and the inner housing **1851**, as shown in FIG. 20G. The foot **1806** can be a molded plastic piece or other non-conductive material anchored to the speaker frame **1859**, the inner housing **1851**, or other component to produce the force of the foot **1806** pressing the flex **1804** against the inner housing as shown. In at least one example, the foot **1806** can also engage the tac switch **1823** such that the foot **1806** presses the tac switch against or toward the flex **1804** and/or the inner housing **1851**. Additionally, or alternatively, one or more adhesives or adhesive layers can be disposed between the flex **1804** and the inner housing **1851**, between the tac switch **1823** and the flex **1804**, and/or between the tac switch **1823** and the inner housing **1851**, to maintain the flex **1804** and tac switch **1823** in position as shown in FIG. 20G.

Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. 20G can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown in the other figures

can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. 20G.

FIG. 20H shows another example of a portion of a device **1800**, including a button **1808** and a button spring **1810**. In at least one example, the button spring **1810** can include upwardly extending spring arms **1812** fixed to the button **1808**. The button spring **1810**, and specifically the spring arms **1812**, can be formed of conductive material, including metal, and shaped so as to provide an upward biasing force against the button **1808**. The button spring **1810** can include a lower portion **1814** anchored to the speaker housing **1859** or other component of the device **1800** relative to which the button **1808** travels when depressed. In at least one example, the button spring **1810** provides a constant force to maintain an upper/outer surface of the button **1808** flush with an outer surface of the housing **1802** of the device **1800** (not shown in FIG. 20H but shown in at least FIGS. 20A and 20B). The materials, shapes, lengths of the spring arms **1814**, and other factors of the button spring **1810** can be tuned to alter the tactile response of the button **1808** when depressed by the user.

Referring briefly to FIG. 20A, when the button **1808** is not depressed, an electrical grounding path can be formed through one or more screws **1816** contacting a portion of the housing **1802**. Accordingly, the screws **1816** and the housing **1802** can be formed of electrically conductive materials. The screws **1816** can act as a stop feature or datum contacting an inner surface of the housing **1802** to prevent the button **1808** from extending beyond the housing **1802** and maintaining a flush external surface with the housing **1802**. Referring again to FIG. 20H, when the button **1808** is partially depressed, the screws **1816** separate from the housing **1802** but the plunger **1837**, which is also in electrical communication with the button **1808**, is not yet in electrical contact with the tac switch **1823**.

As shown in FIG. 20I, in this partially depressed position of the button **1808**, the button spring **1810** can form an electrical grounding pathway between a grounding component or plane of the device **1800** and the button. The button spring **1810** can form such a grounding pathway with the button **1808** whether the button **1808** is fully depressed to contact the plunger against the tac switch **1823**, partially depressed as discussed above, or when not depressed. The lower portion **1814** of the button spring **1810** can electrically contact or couple to a collar **1818**, which can be coupled to ground, or one or more other components forming the grounding pathway. The spring arms **1812** can contact the button **1808** as shown in FIG. 20H to complete the pathway to the button **1808**.

In at least one example, the lower portion **1814** of the button spring **1810** defines an aperture **1820** through which the plunger **1837** extends. In at least one example, the lower portion **1814** of the button spring **1810** forms a bend **1822** that biases portions of the button spring **1810** on either side of the bend **1822** away from each other, contributing to the upward force from the button spring **1810**. For example, the bend **1822** can bias a first portion **1824** on one side of the bend **1822** away from a second portion **1826** on the other side of the bend **1822**.

In at least one example, the first portion **1824** contacts or extends into the housing **1802** at **1828** to complete an electrical pathway from the button **1808**, through the button spring **1810**, to the housing **1802**. In addition, in at least one example, a collar **1818** defines an aperture **1830** and an anti-rotation feature or extension **1832** of the second portion **1826** can extend through the aperture **1830** or at least

partially into the aperture **1830** to prevent the button spring **1810** from rotating out of position as the button **1808** is depressed and travels up and down during use. In at least one example, the anti-rotation feature **1832** engages the collar **1818** without adhesives. In general, the button spring can be disposed and fixed in position as shown without adhesives. The area or volume in which the button spring **1810** is disposed can include an area between the inner housing **1851** and the outer housing **1802** such that any adhesives present could be exposed to chemical aggressors from the external environment, for example through the various vents defined by the housing, including first and second vents **1849** and **1847**, respectively. Thus, the button spring **1810** can be fixed in position via the anti-rotation feature **1832**, an interface with the housing **1802** at **1828**, and/or with the button **1808**.

In at least one example, the device **1800** can include a shim **1834** disposed between the button **1808** or the button cap and the plunger **1837**. In at least one example, the shim **1834** can include a material more elastic or compressible than the button **1808** and/or the plunger **1837**. In one example, the button **1808** and the plunger **1837** include conductive metals and the shim **1834** includes a plastic or rubber material. The shim **1834** can be disposed between and in contact with the button **1808** and the plunger **1837** as shown such that the shim **1834** absorbs forces and movements from the plunger **1837** and the button **1808** as the components of the speakers **1863**, **1861** vibrate and pressure sound waves impinge on the plunger **1837** and button **1808**, the shim **1834** reduces chattering or buzzing caused by the plunger **1837** and button **1808** vibrating against one another. In at least one example, the shim **1834** can include an elastic material. In at least one example, the shim **1834** can include a compressible material. In at least one example, the compressible material can include foam.

Referring back to FIG. 20H, the device **1800** can include speaker meshes **1836a**, **1836b** disposed over speakers **1863**, **1861**, respectively. Each screw **1816** of the button **1808** can include a lower surface **1838** of the screw head facing the mesh **1836a-b**. The lower surface **1838** can be chamfered. The meshes **1836a-b** can be recessed to match or accommodate the curvature of the lower surfaces **1838** of the screws **1816**. The recessed geometry of the meshes **1836a-b** can provide extra space or volume into which the screws **1816** can extend toward the meshes **1836** without the screws **1816** and meshes **1836a-b** contacting or colliding when the button **1808** is depressed. Additionally, the speaker meshes **1836a**, **1836b** can include any number of stacked meshes of varying pore size and material. Pore size can be identified and selected to balance the resistance to ingress of foreign materials, cosmetic benefits, water ejection, and acoustic performance. In some examples, the speaker meshes can both be metal and be welded to the device **1800**. In other examples, the meshes can be metal, fabric, polymer, or a combination thereof, and can be attached to the device **1800** by adhesives, fasteners, welding or other joining methods, and the like.

Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. 20H-20I can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown in the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. 20H-20I.

FIG. 20J shows a cross-sectional view of a portion of a device **1800**, including a button **1808** disposed in an aperture **1833** defined by a housing **1802** and a speaker diaphragm **1836** disposed in an internal volume of the device **1800**. The speaker diaphragm **1836**, the button **1808**, and the housing **1802** can define the front volume **1855**, which is also shown and labeled in FIG. 20A. In at least one example, the device **1800** can include an acoustic gasket **1840** extending between the button **1808** and the housing **1802**. In at least one example, the gasket **1840** is greater than or equal to about 100 microns thick where the gasket **1840** contacts the housing **1802**. In at least one example, an upper surface of the gasket **1840** interfaces at an angle with the vertical surface of the housing **1802** defining the aperture **1833** at greater than 0-degrees, for example at least about 20-degrees or more relative to the horizontal plane orthogonal to the surface of the housing **1802** defining the aperture **1833**. In at least one example, the gasket includes an elastic material. In at least one example, the gasket **1840** includes a material with a Shore-A hardness of between about 30 A and 90 A, or between about 40 A and 80 A, or between about 50 A and 70 A, for example about 60 A.

Accordingly, with a gasket **1840** having the above-noted dimensions and material properties, the gasket **1840** can maintain and rebound to its resting shape after the button **1808** is depressed by the user. In addition, according to the above-noted dimensions and material properties, the gasket **1840** can seal the front volume **1855** such that pressure can build up greater than an atmospheric pressure external to the device **1800**. In this way, the volume of the first speaker **1863** can be increased. In at least one example, the material properties, shape, and dimensions of the gasket **1840** can be tuned to maximize at least one of the resonant frequencies of the first speaker **1863**. In at least one example, the gasket **1840** can be permeable to water but impermeable to dust and debris from the external environment.

The first speaker **1863** can include two peak resonance frequencies, a mechanical resonant frequency generated by shape of the first speaker **1863** itself operating in open air and front port resonance leveraging the length of the first vent **1849** as a tube that creates a higher pitch frequency and lets sound out from the first speaker **1863** through the housing **1802**. The pressure built up in the front volume **1855**, in part due to the seal formed by the gasket **1840**, affects the pressure waves of sound from the first speaker **1863** exiting the first vent **1849**. In this way, the gasket **1840** can tune the sound from the first speaker **1863** and increase the resonant tube frequency. In this way, multiple resonant frequencies (mechanical and tube) can be utilized and a broader range of sound frequencies can be increased.

Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. 20J can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown in the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. 20J.

FIG. 20K illustrates a cross-sectional view of a portion of a device **1800** including a button **1808**, a housing **1802**, and a gasket **1840** extending between the housing **1802** and the button **1808**. The button **1808** can include an exterior color layer **1842** that can be formed by physical vapor deposition. The color layer **1842** can also be referred to as a PVD layer **1842**. The PVD layer can extend over a first surface **1844**

angled or curved at a first angle or curvature and a second surface **1848** angled at a second angle different than the angle of the first surface **1844**. During the PVD process of forming the PVD layer **1842**, the PVD layer **1842** can be deposited onto the button **1808** in a constant direction regardless of the curvature, angles, or portion of the surface being deposited, in order to simplify the PVD process. In one example, the direction of deposition is indicated by the deposition direction **1850**.

Because the first angle of the first surface **1844** is different than the second angle of the second surface **1848**, relative to the deposition direction **1850**, the PVD layer **1842** deposited on the first surface **1844** is formed thicker than the PVD layer **1842** of the second surface **1848**. This can be due to the steeper angle of the second surface **1848** relative to the horizontal plane of FIG. 20K orthogonal to the deposition direction **1850**. As shown, the PVD layer **1842** of the second surface **1848** is thinner than the PVD layer **1842** of the first surface **1844** due to this difference in angle relative to the deposition direction **1850**. The thickness of the PVD layer **1842** affects the color of the PVD layer **1842**. In one example, the thicker PVD layer **1842** at the first surface **1844** can appear red while the color of the PVD layer **1842** at a transition or corner surface **1846** between the first and second surfaces **1844**, **1848** can appear blue and the color of the thinner PVD layer **1842** of the second surface **1848** can shift back toward red, for example appearing orange or red-orange. The thickness of the PVD layer **1842** at the corner **1846** can be thicker than the PVD layer **1842** at the second surface **1848** but thinner than the PVD layer **1842** at the first surface **1844**.

In the example noted above, the blue color of the PVD layer **1842** at the corner **1846** stands out more visually compared to the red color of the PVD layer **1842** at the first surface **1844** than the orange color of the PVD layer **1842** at the second surface **1848**. In order to minimize the contrast of the blue and red between the corner **1846** and the first surface **1844**, the corner **1846** can include a small radius of curvature to minimize the area of the surface of the button **1808** defined by the corner **1846**. While the PVD layer **1842** at the second surface **1848** is thinner than the PVD layer **1842** at the corner **1846**, the orange color of the PVD layer **1842** at the second surface **1848** is closer to red and visually less contrasting or noticeable. Thus, the angle of the second surface **1848** can be chosen to tune the thickness of the PVD layer **1842** on the second surface **1848** relative to the thickness of the PVD layer **1842** of the first surface **1844** in order to minimize the color difference.

In at least one example, the second surface **1848** is angled between about 1-degree and 10-degrees relative to the deposition direction **1850**, or between about 3-degrees and about 7-degrees relative to the deposition direction **1850**, for example at about 5-degrees relative to the deposition angle **1850**. In this way, the thickness of the PVD layer **1842** at the second surface **1848** can be less than about 50% of the thickness or less than the thickness of the PVD layer **1842** of the first surface **1844**. In examples where the PVD layer **1842** at the second surface **1848** is less than about 50% of the thickness of the PVD layer **1842** at the first surface, the color difference between the PVD layer **1842** of the first and second surfaces **1844**, **1848** can be visually minimized.

Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. 20K can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures. Likewise, any of the features, components, and/or parts, including the arrange-

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ments and configurations thereof shown in the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. 20K.

FIG. 21 illustrates a cross-sectional view of the assembly shown in FIGS. 20A-20F, with 1852 showing an internal volume of the device and 1802 being the outer housing. During manufacturing, in order to simplify machining of the housing 1802, an angled receiving cavity for the speaker assembly can be machined into an inner surface of the housing 1802 so that machining tools can reach needed points in the housing to machine the cavity. Thus, in at least one example, the speaker assembly, including the speaker 1863 shown in FIG. 21, can be disposed at an angle relative to the horizontal plane 1831 of the device 1800. In one example, the speaker angle θ can be between about 5-degrees and 10-degrees from the horizontal plane 1831 of the device 1800, for example about 7.5-degrees, and the cavity angle β can be between about 7-degrees and 13-degrees beyond the speaker angle θ , for example about 10-degrees.

Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. 21 can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown in the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. 21.

As noted above, the wearable electronic devices described herein can be configured to be used during any daily activity of the user. Often, the wearable device will rub against other objects, including clothing, or be subjected to wind if outside during use. Typically, these types of interactions, including rubbing, scratching, and wind blowing can cause negatively affect the performance of one or more microphones of the device. For example, some wearable devices can include a microphone to receive the user's voice during a cellular call using the device. However, often when speaking in an outside environment where wind is present, the wind can cause unwanted noise as it passed over the device and specifically as it passed over one or more microphone apertures in the housing of the device, which can create unwanted background noise and unclear voice transmission from the device.

For example, as shown in FIG. 22, a user is wearing a wearable electronic watch 1900 on his/her wrist while riding a bicycle. Such an activity causes wind to pass over and contact the device 1900. The same may be true while walking, jogging, hiking, or any other active and/or outdoor activity. FIG. 23 shows the device 1900 subject to wind flowing toward and around the device 1900. Flow lines 1919 illustrate one possible flow path of wind crossing over the device 1900. In some examples, turbulent currents 1921 can be created at one or more sides of the device 1900. Such wind and turbulent flow can travel over certain components or opening to microphones and speakers of the device 1900 and cause unwanted noise when transmitting the user's voice during a cellular call or while recording his or her voice with the device 1900.

The extent of such noise interference from wind can vary depending on the location of the microphone and the direction of the wind. As noted above, the principles discussed with respect to wind interference can also be true for other types of interference such as water and moisture interference and rubbing or scratching the device 1900 against other

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objects such as clothes. FIG. 24 shows wind 1919 coming from various different directions all around the device 1900. The locations 1917a, 1917b, 1917c, and 1917d on the device 1900 illustrate potential locations of a microphone of the device 1900. Again, the extent of noise interference from wind can vary depending on the location of the microphone and the direction of the wind and those directions and locations can vary from one moment to another during use.

In order to reduce the interference from wind and other objects, wearable electronic devices of the present systems and methods can include three microphones disposed in the internal volume of the device and configured to receive sound through three respective apertures. The location and orientation of the apertures and microphones can be such that while one or two of the microphones may pick up wind interference during use, at least one of the microphones and apertures will be positioned and oriented to pick up less interference noise. In such a configuration, the device can be configured to process the combined noise detected by all three microphones to reduce the noise. In one example, the device can be configured to rely more heavily on microphones picking up less interference noise such that the detected noise is clear and un-affected by the interference noise caused by the wind.

In at least one example, as shown in FIG. 25, a wearable electronic watch 2000 can include a housing sidewall 2028 defining an internal volume 2052 with the sidewall 2028 extending 360-degrees circumferentially around the internal volume 2052. The sidewall 2028 can also define a first aperture 2015, a second aperture 2013 between about 155-degrees and 205-degrees relative to the first aperture 2015, and a third aperture 2011 closer to the second aperture 2013 than the first aperture 2015.

In addition, the device 2000 can include a first microphone 2009 disposed in the internal volume 2052 and configured to receive sound through the first aperture 2015, a second microphone 2007 disposed in the internal volume 2052 and configured to receive sound through the second aperture 2013, and a third microphone 2005 disposed in the internal volume 2052 and configured to receive sound through the third aperture 2011.

In one example, the device 2000 can include a first strap receiving feature 2001 and a second strap receiving feature 2003 opposite the first strap receiving feature 2001. A first sidewall portion 2004 can extend between the first strap receiving feature 2001 and the second strap receiving feature 2003 with the first sidewall portion 2004 defining the first aperture 2015 closer to the first strap receiving feature 2001 than the second strap receiving feature 2003. Further, one example can include a second sidewall portion 2006 disposed opposite the first sidewall portion 2004 and extending between the first strap receiving feature 2001 and the second strap receiving feature 2003, the second sidewall portion 2006 defining a second aperture 2013 and a third aperture 2011, the second aperture 2013 defined closer to the second strap receiving feature 2003 than the first strap receiving feature 2001. In such an example, as shown in FIG. 25, the device 2000 can include a first microphone 2009 disposed in the internal volume 2052 adjacent the first aperture 2015, a second microphone 2007 disposed in the internal volume 2052 adjacent the second aperture 2013, and a third microphone 2005 disposed in the internal volume 2052 adjacent the third aperture 2011. While the present system is described as detecting noises and wind from various side directions, the present system can also include microphones oriented to detect sounds and wind from various orientations including into and out of the page illustrated in FIG. 24.

In one example, as shown in FIG. 25, the electronic device 2000 can include a fourth aperture 2008. The first microphone 2009 can be disposed in the internal volume 2052 adjacent the first aperture 2015, the second microphone 2007 can be disposed in the internal volume 2052 adjacent the second aperture 2013, and the third microphone 2005 can be disposed in the internal volume 2052 adjacent the third aperture 2011. In addition, a speaker 2010 can be disposed in the internal volume 2052 adjacent the fourth aperture 2008 such that a distance along the sidewall 2028 between the first and second apertures 2015, 2013 is larger than a distance along the sidewall 2028 between the second and third apertures 2013, 2011 and the fourth aperture 2008 is adjacent the first aperture 2015.

In at least one example, the second aperture 2013 and the third aperture 2011 can be defined on a distal side of the wearable electronic watch. The distal side of the wearable electronic watch 2000 can include or be defined by the second sidewall portion 2006 where the term “distal” refers to anatomically distal when worn on the wrist of the user. In other words, the distal side of the wearable electronic watch 2000 includes the side facing the hand of the user when worn. Conversely, the proximal side of the wearable electronic watch 2000 can include or be defined by the first sidewall portion 2004 where the term “proximal” refers to anatomically proximal when worn on the wrist of the user. In other words, the proximal side of the wearable electronic watch 200 includes the side facing the forearm of the user when worn. In at least one example, the first aperture 2015 can be defined on the proximal side of the wearable electronic watch 2000.

In at least one example, the second aperture 2013 can be defined between about 170 and 190 degrees relative to the first aperture 2015. In such an example, the third aperture 2011 can be defined between about 30 and 60 degrees counterclockwise along the sidewall 2028 relative to the second aperture 2013. In one example, the third aperture 2011 can be defined between about 40 and 50 degrees counterclockwise along the sidewall 2028 relative to the second aperture 2013.

In at least one example of the wearable electronic watch 2000, the sidewall 2028 defines the strap receiving feature 2001 between the first aperture 2015 and the second aperture 2013. The sidewall 2028 can further define the second strap receiving feature 2003 opposite the first strap receiving feature 2001 and between the third aperture 2011 and the first aperture 2015. In at least one example, the first aperture 2015 can be defined closer to the first strap receiving feature 2001 than the second strap receiving feature 2003 and the second aperture 2013 can be defined closer to the second strap receiving feature 2003 than the first strap receiving feature 2001. In one example, the third aperture 2011 can be defined between the second aperture 2013 and the first strap receiving feature 2001.

In at least one example, the third aperture 2011 is defined between about 30 and 60 degrees counterclockwise from the second aperture 2013 along the housing sidewall 2028. The first aperture 2015 can be defined between about 170 and 190 degrees from the second aperture 2013 along the housing sidewall 2028. In at least one example, the first aperture 2015 can be defined proximally relative to the first strap receiving feature 2001 and the second strap receiving feature 2003 and the second aperture 2013 and the third aperture 2011 can be defined distally relative to the first strap receiving feature 2001 and the second strap receiving feature 2003.

In at least one example, the first microphone 2009 can be oriented to receive sound from a first direction and the second microphone 2007 can be oriented to receive sound from a second direction different than the first direction. In one example, the second direction is opposite the first direction. In such an example, the first aperture 2015 and the fourth aperture 2008 can be defined on a proximal side of the electronic device 2000. In such an example, the second and third apertures 2013, 2011 can be defined on a distal side of the electronic device 2000.

Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. 22-25 can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown in the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. 22-25.

FIG. 26 illustrates another example of a wearable electronic device 2100 including a sidewall 2128 defining an internal volume 2152 and first, second, and third apertures 2115, 2113, and 2111, respectively. A first microphone 2109 is disposed in the internal volume 2152 adjacent the first aperture 2115 and configured to receive sound through the first aperture 2115. A second microphone 2107 is disposed in the internal volume 2152 adjacent the second aperture 2113 and configured to receive sound through the second aperture 2113. A third microphone 2105 is disposed in the internal volume 2152 adjacent the third aperture 2111 and configured to receive sound through the third aperture 2111. In the illustrated example of FIG. 26, the apertures 2115, 2113, and 2111 are defined by a distal sidewall portion 2106 between the first and second strap receiving features 2101 and 2103, respectively.

Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. 26 can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown in the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. 26.

FIG. 27 shows another example of a wearable electronic device 2200 including a sidewall 2228 defining an internal volume 2252 and first and second apertures 2215 and 2213, respectively. A first microphone 2209 is disposed in the internal volume 2252 adjacent the first aperture 2215 and configured to receive sound through the first aperture 2215. A second microphone 2207 is disposed in the internal volume 2252 adjacent the second aperture 2213 and configured to receive sound through the second aperture 2213. In the illustrated example of FIG. 27, the aperture 2215 is defined by a proximal sidewall portion 2204 between the first and second strap receiving features 2201 and 2203, respectively. The second aperture 2213 is defined by a distal sidewall portion 2206 between the first and second strap receiving features 2201 and 2203, respectively.

Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. 27 can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures. Likewise, any of the features, components, and/or parts, including the arrange-

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ments and configurations thereof shown in the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. 27.

FIG. 28 illustrates another example of a wearable electronic device 2300 including a sidewall 2328 defining an internal volume 2352 and first and second apertures 2313 and 2311, respectively. A first microphone 2307 is disposed in the internal volume 2352 adjacent the first aperture 2313 and configured to receive sound through the first aperture 2313. A second microphone 2305 is disposed in the internal volume 2352 adjacent the second aperture 2311 and configured to receive sound through the second aperture 2311. In the illustrated example of FIG. 28, the apertures 2312 and 2111 are defined by a distal sidewall portion 2306 between the first and second strap receiving features 2301 and 2303, respectively.

Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. 28 can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown in the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. 28.

In each example shown in FIGS. 25-28, the device can be configured to receive and process multiple audio signals from the multiple microphones through the multiple apertures and identify the microphone with the lowest perceived wind noise. This microphone can be used as the baseline and data can be extracted from the other locations to process a clear audio signal. This can improve audio transmission and detection performance in windy conditions, when the device is rubbed against another object, or when one or more microphone apertures gets clogged with debris and/or liquid.

FIG. 29A shows a bottom elevation view of an example of a back cover 1114 and an electromagnetically transparent component 1116 assembled together to form the back side or back surface of a device 1100. In at least one example, the back cover 1114 can be secured to the sidewalls 1128 of the device 1100.

In at least one example, the back cover 1114 can be secured to the sidewalls 1128 using one or more fasteners 1192. In the illustrated example of FIG. 29A, four fasteners 1192 are used to secure the back cover 1114 to the sidewalls 1128, with one fastener 1192 disposed at each corner of the device 1100. Using the fasteners shown, the back cover 1114, which in some examples can be made of ceramic, glass, or other brittle material, the back cover 1114 can be secured to the sidewalls 1128 without cracking, separating from the sidewalls 1128, or otherwise being damaged during assembly.

In at least one example, the back cover 1114 can include zirconia, or other brittle material, which is hard to CNC and machine to form intricate connection features. Using the fasteners shown in the figures to secure the back cover 1114 to the sidewalls 1128 of the device 1100 can simplify the geometry of the back cover 1114 in order to simplify the manufacturing process thereof. For example, as shown in the cross-sectional view of FIG. 15B, the back cover 1114 can be formed of a simple geometry extending around the electromagnetically transparent component 1116 and defining a through hole 1194 for each fastener 1192 to pass through.

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In at least one example, each fastener 1192 can be disposed through the back cover 1114 a certain distance away from an outer peripheral edge of the back cover 1114 such that enough material is present between the fastener 1192 and the outer peripheral edge of the back cover 1114 to prevent cracking of the back cover between the fastener and the outer edge. This distance is also designed to reduce any stress concentrations in the back cover 1114 during and after assembly as the fastener 1192 presses the material of the back cover 1114 against the sidewalls 1128.

As shown in FIGS. 30 and 31, in at least one example, the through hole 1194 defined by the back cover 1114 can include a counter bore in which the head 1196 of the fastener 1192 is disposed when assembled. In addition, in at least one example, the head 1196 of the fastener 1192 can include an outwardly extend flange 1198 with a gasket 1199 pressed between the flange 1198 and the back cover 1114. In one example, the fastener 1192 can include a threaded screw. When assembled, the threaded screw can be threaded into a threaded receiving hole defined by the sidewalls 1128 such that the head 1196 of the fastener 1192 presses the back cover 1114 against the sidewalls 1128.

The flange 1198 thus presses against the gasket 1199 forming an environmental seal against external moisture and other debris from entering the through hole 1194. This environmental seal can also reduce corrosion of the fastener itself as it prevent water or other moisture/fluid from entering the through hole 1194 and coming into contact with the fastener 1192 disposed inside the through hole 1194. FIG. 31 shows an O-ring seal 1197 instead of the gasket shown in the example of FIG. 30, and FIG. 32 shows a side view of the fastener 1192 with the gasket 1199 disposed underneath the flange 1198.

Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. 29A-32 can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown in the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. 29A-32.

In at least one example, as shown in FIG. 33A, the fastener 1292 can include an eave feature 1295 defined by the flange 1298 on a lower side of the head 1296 configured to constrain an O-ring or gasket laterally as the fastener 1292 pressed downward on the O-ring or gasket. FIG. 33B shows an example of an O-ring 1297 disposed in the eave feature 1295. Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. 33A and 33B can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown in the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. 33A and 33B.

FIG. 34 illustrates a five-pointed punch 1393 indented into a top surface of the head 1396 of a fastener 1392 with convex transition edges between each of the five points of the punch shape. FIG. 35 illustrates a five-pointed punch 1493 indented into a top surface of the head 1496 of a fastener 1492 with five concave points to form a five-leaf clover shape of the punch 1493. These punches 1393, 1493 can provide aesthetically pleasing punch designs and tool-

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specific mating features for assembly and disassembly that increase surface area and engagement for secure fastening and removal thereof.

Further, as shown in the top and side views of FIGS. 36A and 36B, respectively, the top surface of a fastener head 1596 can include patterns and lines 1591 to further improve the aesthetic appeal, while improving surface engagement and secure removal, of the fastener 1592 and other fasteners described herein and shown in other figures. In at least one example, the lines 1591 can be scored, machined, etches, or otherwise physically formed into the surface of the head 1596. As shown from the side view of FIG. 36B, the lines or scoring features 1591 can be formed to a certain depth into the head 1596 of the fastener 1592.

FIG. 37 illustrates a flow chart of a method 1600 of forming the score lines and/or machined features 1591 shown in FIGS. 36A and 36B. In a first step of the method 1600, a 50-degree feature is machined or otherwise formed into the surface at a depth of approximately 0.01 mm. Next, in step 1687 of the method 1600, the feature can be widened to approximately 130-degrees at the same depth. Next, at step 1685, the feature can be increased to a depth of approximately 0.05 mm to form an approximately 45-degree feature. Then, at step 1683, the feature can be widened to approximately 60-degrees at the same depth of approximately 0.05 mm. Next, the depth of the feature can be increased to approximately 0.10 mm to form an approximately 45-degree feature. The angles and depth dimensions shown in FIG. 37 and described herein are exemplary only and can vary to form features of different sizes, shapes, number, and depth. In general, the depth and angle of the features can be iteratively widened and deepened as described until the desired depth and angle of each feature is accomplished.

Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. 33A-37 can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown in the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. 33A-37.

Wearable electronic devices currently on the market, including current wearable electronic watches, cannot accurately detect environmental pressures in both submerged environments and above water. Usually, this is because the scale of pressures is so different between air pressure above water and fluid pressure below water. It can be especially difficult to configure a single pressure sensor into such a device that is sensitive enough to detect changes in air pressure above water but robust enough to detect pressure changes under water, for example up to 10-bar under water.

However, devices of the present disclosure, including the wearable electronic devices and watches described herein, can include a single pressure sensor to detect pressure above water and below water up to 10-bar. In at least one example, the pressure sensor can be electrically connected to an ASIC switch and associated circuitry and processors to switch pressure scales when high pressures are detected when the device is submerged under water.

For example, such an ASIC circuitry connected to the sensor can include a low gain mode that measures depth and a high gain mode that measures depth and elevation. This change in gain can be switched with the ASIC to tune the sensor between water and air sensitivities. The processor of

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the device can also receive temperature measurement from a temperature sensor of the device to take into account the temperature of the external environment, which can affect the pressure sensor reading and sensitivity. Along these lines, at least one example of such a device can also include a heater to apply heat to the pressure sensor in order to perform a health check on the sensor to calibrate the sensor to its original calibration that was performed or set at the same temperature to which it is heated by the heater.

To the extent applicable to the present technology, gathering and use of data available from various sources can be used to improve the delivery to users of invitational content or any other content that may be of interest to them. The present disclosure contemplates that in some instances, this gathered data may include personal information data that uniquely identifies or can be used to contact or locate a specific person. Such personal information data can include demographic data, location-based data, telephone numbers, email addresses, TWITTER® ID's, home addresses, data or records relating to a user's health or level of fitness (e.g., vital signs measurements, medication information, exercise information), date of birth, or any other identifying or personal information.

The present disclosure recognizes that the use of such personal information data, in the present technology, can be used to the benefit of users. For example, the personal information data can be used to deliver targeted content that is of greater interest to the user. Accordingly, use of such personal information data enables users to calculated control of the delivered content. Further, other uses for personal information data that benefit the user are also contemplated by the present disclosure. For instance, health and fitness data may be used to provide insights into a user's general wellness, or may be used as positive feedback to individuals using technology to pursue wellness goals.

The present disclosure contemplates that the entities responsible for the collection, analysis, disclosure, transfer, storage, or other use of such personal information data will comply with well-established privacy policies and/or privacy practices. In particular, such entities should implement and consistently use privacy policies and practices that are generally recognized as meeting or exceeding industry or governmental requirements for maintaining personal information data private and secure. Such policies should be easily accessible by users, and should be updated as the collection and/or use of data changes. Personal information from users should be collected for legitimate and reasonable uses of the entity and not shared or sold outside of those legitimate uses. Further, such collection/sharing should occur after receiving the informed consent of the users. Additionally, such entities should consider taking any needed steps for safeguarding and securing access to such personal information data and ensuring that others with access to the personal information data adhere to their privacy policies and procedures. Further, such entities can subject themselves to evaluation by third parties to certify their adherence to widely accepted privacy policies and practices. In addition, policies and practices should be adapted for the particular types of personal information data being collected and/or accessed and adapted to applicable laws and standards, including jurisdiction-specific considerations. For instance, in the US, collection of or access to certain health data may be governed by federal and/or state laws, such as the Health Insurance Portability and Accountability Act (HIPAA); whereas health data in other countries may be subject to other regulations and policies and should

be handled accordingly. Hence different privacy practices should be maintained for different personal data types in each country.

Despite the foregoing, the present disclosure also contemplates embodiments in which users selectively block the use of, or access to, personal information data. That is, the present disclosure contemplates that hardware and/or software elements can be provided to prevent or block access to such personal information data. For example, in the case of advertisement delivery services, the present technology can be configured to allow users to select to “opt in” or “opt out” of participation in the collection of personal information data during registration for services or anytime thereafter. In another example, users can select not to provide mood-associated data for targeted content delivery services. In yet another example, users can select to limit the length of time mood-associated data is maintained or entirely prohibit the development of a baseline mood profile. In addition to providing “opt in” and “opt out” options, the present disclosure contemplates providing notifications relating to the access or use of personal information. For instance, a user may be notified upon downloading an app that their personal information data will be accessed and then reminded again just before personal information data is accessed by the app.

Moreover, it is the intent of the present disclosure that personal information data should be managed and handled in a way to minimize risks of unintentional or unauthorized access or use. Risk can be minimized by limiting the collection of data and deleting data once it is no longer needed. In addition, and when applicable, including in certain health related applications, data de-identification can be used to protect a user’s privacy. De-identification may be facilitated, when appropriate, by removing specific identifiers (e.g., date of birth, etc.), controlling the amount or specificity of data stored (e.g., collecting location data a city level rather than at an address level), controlling how data is stored (e.g., aggregating data across users), and/or other methods.

Therefore, although the present disclosure broadly covers use of personal information data to implement one or more various disclosed embodiments, the present disclosure also contemplates that the various embodiments can also be implemented without the need for accessing such personal information data. That is, the various embodiments of the present technology are not rendered inoperable due to the lack of all or a portion of such personal information data. For example, content can be selected and delivered to users by inferring preferences based on non-personal information data or a bare minimum amount of personal information, such as the content being requested by the device associated with a user, other non-personal information available to the content delivery services, or publicly available information.

The foregoing description, for purposes of explanation, used specific nomenclature to provide a thorough understanding of the described embodiments. However, it will be apparent to one skilled in the art that the specific details are not required in order to practice the described embodiments. Thus, the foregoing descriptions of the specific embodiments described herein are presented for purposes of illustration and description. They are not intended to be exhaustive or to limit the embodiments to the precise forms disclosed. It will be apparent to one of ordinary skill in the art that many modifications and variations are possible in view of the above teachings.

What is claimed is:

1. An electronic device, comprising:

a housing including a sidewall defining an upper peripheral edge surrounding an opening; and
a transparent display cover disposed in the opening, the transparent display cover including an outward facing surface;

wherein;

the transparent display cover is positioned such that the upper peripheral edge is level with the outward facing surface; and

the transparent display cover is spaced apart from the sidewall to define a gap forming a cavity extending below the transparent display cover.

2. The electronic device of claim 1, wherein the electronic device includes a display assembly comprising the transparent display cover.

3. The electronic device of claim 2, wherein the outward facing surface is planar.

4. The electronic device of claim 1, wherein the display assembly further comprises a touch interface.

5. The electronic device of claim 1, further comprising a protrusion extending from the sidewall, the protrusion defining a recess.

6. The electronic device of claim 5, further comprising a crown disposed in the recess.

7. The electronic device of claim 6, wherein the protrusion surrounds a portion of the crown.

8. The electronic device of claim 5, wherein:

the recess is a first recess;

the protrusion further defines a second recess; and

the electronic device further comprises a button disposed in the second recess.

9. An electronic device, comprising:

a housing sidewall defining:

a first speaker vent;

a second speaker vent; and

an aperture between the first speaker vent and the second speaker vent; and

a button disposed between the first speaker vent and the second speaker vent in the aperture.

10. The electronic device of claim 9, wherein:

the housing sidewall defines an internal volume;

the electronic device further comprises a first speaker disposed within the internal volume, the first speaker positioned to direct sound through the first speaker vent; and

the electronic device further comprises a second speaker disposed within the internal volume, the second speaker positioned to direct sound through the second speaker vent.

11. The electronic device of claim 10, wherein the housing sidewall further defines a front speaker volume disposed between the housing sidewall and the first and second speakers.

12. The electronic device of claim 9, wherein:

the first speaker vent comprises at least two openings defined by the housing sidewall; and

the second speaker vent consists of a single opening defined by the housing sidewall.

13. The electronic device of claim 9, wherein the housing sidewall defines:

a first side of the electronic device; and

a second side of the electronic device opposite the first side;

wherein the first side defines the first and second speaker vents and the aperture.

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14. The electronic device of claim 13, wherein the second side defines a protrusion defining a recessed feature.

15. The electronic device of claim 14, further comprising a crown disposed in the recessed feature.

16. An electronic device, comprising:

a housing defining an internal volume and an upper peripheral edge, the housing including:

a first sidewall defining a first speaker vent and a second speaker vent; and

a second sidewall opposite the first sidewall, the second sidewall including a protrusion defining a recessed feature;

a transparent display cover flush with the upper peripheral edge;

a button disposed between the first speaker vent and the second speaker vent; and

a dial disposed in the recessed feature.

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17. The electronic device of claim 16, wherein the upper peripheral edge of the housing is configured to protect the transparent display cover from inadvertent contact during use.

18. The electronic device of claim 16, wherein the protrusion is configured to protect the dial from inadvertent contact during use.

19. The electronic device of claim 16, wherein:
the first sidewall defines an aperture between the first speaker vent and the second speaker vent; and
the button is disposed in the aperture.

20. The electronic device of claim 16, wherein:
the protrusion is a first protrusion;
the housing includes a second protrusion extending upward toward the upper peripheral edge, the second protrusion including an upper portion separated from a lower portion by a non-metallic material portion, the upper portion defining the upper peripheral edge; and
the upper peripheral edge surrounds the display cover.

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